

Epigenomes, DNA Modifications, and Single-Molecule genome sequencing



@mason_lab

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Director of the Initiative for Quantitative Prediction

Department of Physiology and Biophysics &

The Institute for Computational Biomedicine (ICB),

Meyer Cancer Center, Feil Family Brain and Mind Research Institute,

at Weill Cornell Medicine,

Fellow of the Information Society Project, Yale Law School

April 29th, 2025

Precision Medicine



Weill Cornell Medicine

Englander Institute
for Precision Medicine

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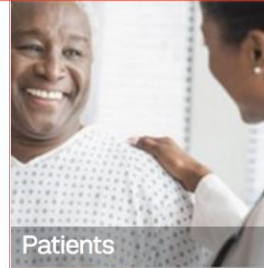
Resources

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The Precision
THE PRECISION ME Medicine Initiative



Patients



Clinicians



Publications



All of Us Program

to Predictive Medicine



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WorldQuant Initiative for Quantitative Prediction Established at Weill Cornell Medicine

Initiative Combines State-of-the-Art Molecular Profiling Technologies with Advanced Financial Algorithms and Computational Methods to Enhance Precision Medicine

APRIL 28, 2017



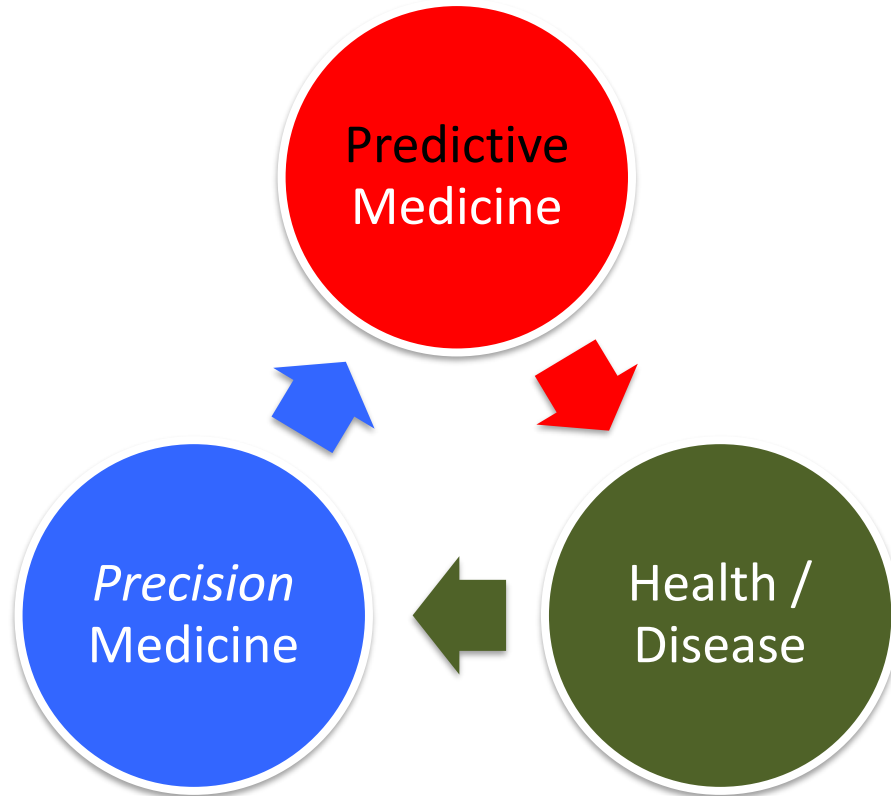
From left: Igor Tulchinsky, Dr. Christopher Mason and Dr. Olivier Elemento.

[Back to News](#)



Weill Cornell Medicine
Office of External Affairs
1300 York Ave.
Box 314
New York, NY 10065
Phone: (646) 962-9564

The model



ALL OF USSM RESEARCH PROGRAM

All of Us Research Program

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October 12, 2016

PMI Cohort Program announces new name: the All of Us Research Program

The Precision Medicine Initiative® (PMI) Cohort Program will now be called the *All of Us* Research Program and will be the largest health and medical research program on precision medicine. A set of core values is guiding its development and implementation:

- Participation is open to all.
- Participants reflect the rich diversity of the U.S.
- Participants are partners.



1 million U.S. Veterans WGS and epigenomes



U.S. Department
of Veterans Affairs

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Health Disparities & Minority Health

▼ Million Veteran Program (MVP)

Million Veteran Program (MVP)

MVP is a national, **voluntary** research program funded entirely by the Department of Veterans Affairs Office of Research & Development. The goal of MVP is to partner with Veterans receiving their care in the VA Healthcare System to study how genes affect health. To do this, MVP will build one of the world's largest medical databases by safely collecting blood samples and health information from one million Veteran volunteers. Data collected from MVP will be stored anonymously for research on diseases like diabetes and cancer, and military-related illnesses, such as post-traumatic stress disorder. [Learn more.](#)

[Frequently Asked Questions](#)

- [How do I participate?](#)
- [Do I need to schedule an appointment to participate?](#)

Text size: [+](#) [-](#)

CONTACT MVP

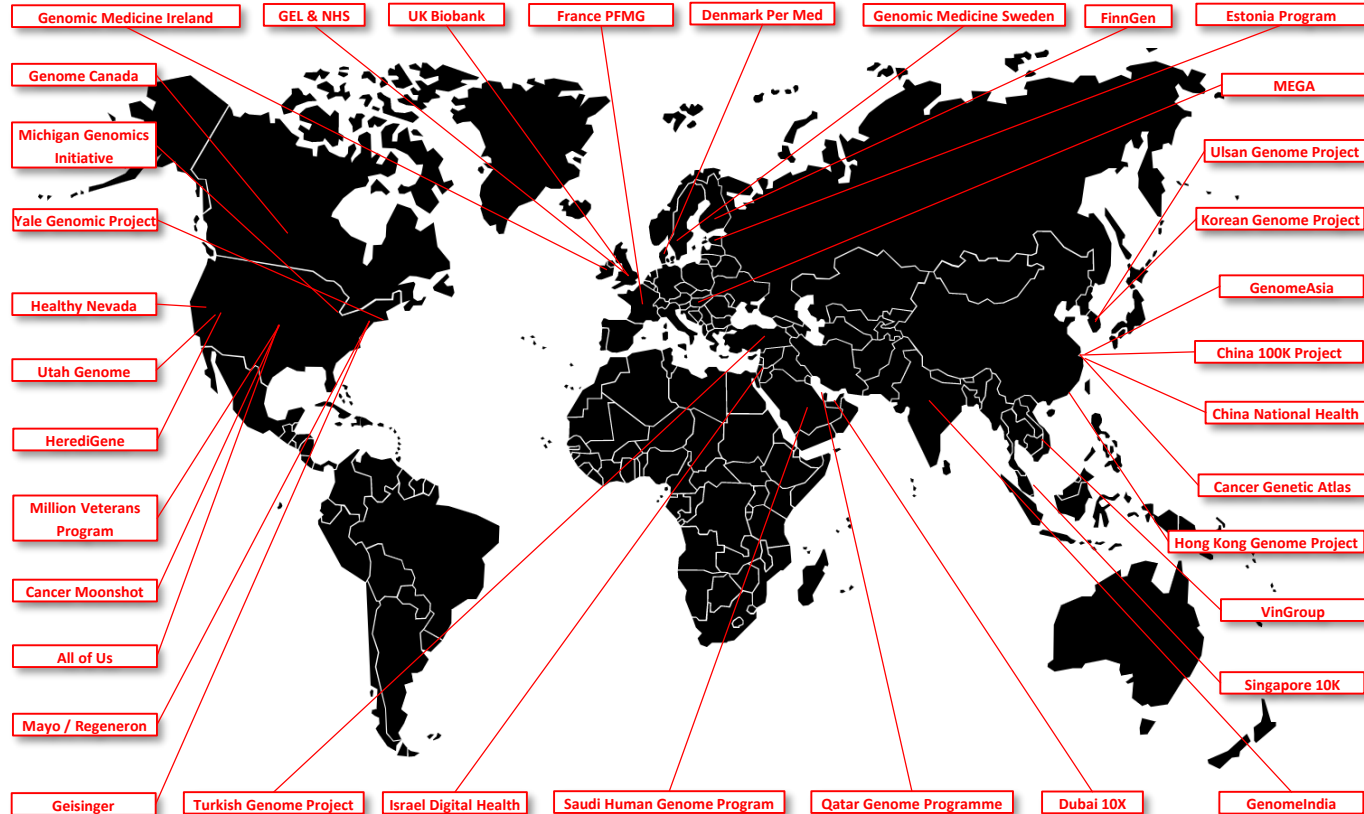
Contact the MVP Information Center toll-free at:

866-441-6075

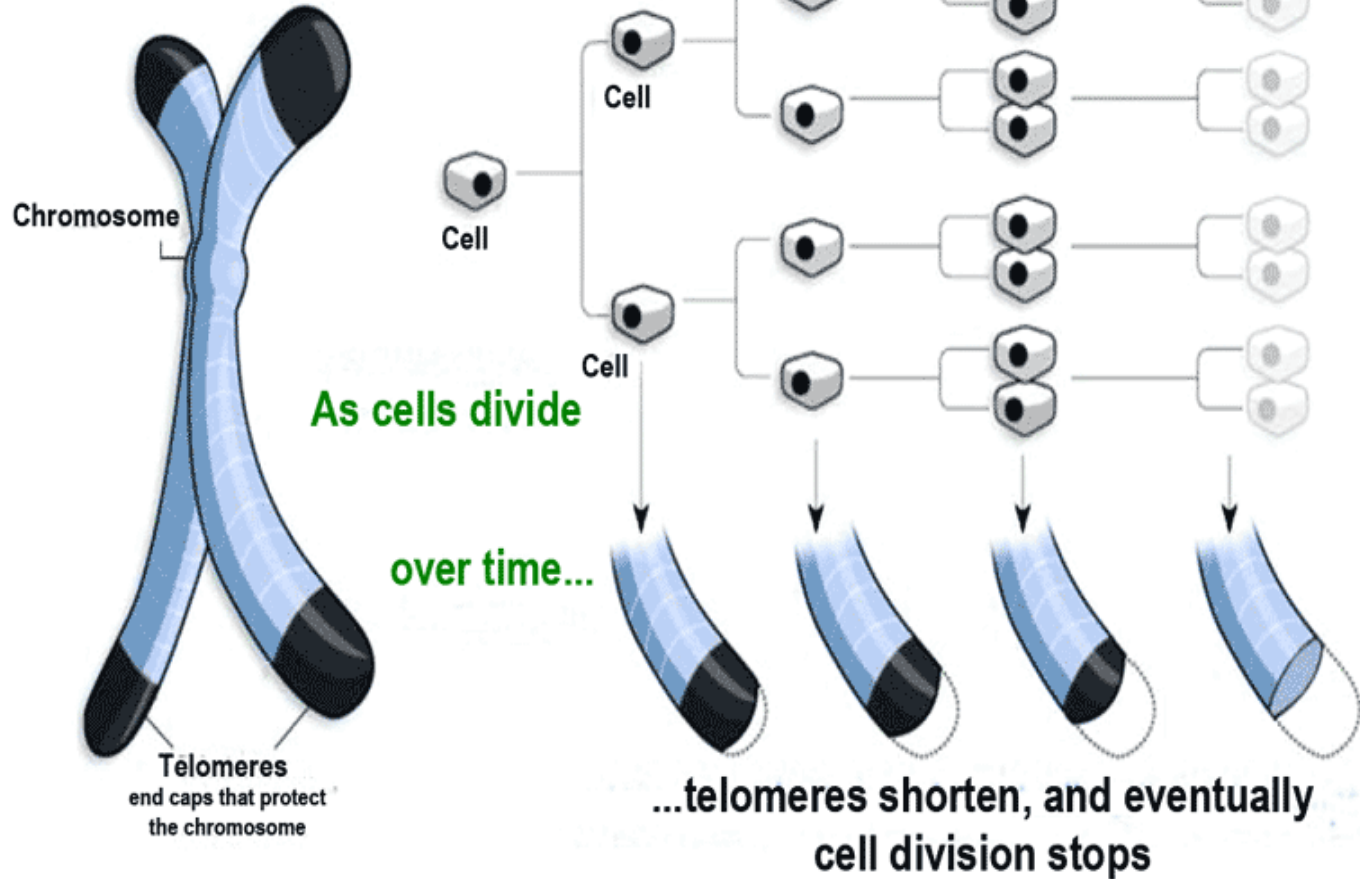
INFORMED CONSENT



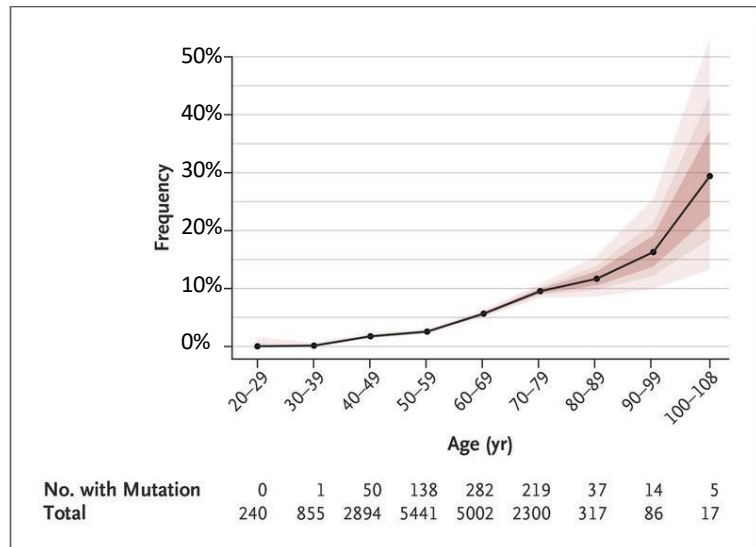
POPULATION-SCALE NGS 2024 IS GLOBAL



What We Lose With Age



What we gain with age: Clonal Hematopoiesis of Indeterminate Potential (CHIP)



▼ nature
medicine

Article | Published: 09 July 2018

Somatic mutations precede acute myeloid leukemia years before diagnosis

Pinkal Desai ✉, Nuria Mencia-Trinchant, Oleksandr Savenkov, Michael S. Simon, Gloria Cheang, Sangmin Lee, Michael Samuel, Ellen K. Ritchie, Monica L. Guzman, Karla V. Ballman, Gail J. Roboz & Duane C. Hassane ✉

Nature Medicine **24**, 1015–1023 (2018) | [Download Citation](#) ↓

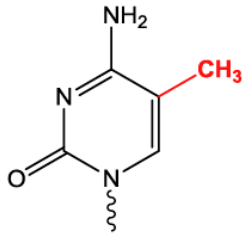
Epigenome

DNA $\xrightarrow{\text{transcription}}$ RNA $\xrightarrow{\text{translation}}$ Protein

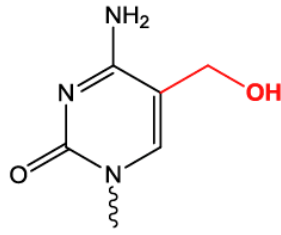
```
graph LR; Epigenome[Epigenome] --- DNA; DNA -- transcription --> RNA; RNA -- translation --> Protein
```

The diagram illustrates the central dogma of molecular biology. A vertical blue bar on the left represents the genome, with the top section labeled 'Epigenome'. The main body of the bar is labeled 'DNA'. An arrow labeled 'transcription' points from the DNA section to the label 'RNA'. Another arrow labeled 'translation' points from 'RNA' to the label 'Protein'.

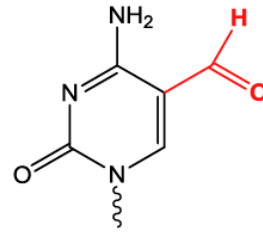
The four-base genome (black) and epigenome marks (red)



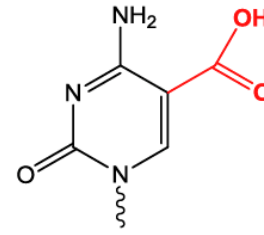
5-mC



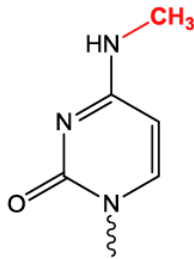
5-hmC



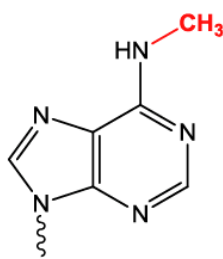
5-fC



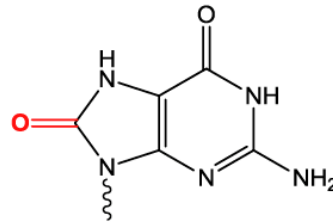
5-caC



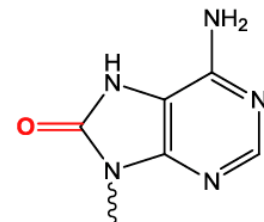
4-mC



6-mA

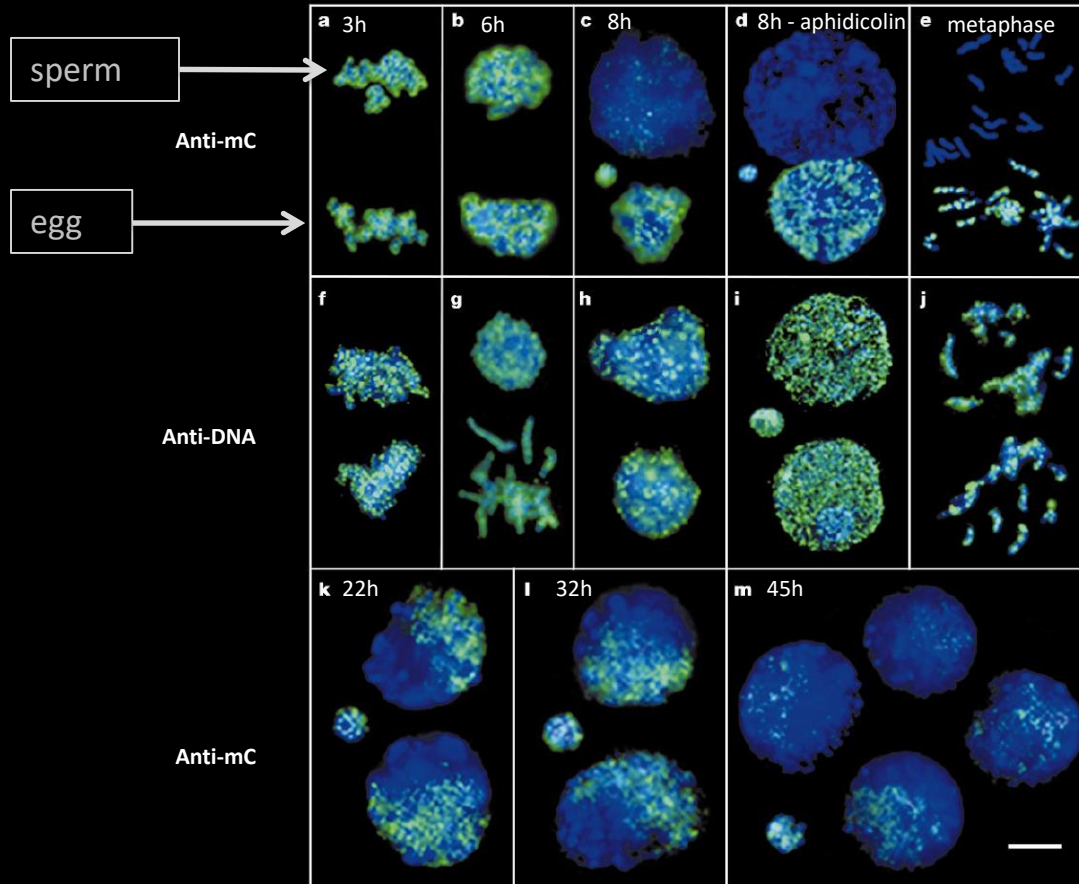


8-oxoG

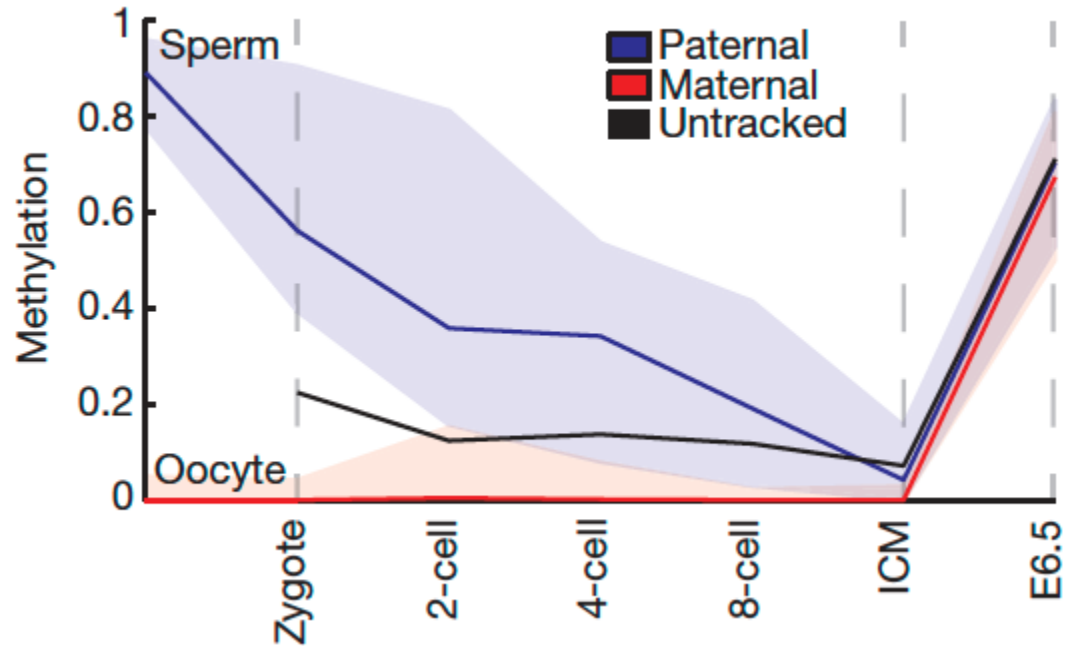


8-oxoA

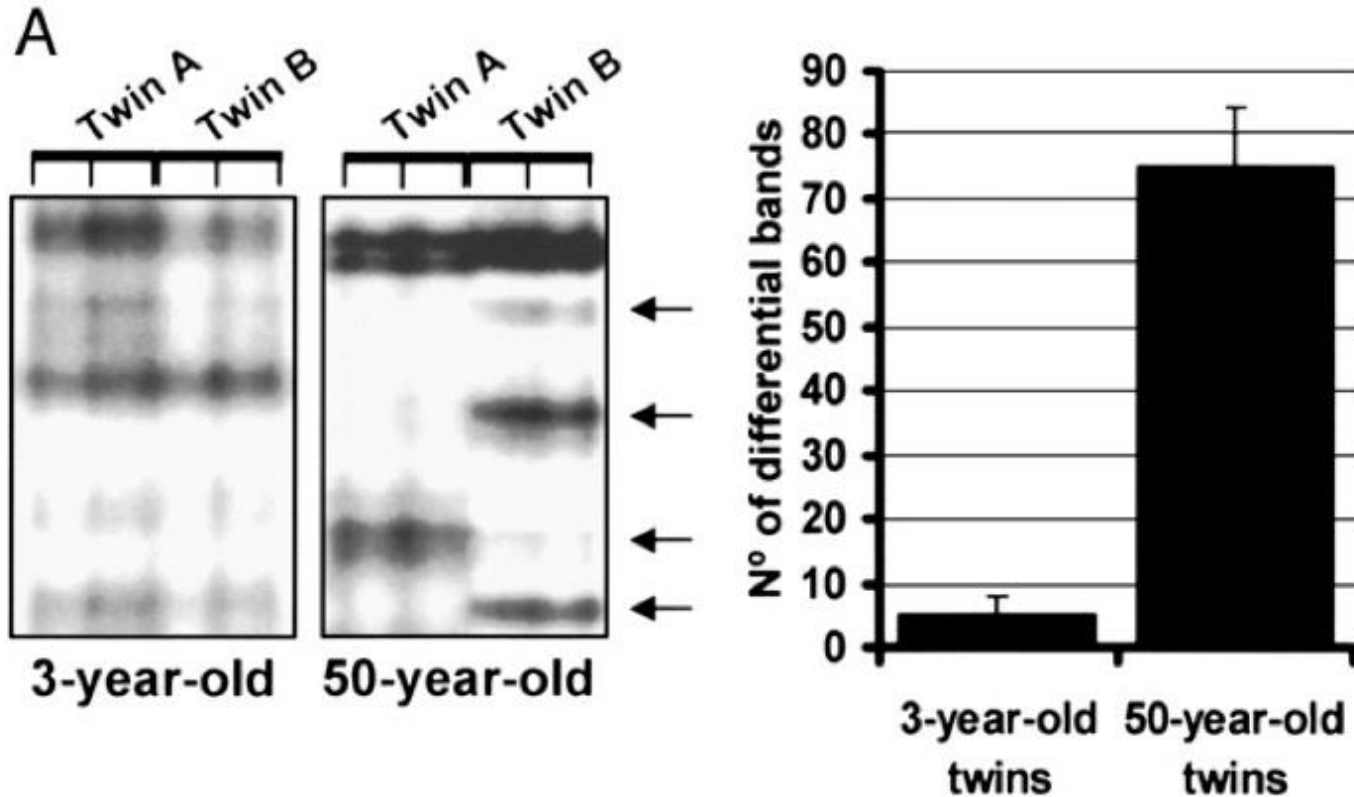
mC is important on day one



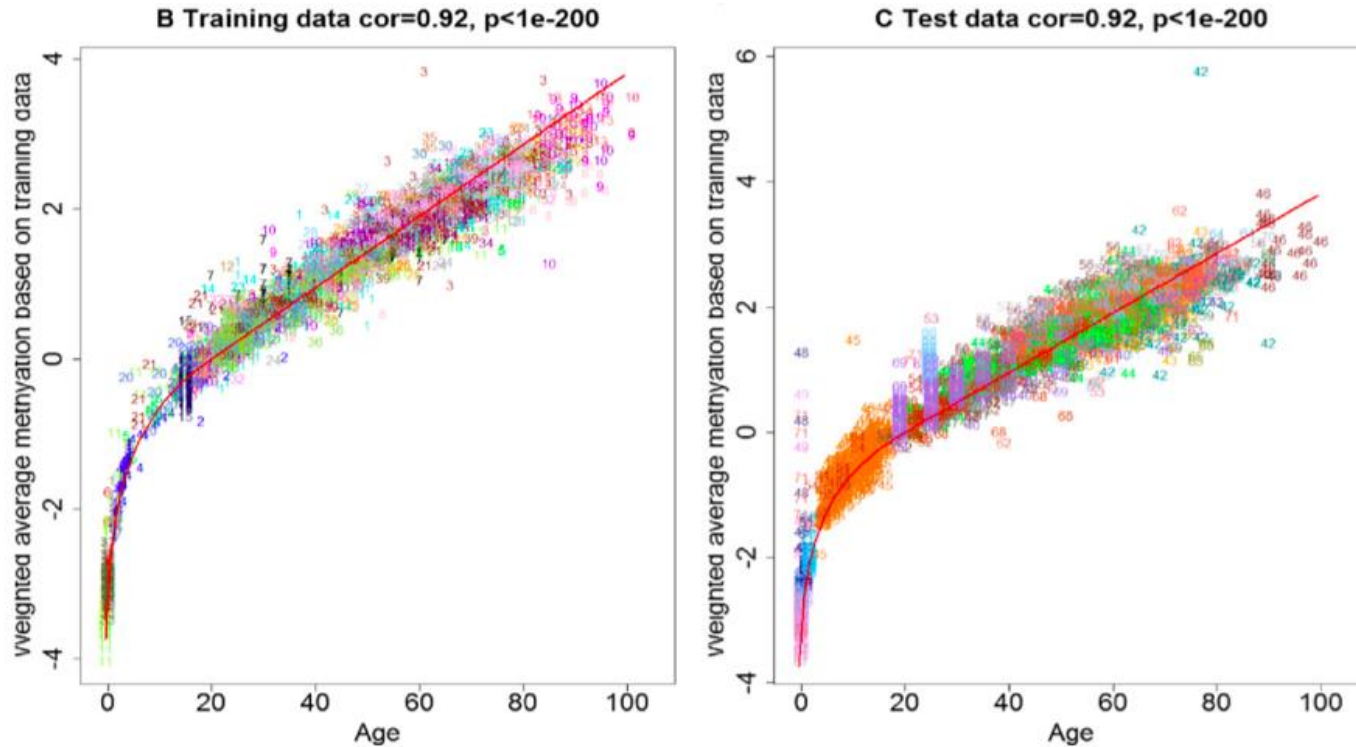
Methylation is critical for specification of embryonic cells after the inner cell mass



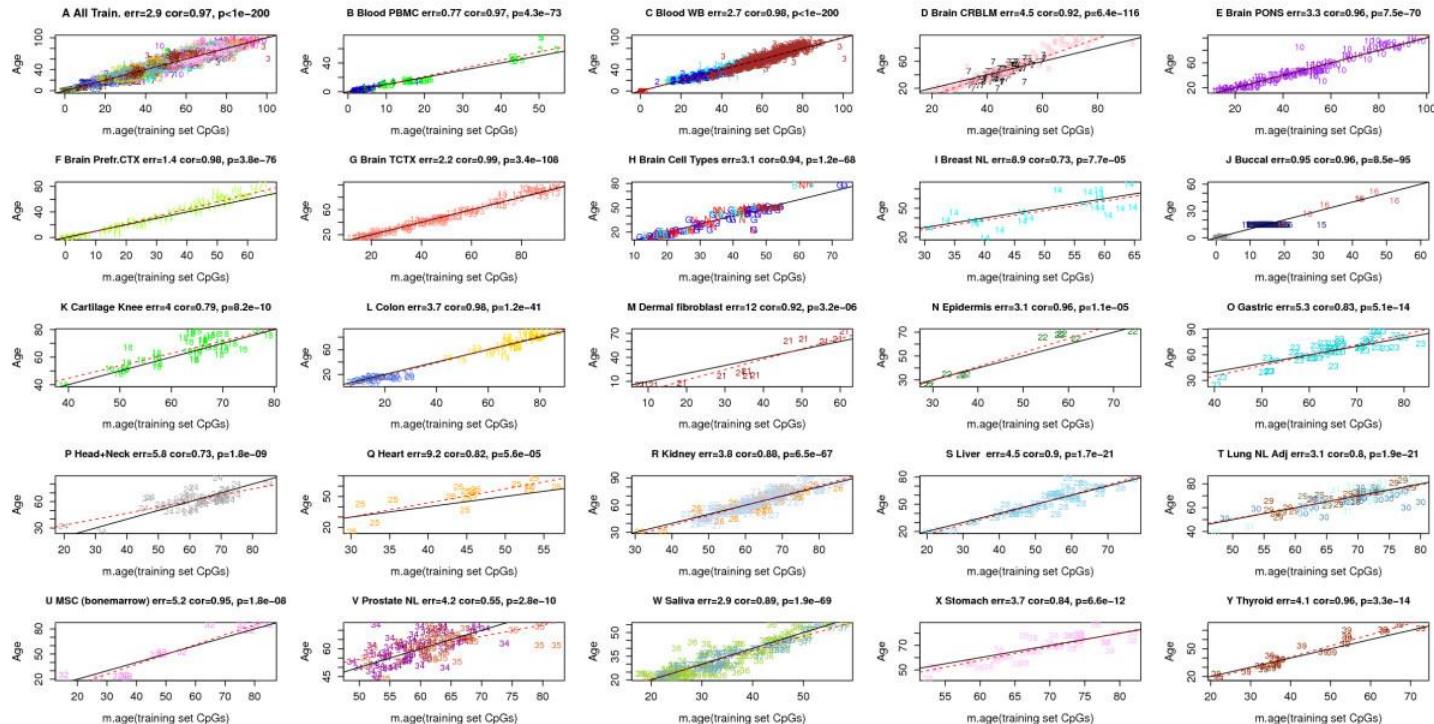
Epigenetic Drift in Twins (5mC)

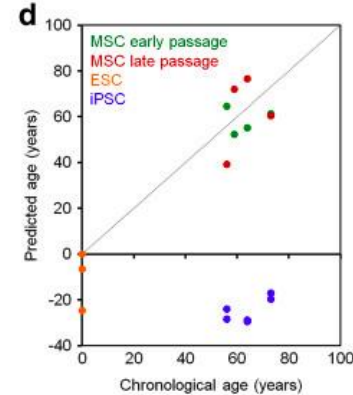
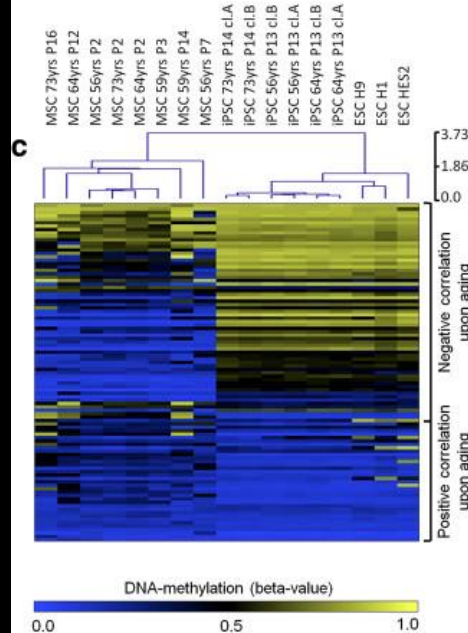
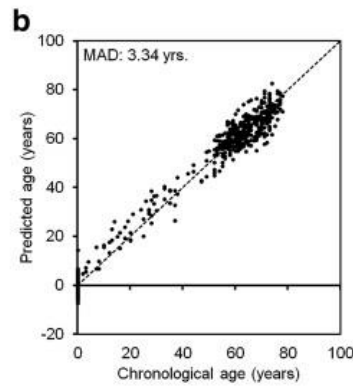
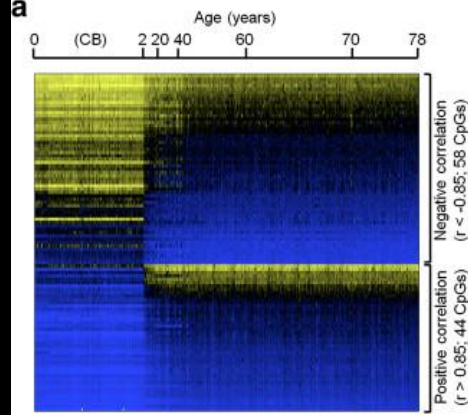


DNA Methylation can predict your age



“Epigenetic age” increases in all tissues as time passes





Epigenetic marks
(DNA methylation)
can reveal
your age.

Weidner CI *et al.*,
“Aging of blood can be tracked by
DNA methylation changes at just
three CpG sites.”
Genome Biology 2014
Feb 3;15(2):R24.

Epigenetic age

WIRED

BACKCHANNEL BUSINESS CULTURE GEAR IDEAS SCIENCE SECURITY

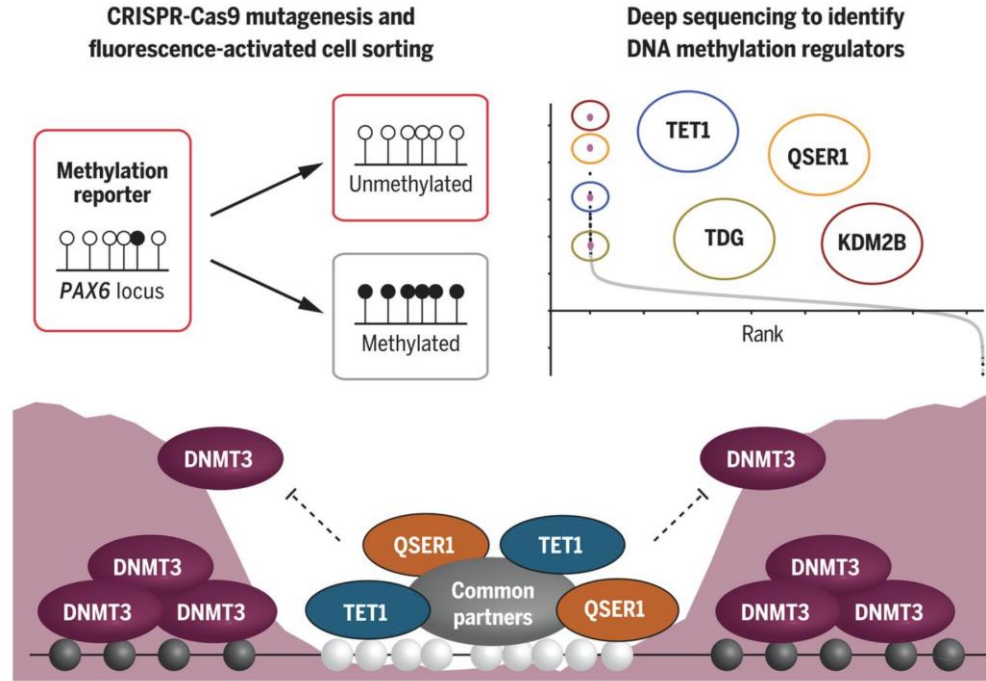
How Old Are You, Really? New Tests Want to Tell You

About a dozen such consumer tests are now on the market, but the science of reading DNA for insights about longevity is still young.



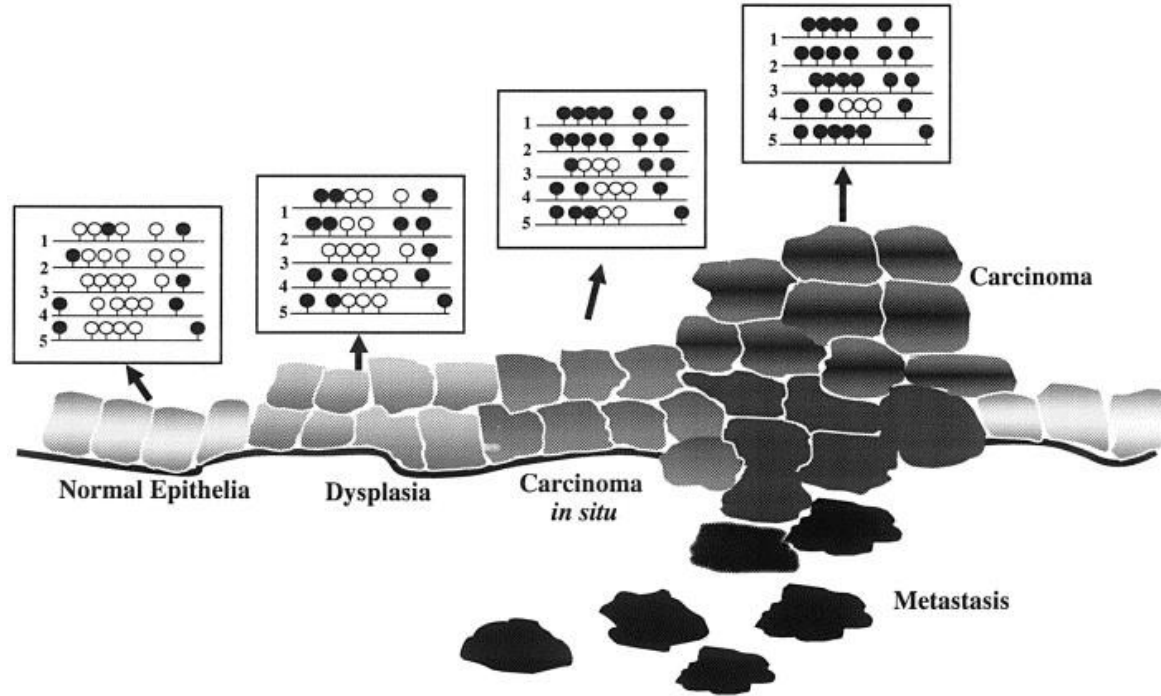
<https://www.wired.com/story/how-old-are-you-really-new-tests-want-to-tell-you/>

We are still finding new tools in the molecular toolbox



In disease

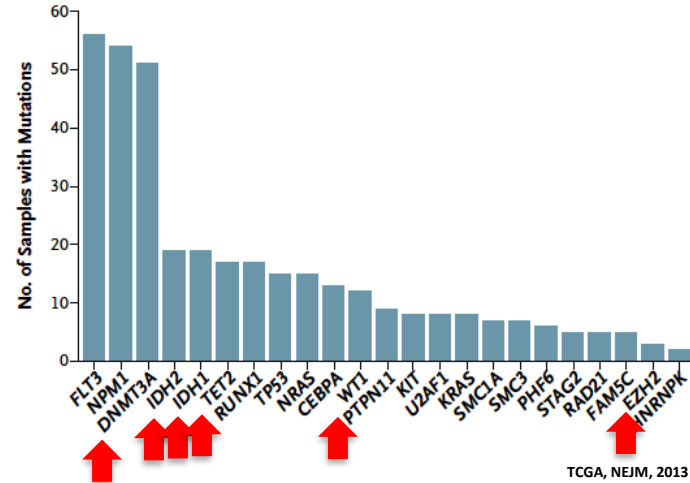
Methylation changes are also a hallmark of cancer



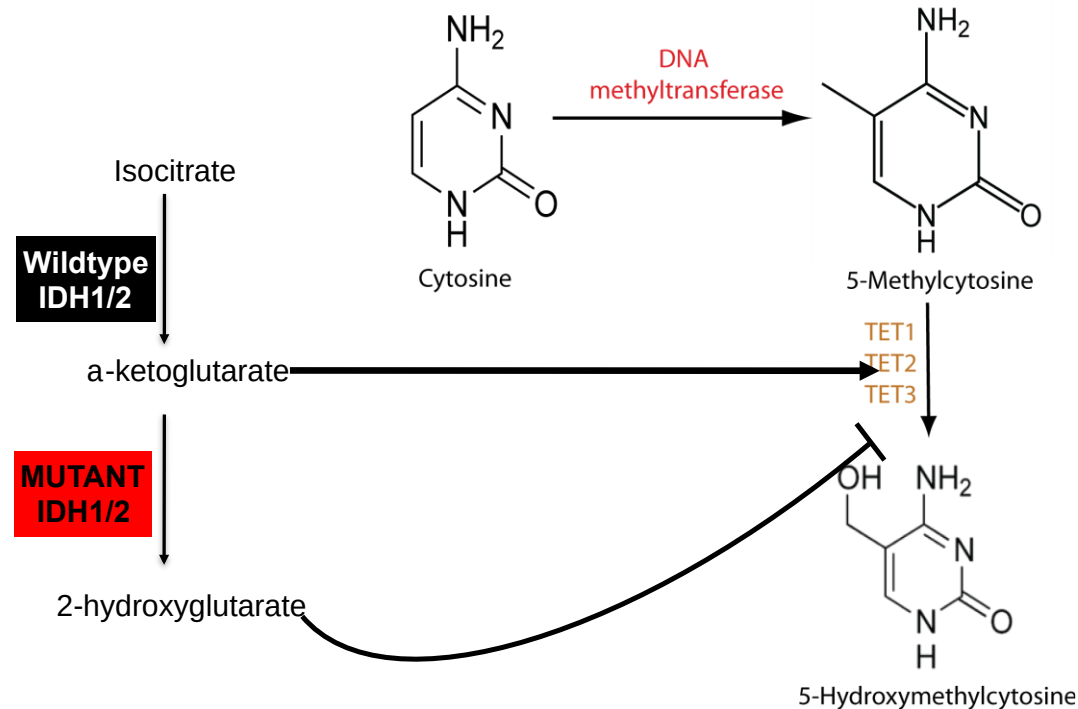
Recurrent somatic mutations in AML are epigenetic modifiers



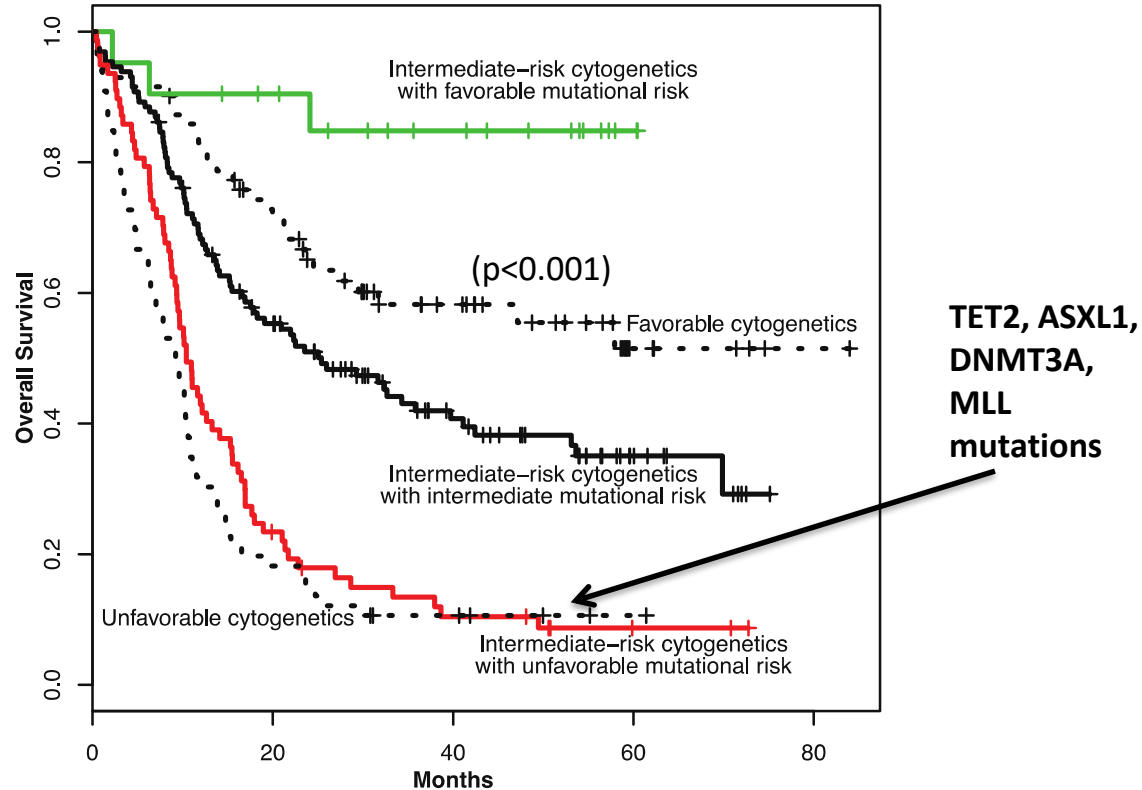
200 patients



Mutations in *IDH1/2* and *TET2* lead to impaired DNA Hydroxymethylation and Increased DNA Methylation



Adverse Outcome in AML Patients With Mutations in Specific Epigenetic Modifiers



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This article is part of a [special issue on epigenomics](#).

Software

[Highly accessed](#)[Open Access](#)

methyKit: a comprehensive R package for the analysis of genome-wide DNA methylation profiles

Altuna Akalin^{1,2*}, Matthias Kormaksson³, Sheng Li^{1,2}, Francine E Garrett-Bakelman⁴, Maria E Figueroa⁵, Ari Melnick^{4,6} and Christopher E Mason^{1,2*}

* Corresponding authors: Altuna Akalin ala2027@med.cornell.edu -
Christopher E Mason chm2042@med.cornell.edu

[▼ Author Affiliations](#)

Proceedings

[Open Access](#)

An optimized algorithm for detecting and annotating regional differential methylation

Sheng Li^{1,2}, Francine E Garrett-Bakelman³, Altuna Akalin^{1,2}, Paul Zumbo^{1,2}, Ross Levine⁴, Bik L To⁵, Ian D Lewis⁵, Anna L Brown⁶, Richard J D'Andrea^{6,7}, Ari Melnick^{3,8} and Christopher E Mason^{1,2*}

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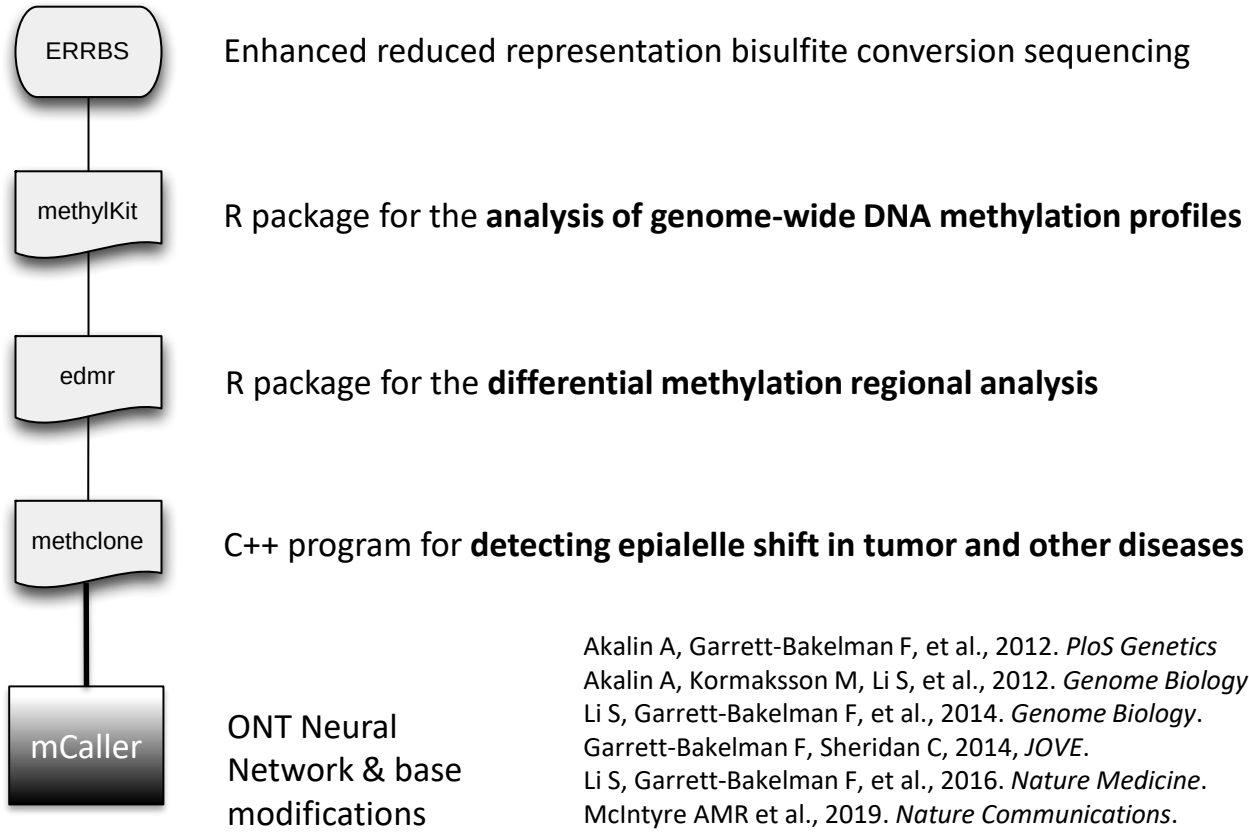
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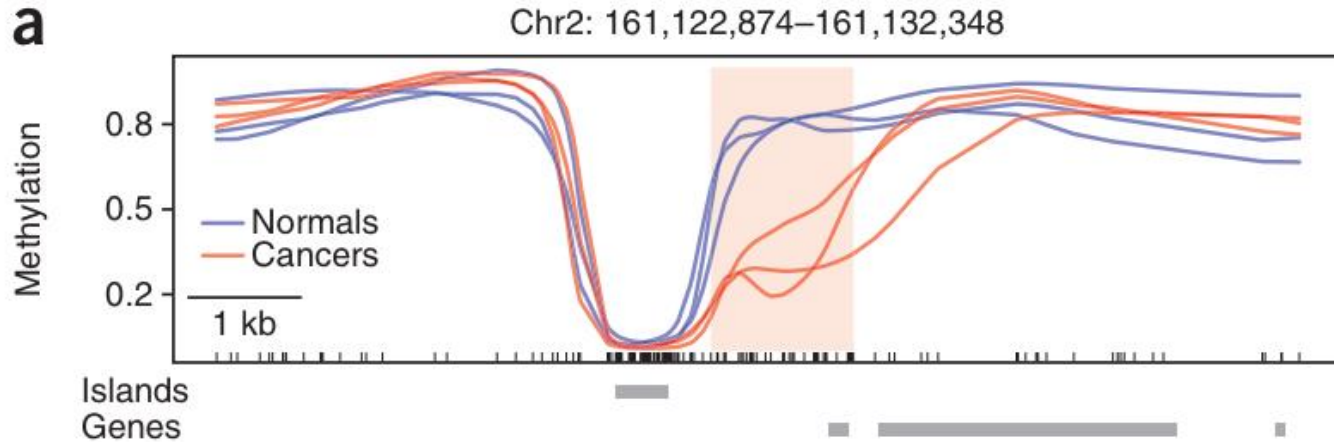
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EpiSuite:

Epigenetics sequencing and analysis suite



Methylation can shift in cancers, altering the “thumb of evolution”



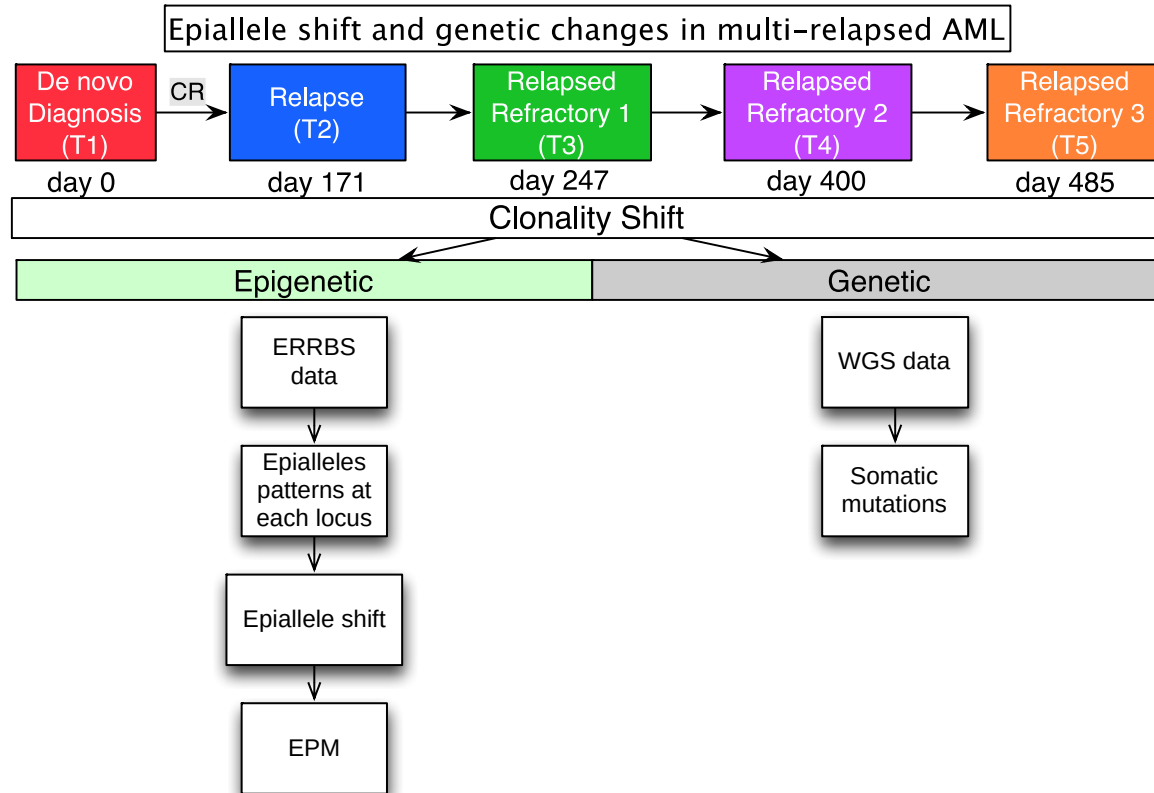
Nat Genet. 2011 Jun 26;43(8):768-75.

Longitudinal changes

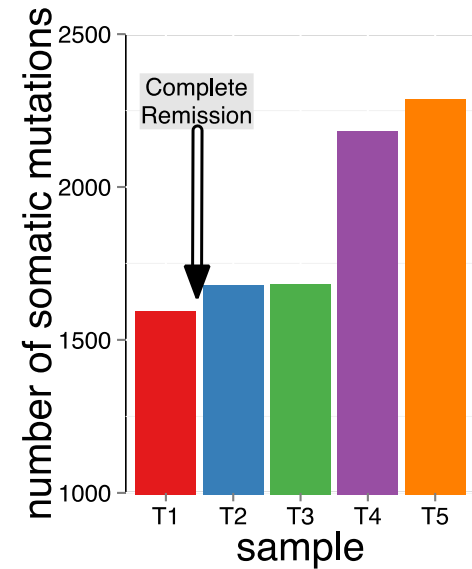
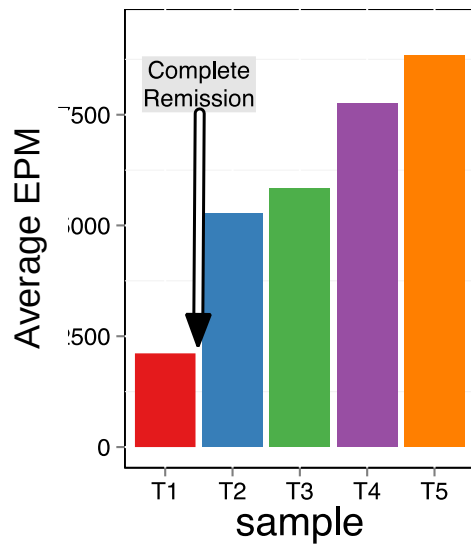
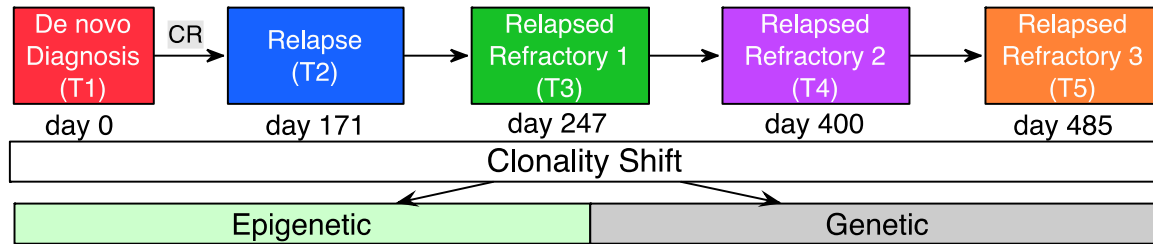
Chernobyl, 1986



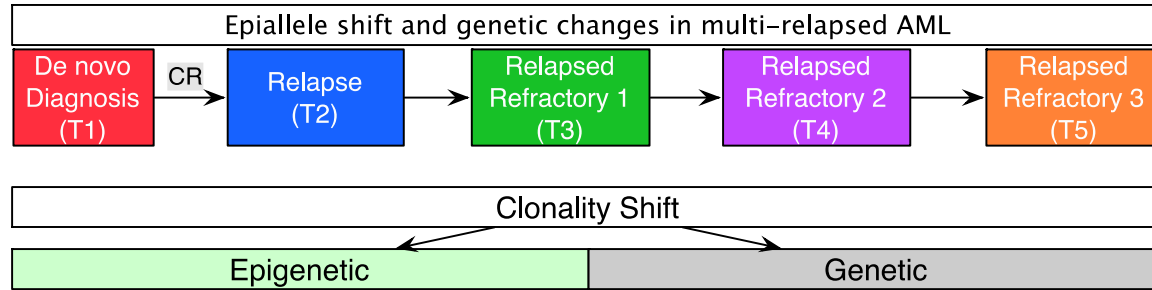
Chernobyl man



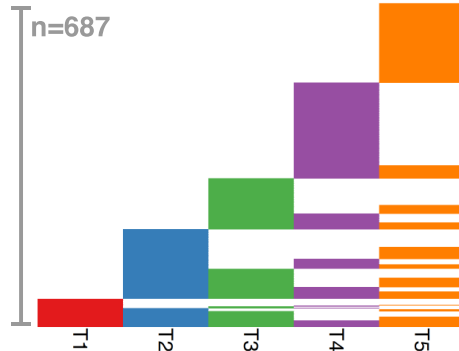
Epiallele shift and genetic changes in multi-relapsed AML



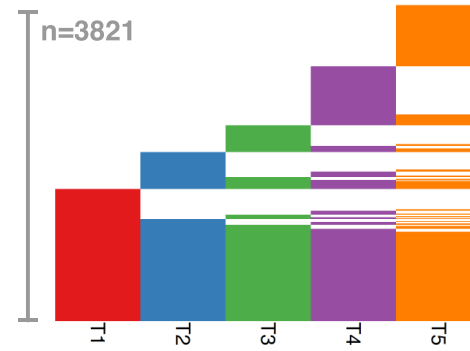
The eloci/mutations shared among time points were different



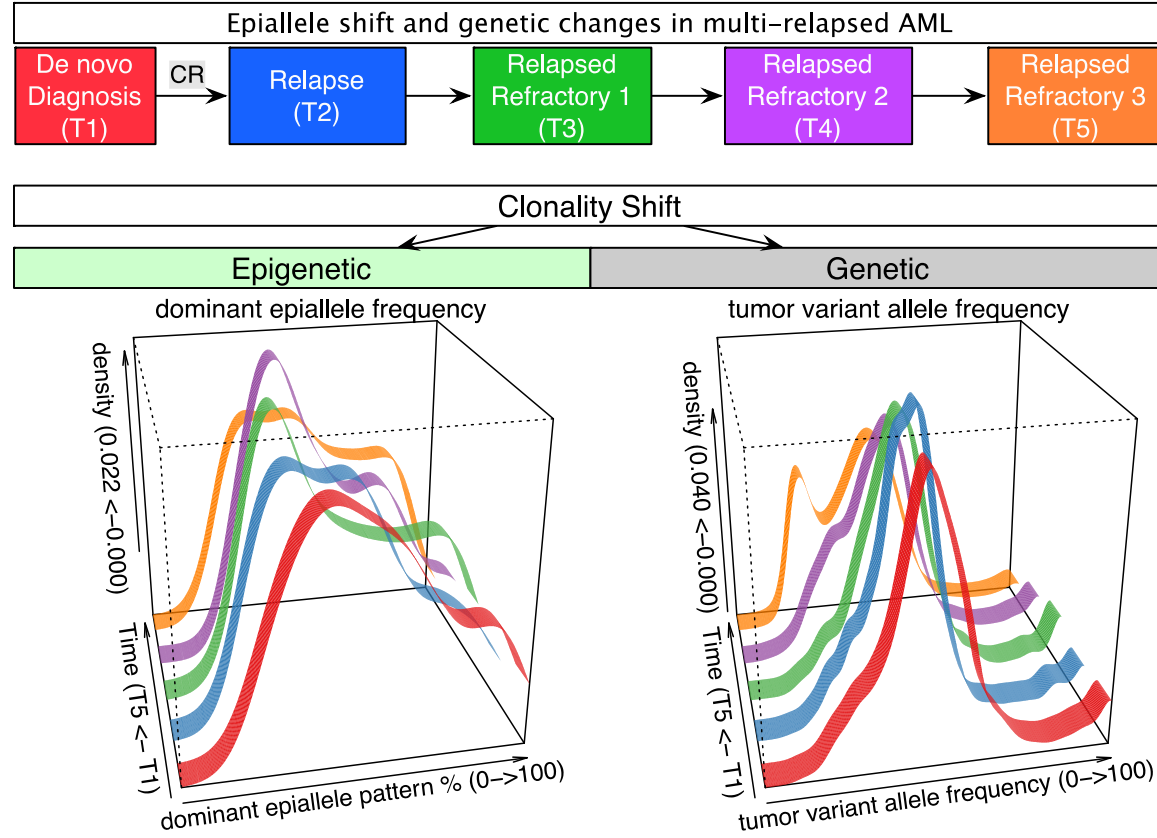
eloci: loci with high epiallele shift



somatic mutations

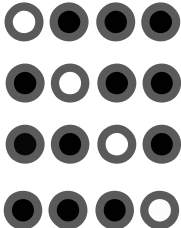
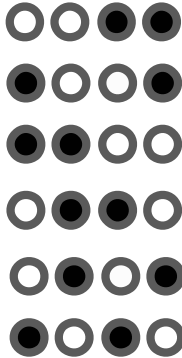
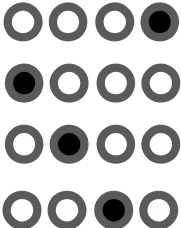


Low EAF (epigenetic evolution) is antecedent to low VAF (genetic evolution)



From a single read...

16 Possible Patterns for Epiallele of 4 CpGs

4	3	2	1	0
				

Number on the top of each column stands for the number of methylated CpG site(s).
 Each circle stands for a CpG site.
 Open circle stands for a unmethylated CpG site.
 Fill circle stands for a methylated CpG site.

Entropy measurement

- Entropy is an expression of the disorder, or randomness of a system.

- Boltzmann's entropy formula (1875):

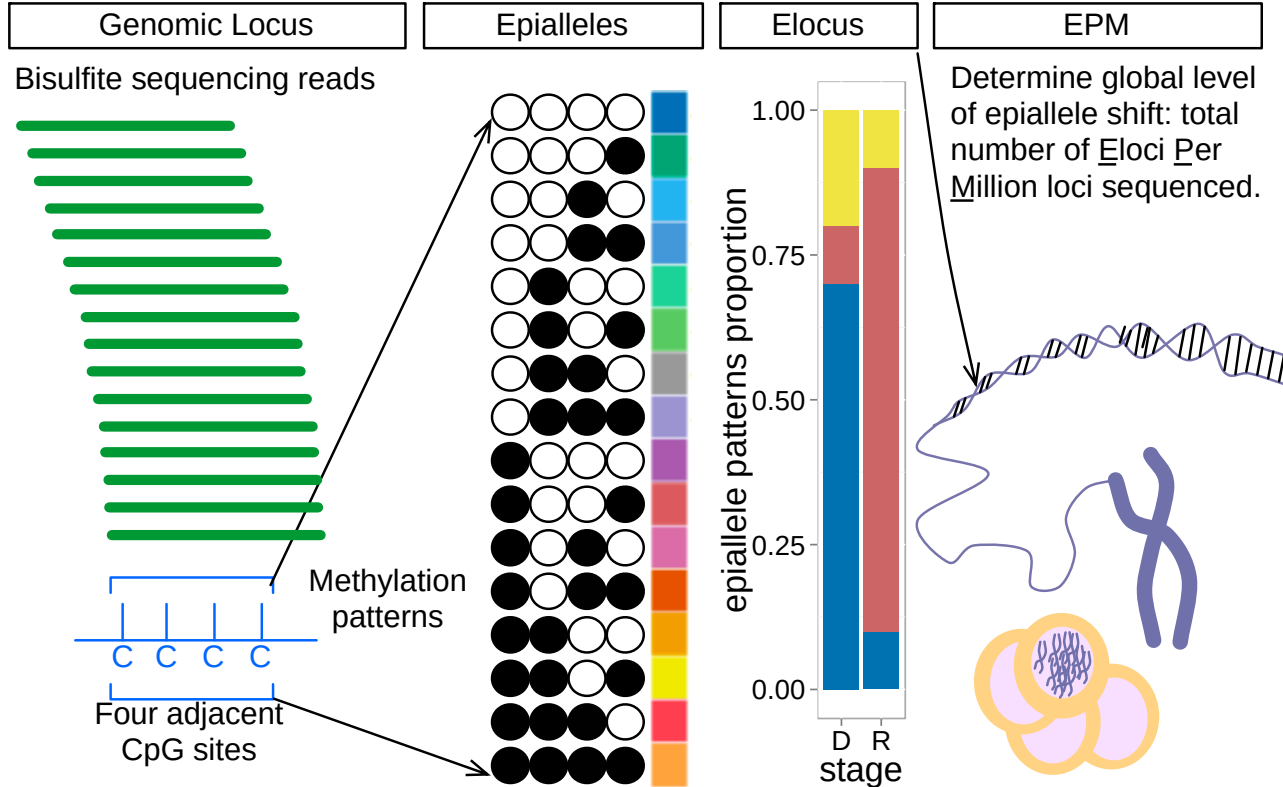
$$S = k_B \ln W$$

W is the number of microstates corresponding to a given macrostate, and can be counted using the formula for permutations: $W = N! / \prod_i N_i!$

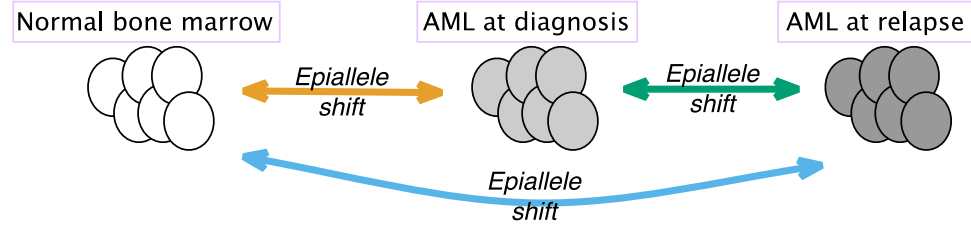
- Previously being applied to amino acid sequence classification as combinatorial entropy.



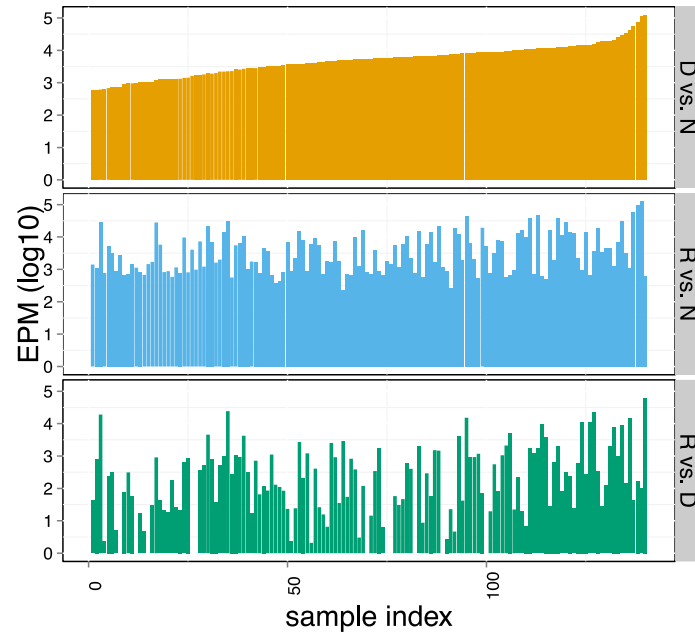
Epiallele dynamics calculation



EPM: total number of Eloci Per Million loci sequenced

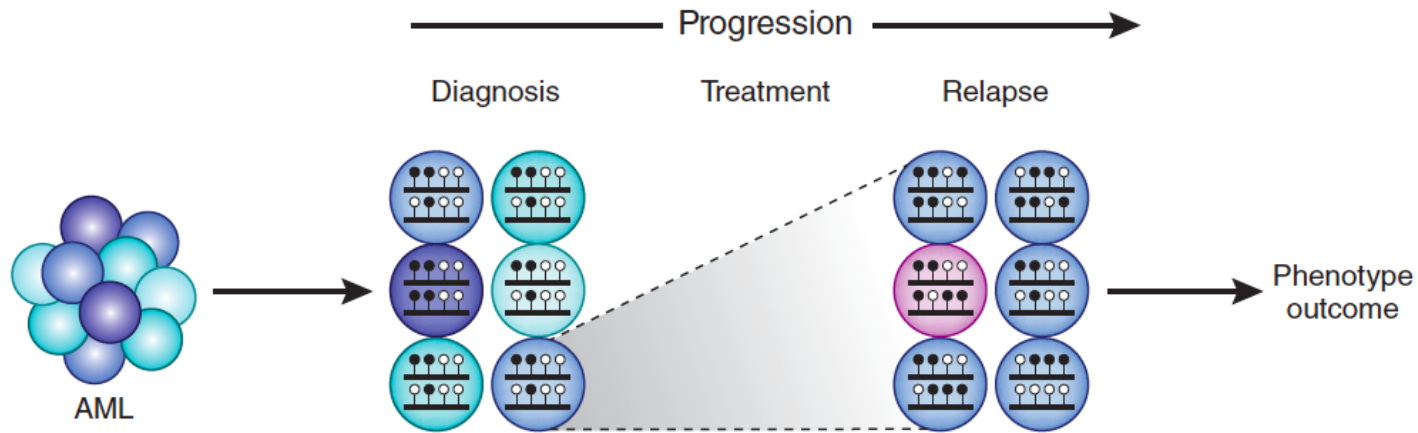


EPM distribution of 140 paired AML

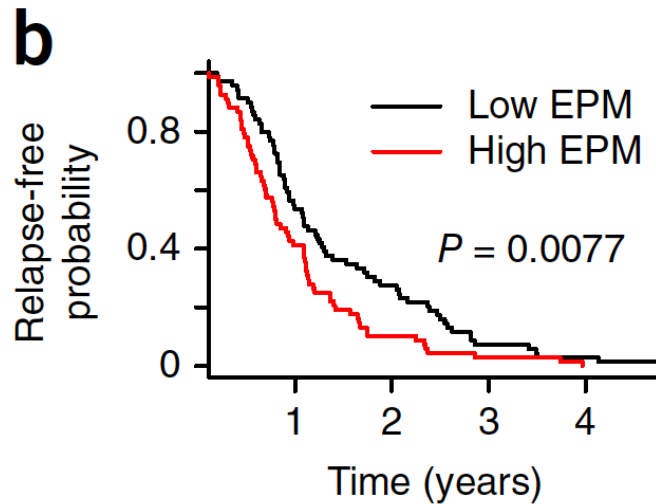


diagnosis vs. NBM relapse vs. NBM relapse vs. diagnosis





Intermediate risk patients:



Li S, *et al.*, "Dynamic Evolution of Clonal Epialleles Revealed by Methclone." *Genome Biology*, 2014.

Li S, Garrett-Bakelman F, *et al.*, *Distinct Evolution and dynamics of epigenetic and genetic heterogeneity in AML.* *Nature Medicine*, 2016.

Distinct evolution and dynamics of epigenetic and genetic heterogeneity in acute myeloid leukemia

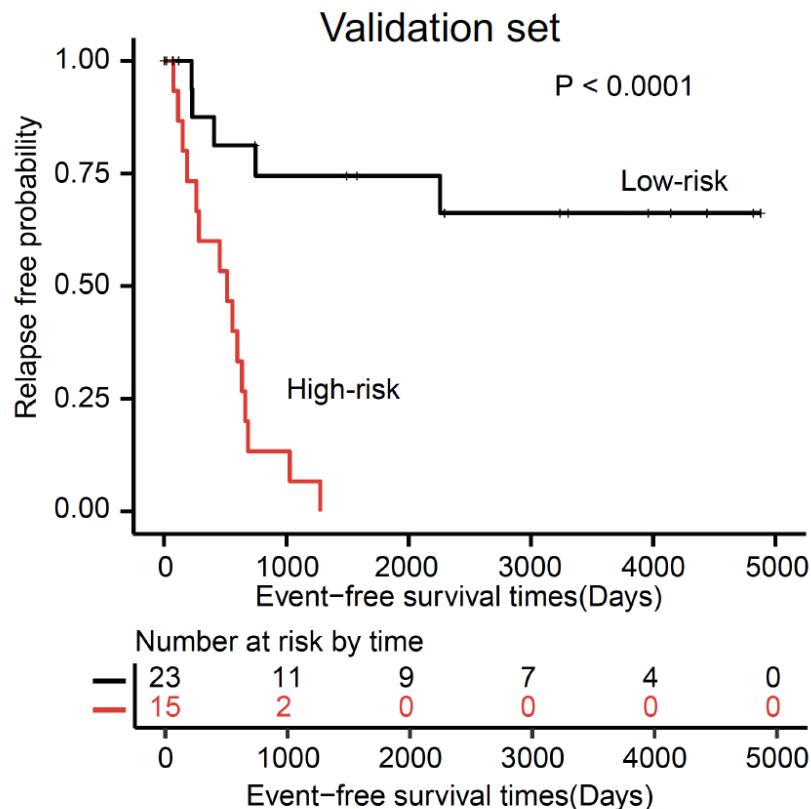
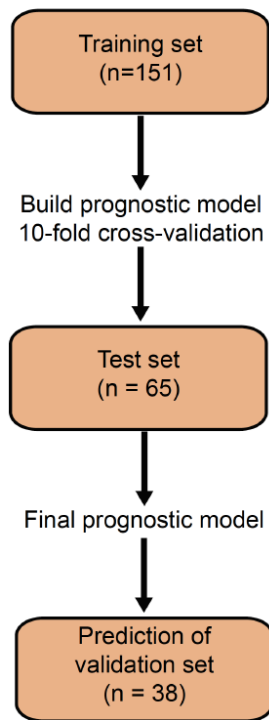
Sheng Li^{1,18,19}, Francine E Garrett-Bakelman^{2,19}, Stephen S Chung³, Mathijs A Sanders⁴, Todd Hricik³, Franck Rapaport³, Jay Patel³, Richard Dillon⁵, Priyanka Vijay⁶, Anna L Brown⁷⁻⁹, Alexander E Perl¹⁰, Joy Cannon¹⁰, Lars Bullinger¹¹, Selina Luger¹⁰, Michael Becker¹², Ian D Lewis^{7,9,13}, Luen Bik To^{9,13}, Ruud Delwel⁴, Bob Löwenberg⁴, Hartmut Döhner¹¹, Konstanze Döhner¹¹, Monica L Guzman², Duane C Hassane², Gail J Roboz², David Grimwade⁵, Peter J M Valk⁴, Richard J D'Andrea⁷⁻⁹, Martin Carroll¹⁰, Christopher Y Park^{14,15}, Donna Neuberg¹⁶, Ross Levine³, Ari M Melnick² & Christopher E Mason^{1,17}



Separate paths for epigenomes
and genomes in cancer evolution?

NATURE REVIEWS | GENETICS

An epiallele prognostic classifier predicts clinical outcome in AML



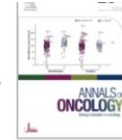
cfDNA methylation also holds signatures of disease



Annals of Oncology

Volume 32, Issue 9, September 2021, Pages 1167-

1177



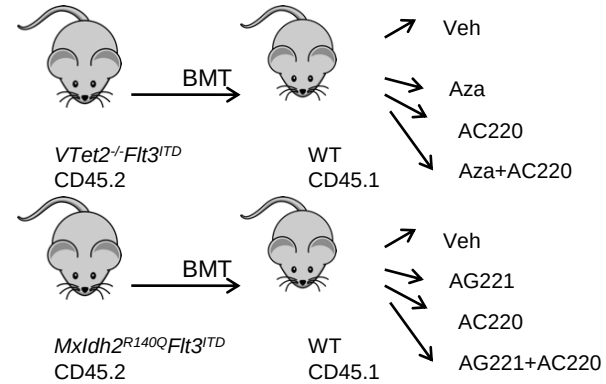
Original Article

Clinical validation of a targeted methylation-based multi-cancer early detection test using an independent validation set

Can we target
them better?

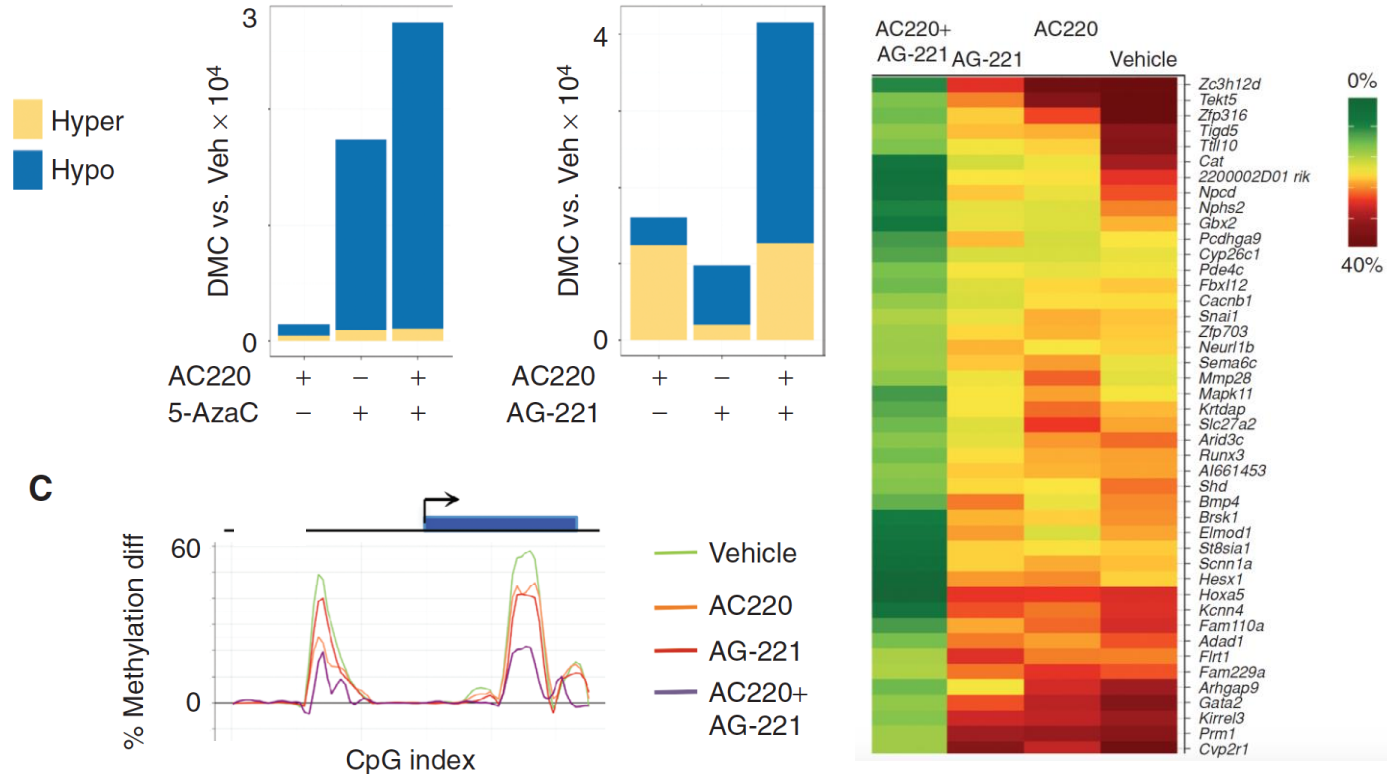
Combination Therapy

- Two models
 - TET2+FLT3
 - IDH2+FLT3
- Drugs
 - AC220 – FLT3 inhibitor (Ambit)
 - AG221 – IDH inhibitor (Agiros)
 - AzaC – hypomethylating agent



- TET2+FLT3 → Treat with AC220, AzaC or both
- IDH2+FLT3 → Treat with AC220, AG221 or both

Reversal of hypomethylation improved with combination therapy

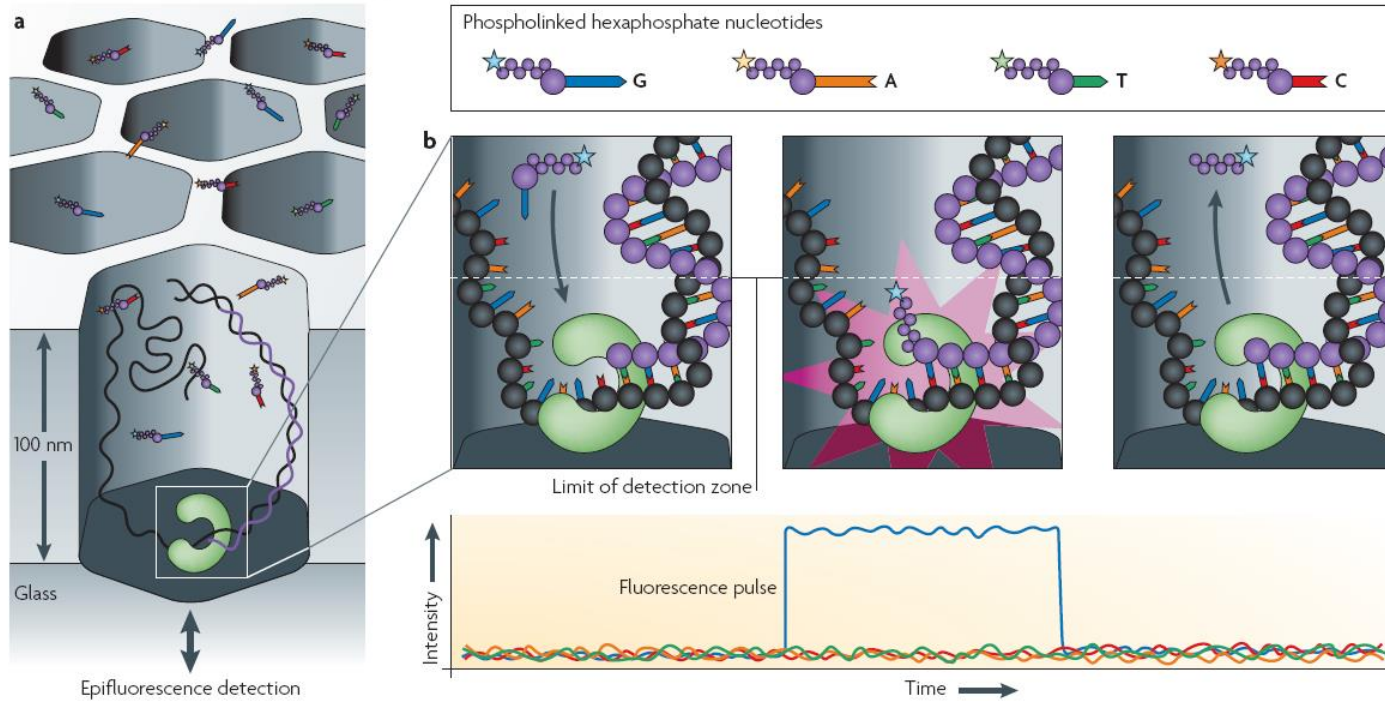


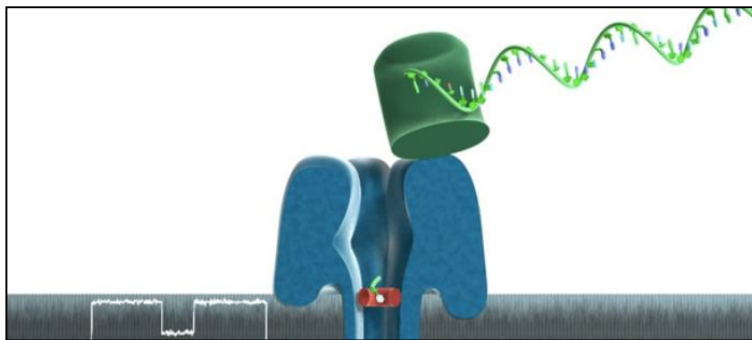
Can we detect
them better?

Pacific Biosciences

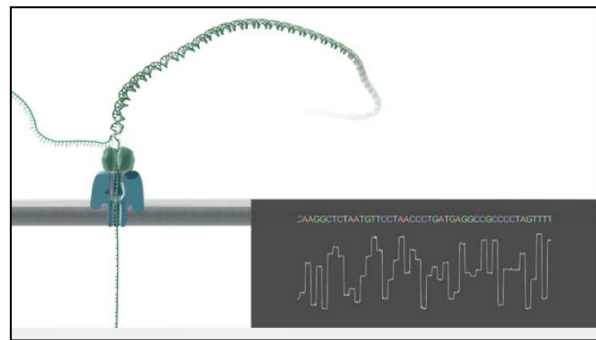
Single Molecule Real-Time (SMRT) Sequencing

Pacific Biosciences — Real-time sequencing





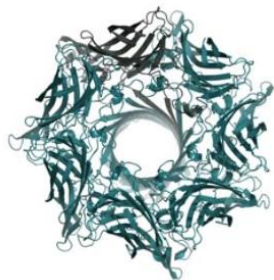
Exonuclease-Seq



Strand-Seq



MinION



PromethION



New Methods
need
Standards

To know what's real, we will need standards



<http://genomeinabottle.org>

SEQC2!



On Earth
and in Space

Are you lonely?

Tired of working on your own?

Do you hate making decisions?

HOLD A MEETING!

You can —

- See people
- Show charts
- Feel important
- Point with a stick
- Eat donuts
- Impress your colleagues

All on company time!



MEETINGS

THE PRACTICAL ALTERNATIVE TO WORK

Products to ensure meetings are not just loneliness'
avoidance:

National Institute of Standards and Technology (NIST), SEQC
Genome in a Bottle Consortium (GIAB)

- *Genome in a Bottle Consortium* is developing standards to address:
 - well-characterized human genomes as *Reference Materials* (RMs)
 - tools and methods to use these RMs
 - Work with GA4GH
- Testing protocols and methods as part of the ABRF-NGS Phase II Project and the FDA's Sequencing Quality Control Consortium (SEQC)



Genome in a Bottle
Consortium

<http://genomeinabottle.org>

SEQC2!

History

In 2014, the FDA and the ABRF-NGS groups completed three large-scale studies on RNA-seq (Sequencing Quality Control, or SEQC), which compared all major platforms for NGS and RNA-seq, including Illumina, Proton, PGM, 454, and Pacbio:

<http://www.nature.com/nbt/journal/v32/n9/full/nbt.2972.html>,

<http://www.nature.com/nbt/journal/v32/n9/full/nbt.3000.html>,

<http://www.nature.com/nbt/journal/v32/n9/full/nbt.2957.html>.

Then came the **FDA SEQC2** and ABRF-NGS Phase 2 Study and EpiQC Study....





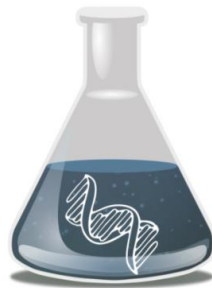
SEXYBACK

JUSTIN TIMBERLAKE

netmen

GENOME IN A BOTTLE

Our mission is to provide the authoritative characterization of human genomes.



OVERVIEW

WORKSHOPS

NEWS

RESOURCES

GOOGLE GROUP

ANALYSIS TEAM

Reference Materials and Data

The Genome in a Bottle Consortium has selected several genomes to produce and characterize as reference materials. The National Institute of Standards and Technology (NIST) is developing NIST Reference Materials from these genomes, which are DNA extracted from a large homogenized growth of B lymphoblastoid cell lines from the Coriell Institute for Medical Research.

Note that there may be small differences between the NIST DNA and the Coriell DNA since they come from different growths of cells, though we do not expect these differences to be large for most applications.

The NIST Reference Materials available and planned are listed below, along with links to their data.

A description of data generated by GIAB for all the genomes below is published [here](#), and characterization of small variants is published [here](#). Ongoing work to characterize more difficult variants and regions is announced in the [GIAB Analysis Team google group](#).

[Why GitHub?](#)[Team](#)[Enterprise](#)[Explore](#)[Marketplace](#)[Pricing](#)[Search](#)[Sign in](#)[Sign up](#)

genome-in-a-bottle

A public-private-academic consortium hosted by NIST to develop reference materials and standards for clinical sequencing

[Repositories](#) 9[Packages](#)[People](#)[Projects](#)

Type ▾

Language ▾

Sort ▾

giab_publications

This repository contains a list of publications pertinent to the Genome in a Bottle project

👤 3 ⭐ 6 🚫 0 📄 0 Updated on Nov 5, 2020

Top languages

● Jupyter Notebook ● Roff

genome-stratifications

Stratification BED files from the Global Alliance for Genomics and Health (GA4GH) Benchmarking Team and the Genome in a Bottle Consortium. These files can be used as standard resource of BED files for use with GA4GH benchmarking tools such as hap.py to stratify true positive, false positive, and false negative variant calls based on genomic cont...

● Jupyter Notebook 👤 0 ⭐ 13 🚫 0 📄 0 Updated on Jul 20, 2020

People

This organization has no public members. You must be a member to see who's a part of this organization.

giab_data_indexes

This repository contains data indexes from NIST's Genome in a Bottle project.

● Roff 👤 44 ⭐ 122 🚫 2 📄 0 Updated on Apr 20, 2020

<https://github.com/genome-in-a-bottle>

Table 2 Genome in a Bottle Consortium Genomes.

From: [Extensive sequencing of seven human genomes to characterize benchmark reference materials](#)

Genome	Coriell cell line ID	NIST ID	NIST RM #	NCBI BioSample	PGP ID
CEPH Mother/Daughter	GM12878	HG001	RM8398	SAMN03492678	Not PGP
AJ Son	GM24385	HG002	RM8391* (son)/RM8392* (trio)	SAMN03283347	huAA53E0
AJ Father	GM24149	HG003	RM8392* (trio)	SAMN03283345	hu6E4515
AJ Mother	GM24143	HG004	RM8392* (trio)	SAMN03283346	hu8E87A9
Chinese Son	GM24631	HG005	RM8393*	SAMN03283350	hu91BD69
Chinese Father	GM24694	N/A [†]	N/A [†]	SAMN03283348	huCA017E
Chinese Mother	GM24695	N/A [†]	N/A [†]	SAMN03283349	hu38168C

Identifiers associated with genomes currently being characterized by GIAB. The NCBI BioProject for GIAB is PRJNA200694.

* Not currently available. Planned release as NIST RMs in mid-2016.

† NIST Reference Materials are not planned for the Chinese parents, but cells and DNA are available from Coriell.

Zook, J., Catoe, D., McDaniel, J. *et al.* Extensive sequencing of seven human genomes to characterize benchmark reference materials. *Sci Data* **3**, 160025 (2016). <https://doi.org/10.1038/sdata.2016.25>

EpiQC

Genome Biology

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[Submit manuscript](#) 

Research | [Open access](#) | Published: 06 December 2021

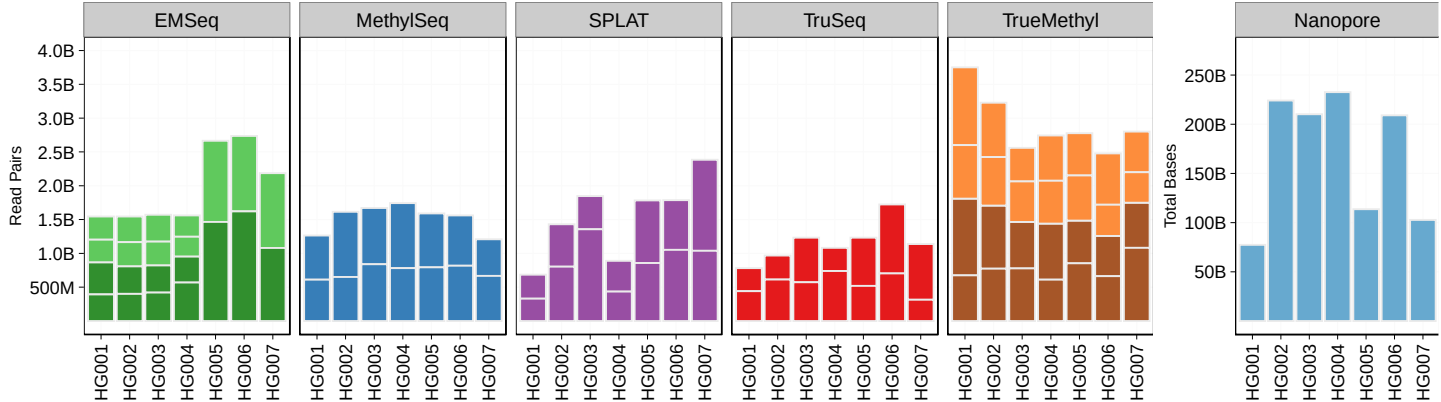
The SEQC2 epigenomics quality control (EpiQC) study

[Jonathan Foox](#), [Jessica Nordlund](#), [Claudia Lalancette](#), [Ting Gong](#), [Michelle Lacey](#), [Samantha Lent](#), [Bradley W. Langhorst](#), [V. K. Chaithanya Ponnaluri](#), [Louise Williams](#), [Karthik Ramaswamy Padmanabhan](#), [Raymond Cavalcante](#), [Anders Lundmark](#), [Daniel Butler](#), [Christopher Mozsary](#), [Justin Gurvitch](#), [John M. Greally](#), [Masako Suzuki](#), [Mark Menor](#), [Masaki Nasu](#), [Alicia Alonso](#), [Caroline Sheridan](#), [Andreas Scherer](#), [Stephen Bruinsma](#), [Gosia Golda](#), ... [Christopher E. Mason](#) 

[+ Show authors](#)

[Genome Biology](#) **22**, Article number: 332 (2021) | [Cite this article](#)

Genome	Coriell ID	NIST ID	NCBI BioSample	Whole Genome												Targeted
				EM-Seq						MethylS eq	PromethION	SPLAT	TrueMethyl		TruSeq	EPIC
				100ng		50ng		10ng					Bisulfite	Oxidative		
				Lab 1	Lab 2	Lab 1	Lab 2	Lab 1	Lab 2							
CEPH Mother/Daughter	GM12878	HG001	SAMN03492678	340 337	468 392					652 609	7.8 (4085) 5.1 (6117) 2.5 (5583)	353 329	1093 395	514 508	338 437	267 326
AJ Son	GM24385	HG002	SAMN03283347	379 357	403 399					960 650	13.3 (3867) 4.5 (7346) 1.6 (5126) 1.4 (5064)	625 801	901 504	508 447	351 609	239 335
AJ Father	GM24149	HG003	SAMN03283345	77 354	397 419					829 838	17.5 (3533) 4.8 (3760) 1.1 (5162) 1.4 (5231)	484 1353	664 367	272 344	654 568	288 337
AJ Mother	GM24143	HG004	SAMN03283346	313 294	381 173					959 779	17.4 (3492) 6.0 (4315) 1.1 (5821) 1.5 (5590)	453 433	802 321	519 345	340 733	235 339
Chinese Son	GM24631	HG005	SAMN03283350	89	451	430	497	313	244	796 791	2.5 (2984) 7.0 (5087) 13.8 (5073)	922 855	605 447	360 450	709 514	243 321
Chinese Father	GM24694	HG006	SAMN03283348	359	451	344	422	412	186	741 815	2.0 (3987) 1.4 (5197) 1.0 (5505)	733 1050	573 631	730 220	1012 698	247 265
Chinese Mother	GM24695	HG007	SAMN03283349	352	466	365	480	387	176	714 665	4.5 (4907) 16.1 (5022)	1343 1035	638 1015	575 199	993 312	234 243



5mC Detection — Bisulfite conversion

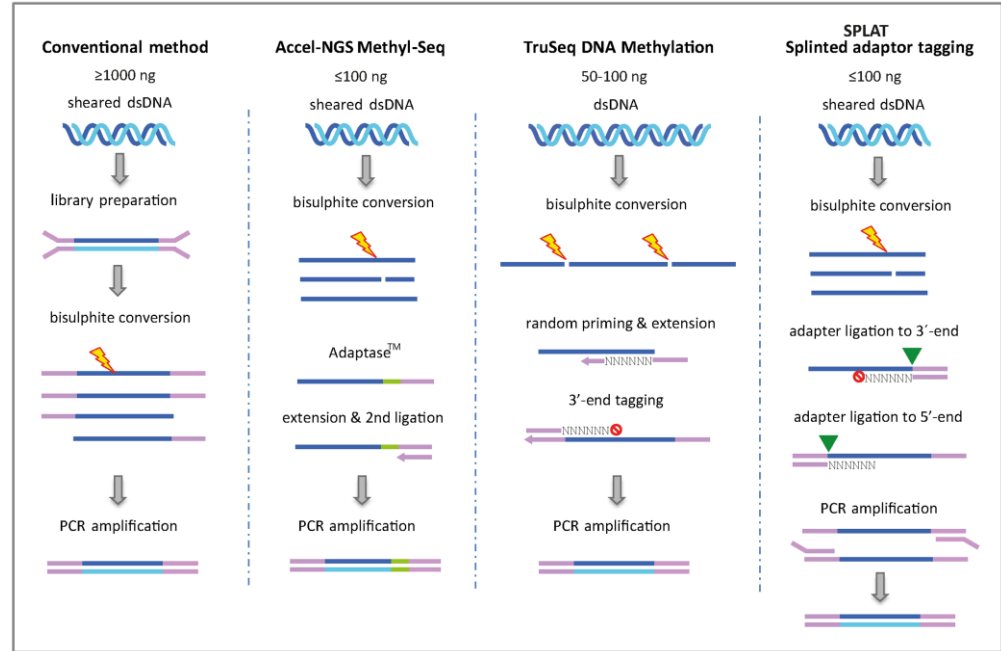
GCAACG"mC"GAACG"mC"GCAAGCG

Base conversion

GUAAUG"mC"GAAUT"mC"GUAAGUG

PCR

GTAATG"C"GAATT"C"GTAAGTG

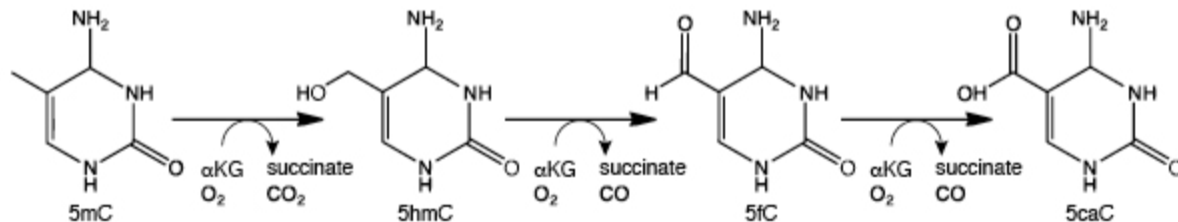


Amanda Raine, Erika Manlig, Per Wahlberg, Ann-Christine Syvänen, Jessica Nordlund, SPLinted Ligation Adapter Tagging (SPLAT), a novel library preparation method for whole genome bisulphite sequencing, *Nucleic Acids Research*, Volume 45, Issue 6, 7 April 2017, Page e36, <https://doi.org/10.1093/nar/gkw1110>

Enzymatic Methyl-seq – A New Approach

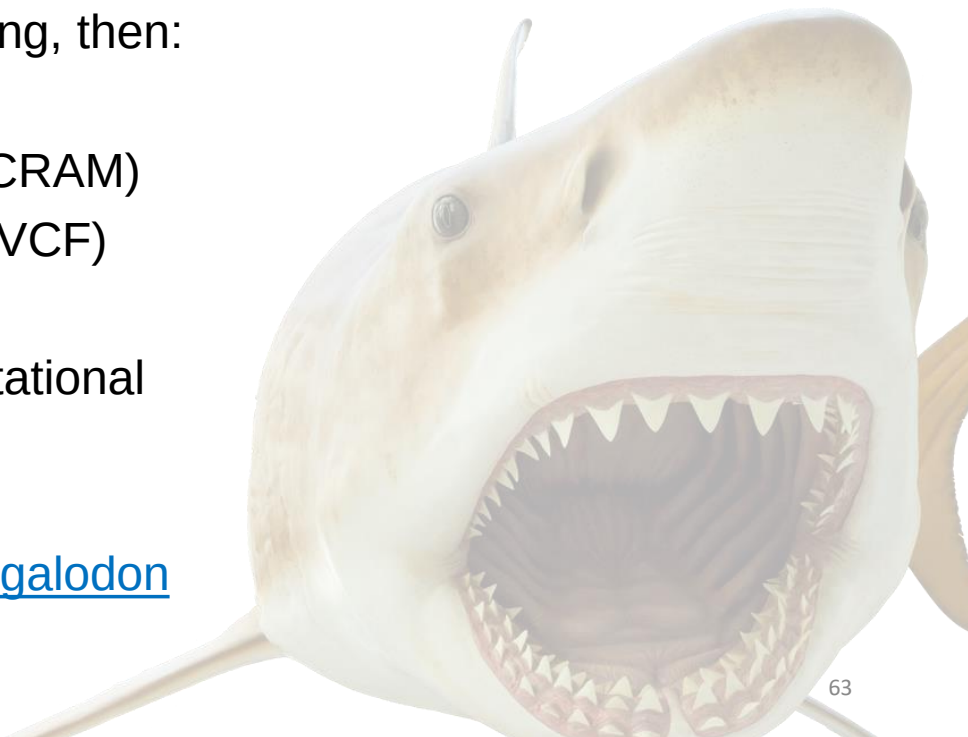
The enzymatic methyl-seq workflow developed at NEB provides a much-needed alternative to bisulfite sequencing. This method relies on the ability of APOBEC to deaminate cytosines to uracils. Unfortunately, APOBEC also deaminates 5mC and 5hmC, making it impossible to differentiate between cytosine and its modified forms (7,8). In order to detect 5mC and 5hmC, this method also utilizes TET2 and an Oxidation Enhancer, which enzymatically modifies 5mC and 5hmC to forms that are not substrates for APOBEC. The TET2 enzyme converts 5mC to 5caC (Figure 2) and the Oxidation Enhancer converts 5hmC to 5ghmC (9,10,11). Ultimately, cytosines are sequenced as thymines and 5mC and 5hmC are sequenced as cytosines, thereby protecting the integrity of the original 5mC and 5hmC sequence information.

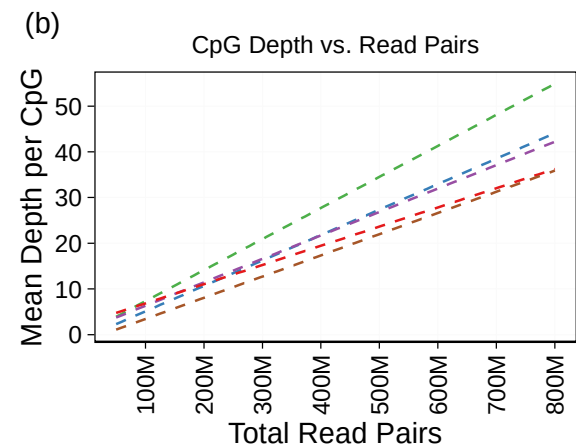
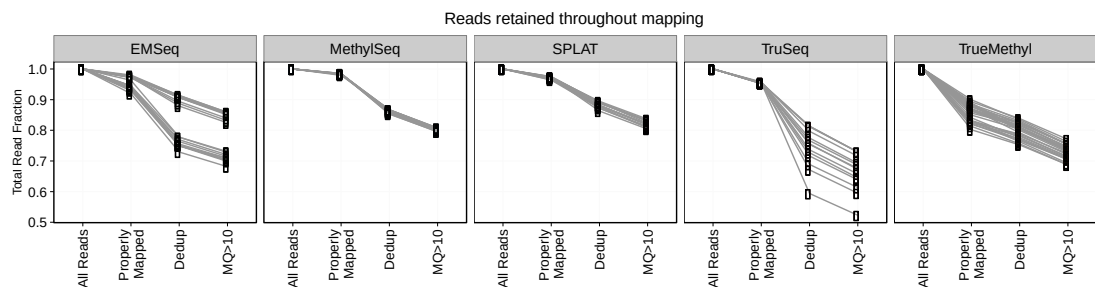
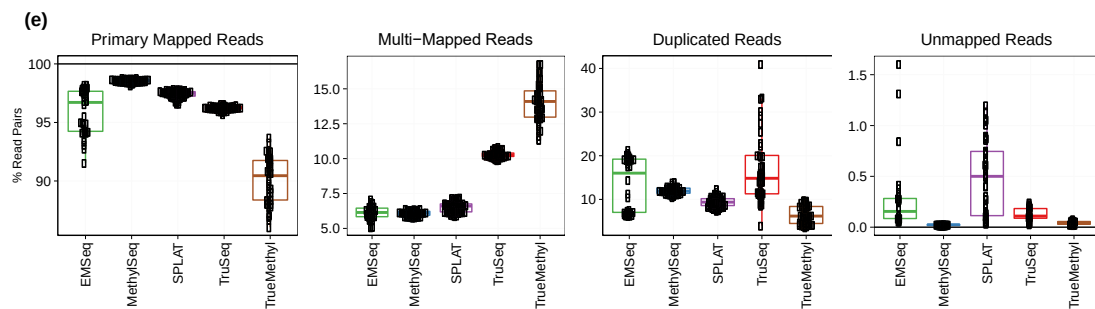
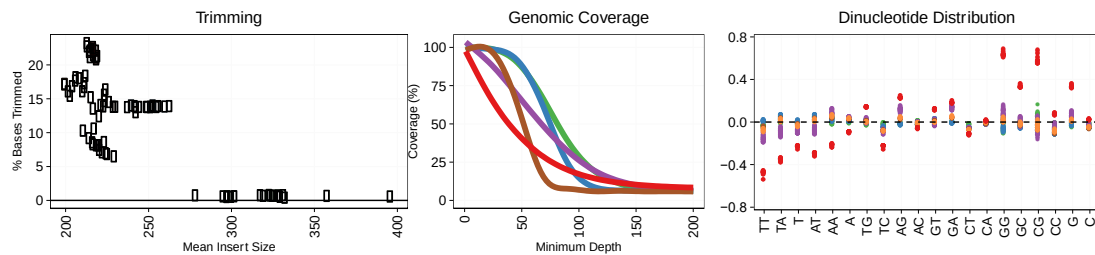
FIGURE 2: Enzymatic modification of cytosine



Megalodon (v2.2)

- Built on guppy (min. version 4.0.15)
- Guppy under the hood for base calling, then:
 - base calls (FASTA/Q)
 - reference mapping (SAM/BAM/CRAM)
 - base mod calls (bedMethyl/modVCF)
- Highest accuracy, optimized computational performance
- <https://github.com/nanoporetech/megalodon>

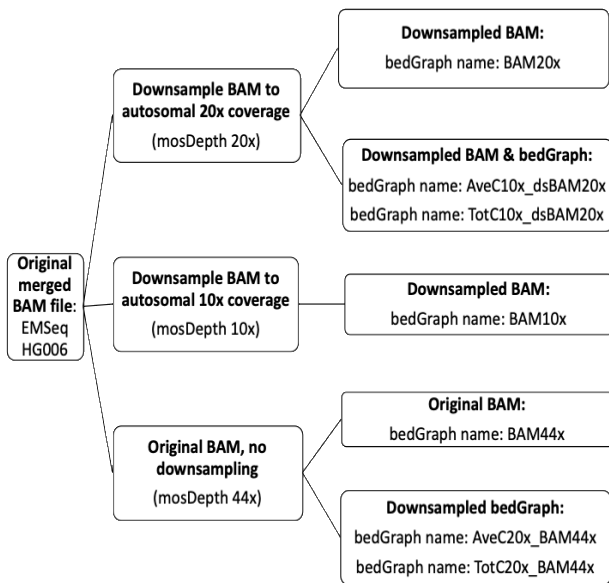




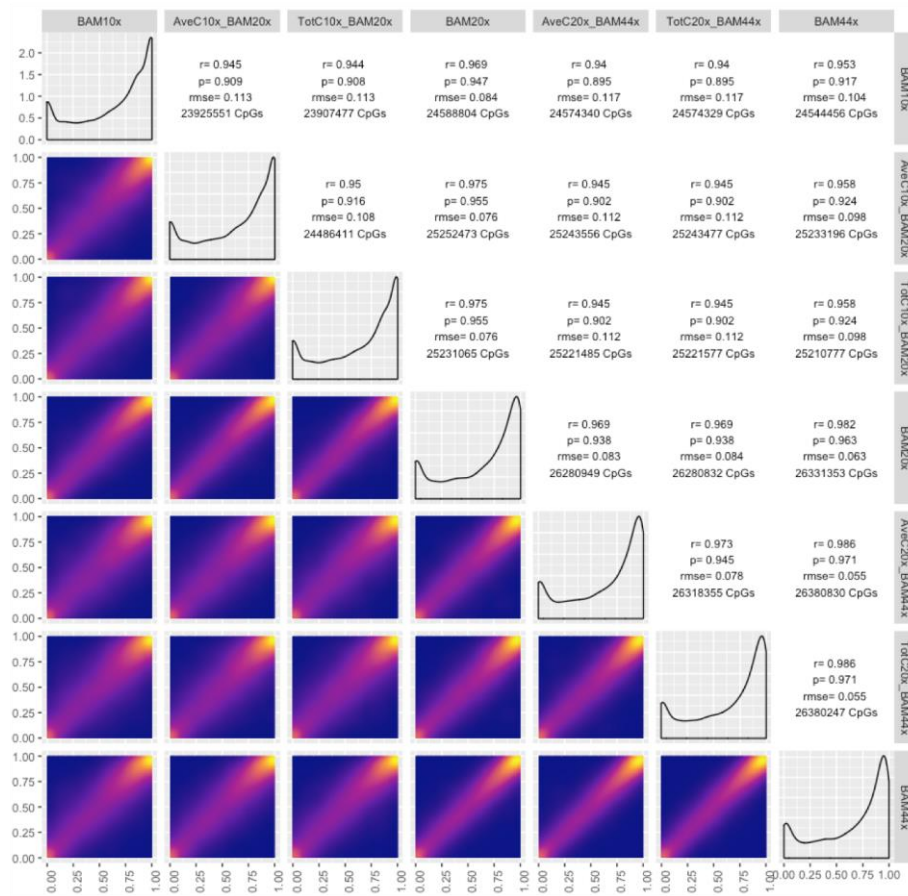
	EMSeq Lab 1	EMSeq Lab 2	Methyl Seq	SPLAT	True Methyl	TruSeq
Insert Size (bp)	299	327	250	221	217	215
Mapping Rate (%)	97	93	98	97	86	95
Duplicate Rate (%)	9	25	12	8	6	21
Dinucleotide Bias Score	3	1	4	10	2	27
Useable Bases (%)	90	77	74	81	69	60
Reads to reach 20x CpG coverage (M)	275	303	366	369	461	692
% CpGs $\geq$ 1x coverage	100	100	100	100	100	100
% CpGs $\geq$ 10x coverage	94	92	91	89	94	74

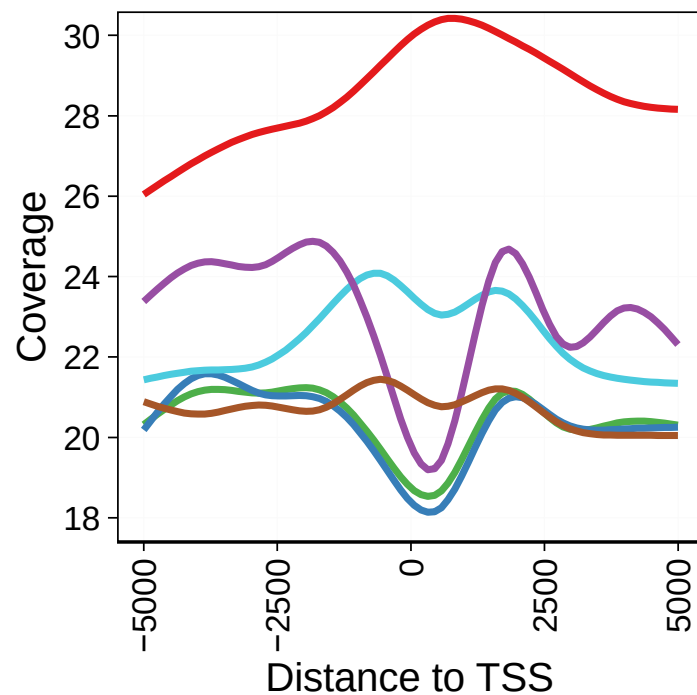
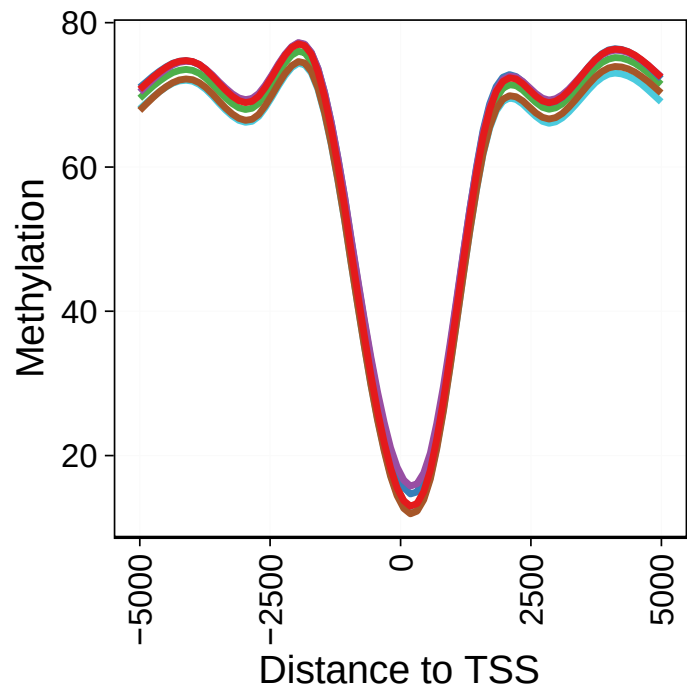
Downsampling at BAM and BED are comparable

a)



c)

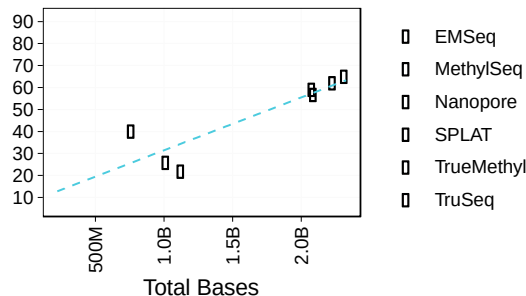




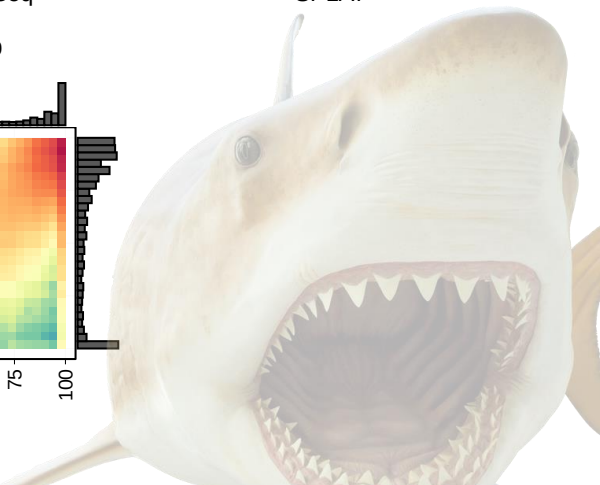
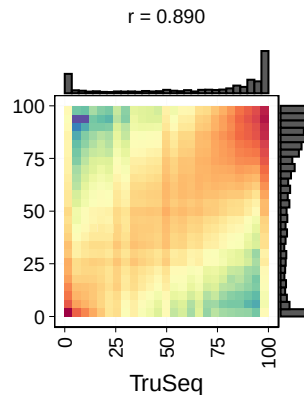
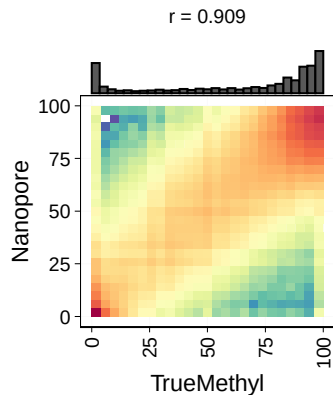
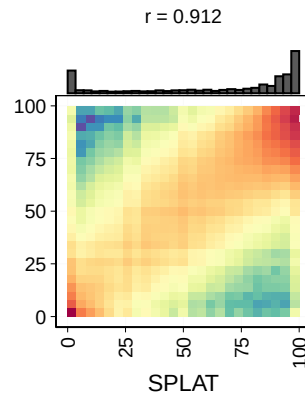
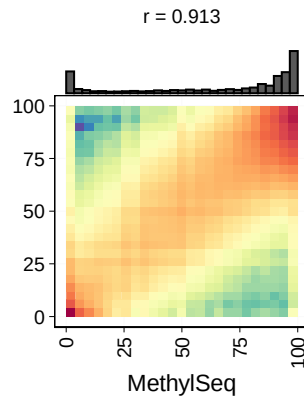
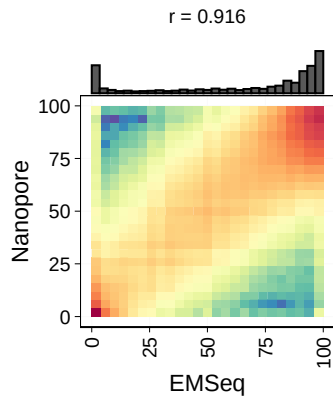
- EMSeq
- MethylSeq
- Nanopore
- SPLAT
- TrueMethyl
- TruSeq

Nanopore 5mC calls stacking up against WGBS

CpG Depth vs. Total Bases

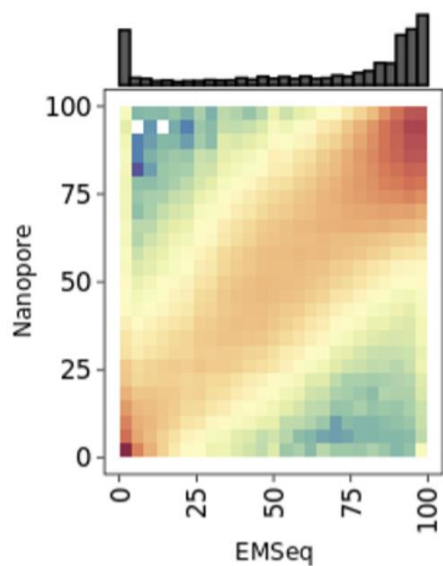


	TrueMethyl	TruSeq	MethylSeq	SPLAT	EMSeq	Nanopore
Nanopore	0.92	0.89	0.92	0.92	0.92	1
EMSeq	0.95	0.93	0.95	0.95	1	0.92
SPLAT	0.95	0.93	0.95	1	0.95	0.92
MethylSeq	0.95	0.93	1	0.95	0.95	0.92
TruSeq	0.93	1	0.93	0.93	0.93	0.89
TrueMethyl	1	0.93	0.95	0.95	0.95	0.92

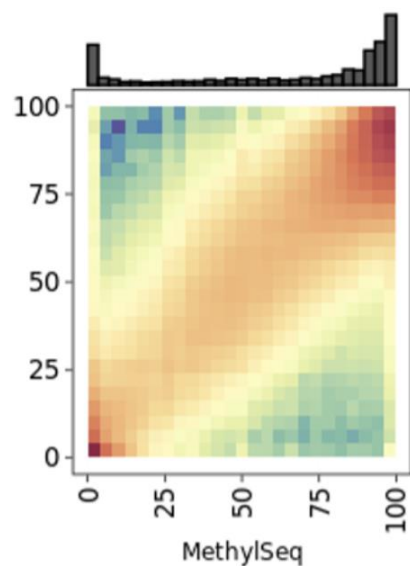


At 20x

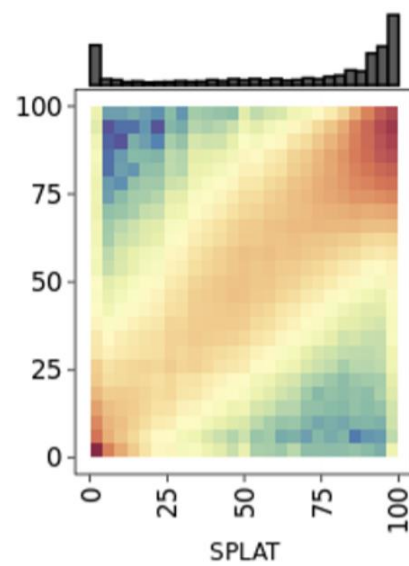
$r = 0.933$



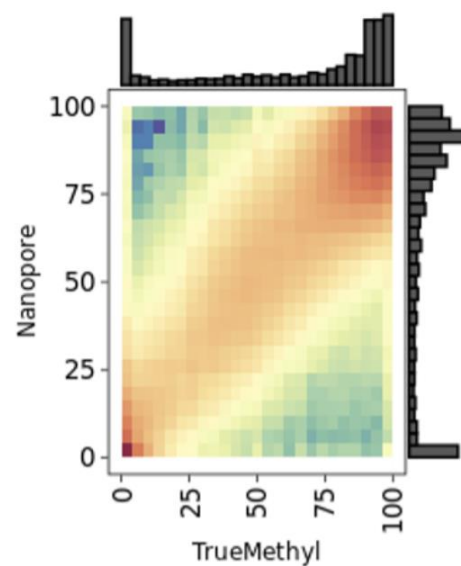
$r = 0.931$



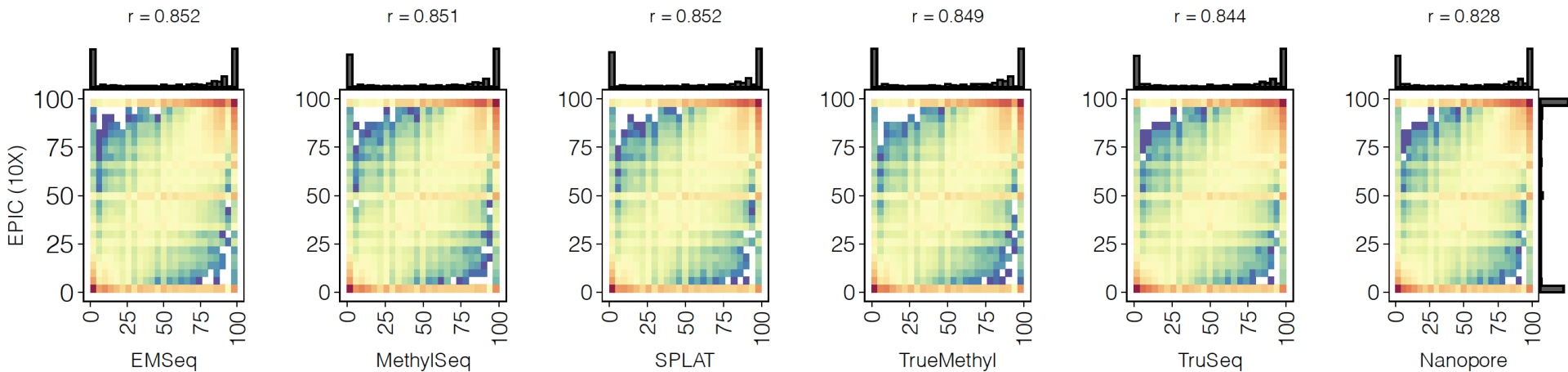
$r = 0.930$



$r = 0.927$



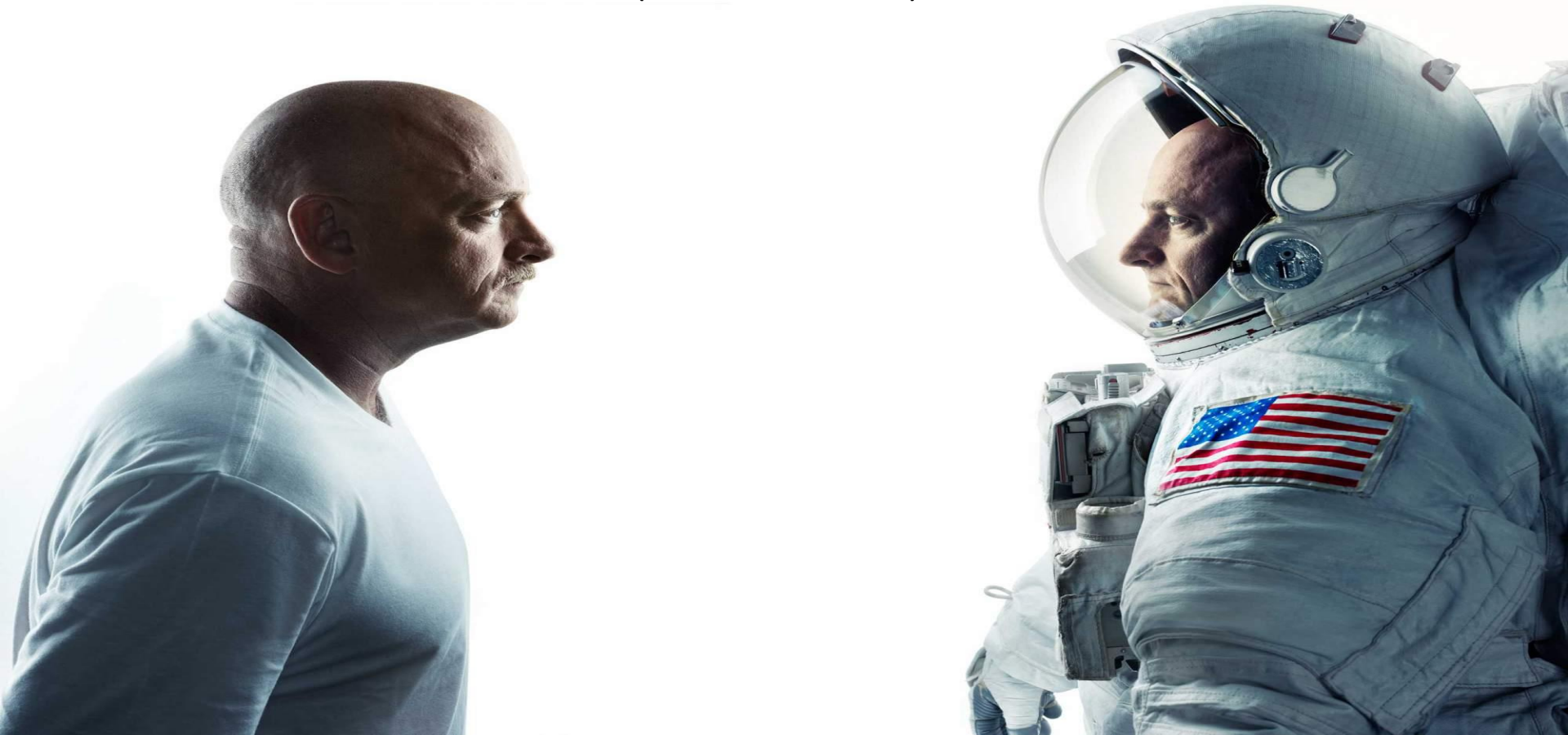
Or even 10x



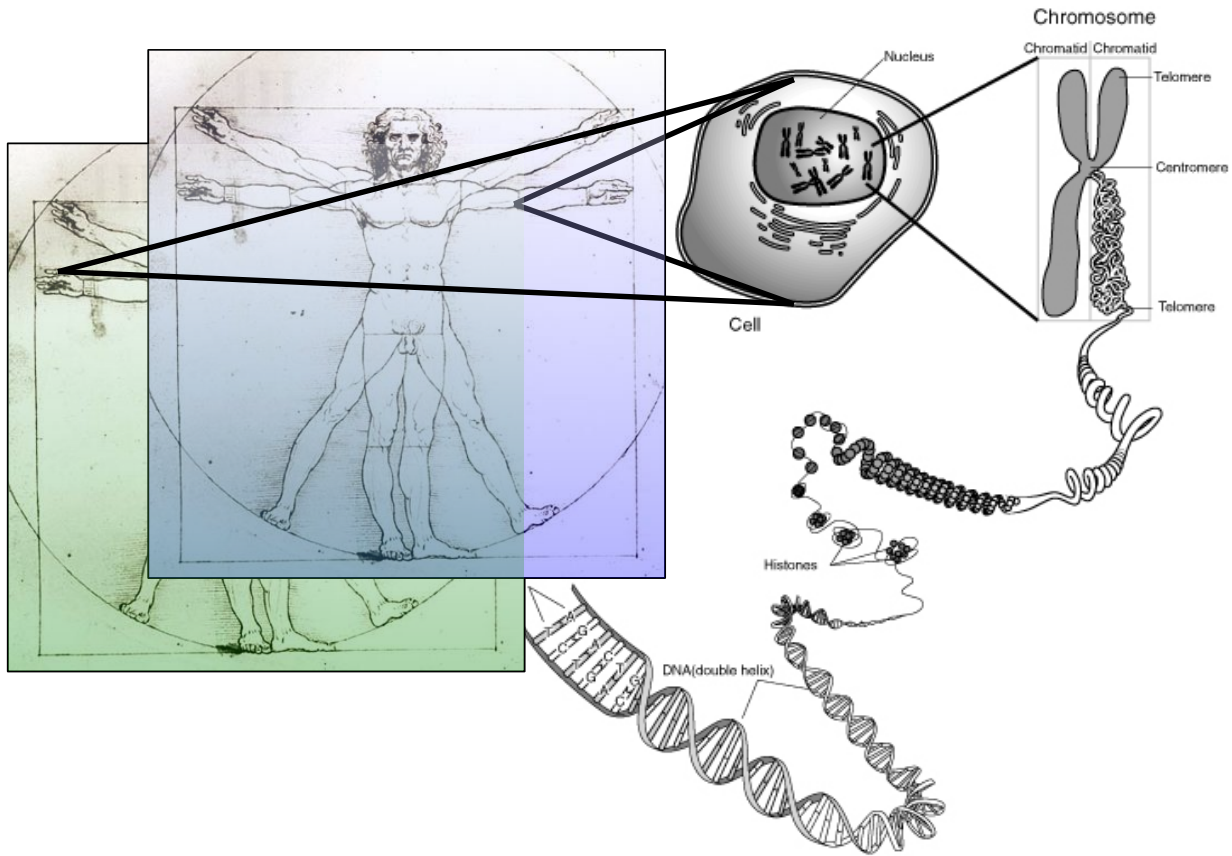
What about in
Space?

Integrative medicine with twin astronauts

(and one Senator)









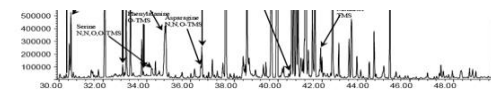
Vasculature



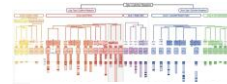
Cognition



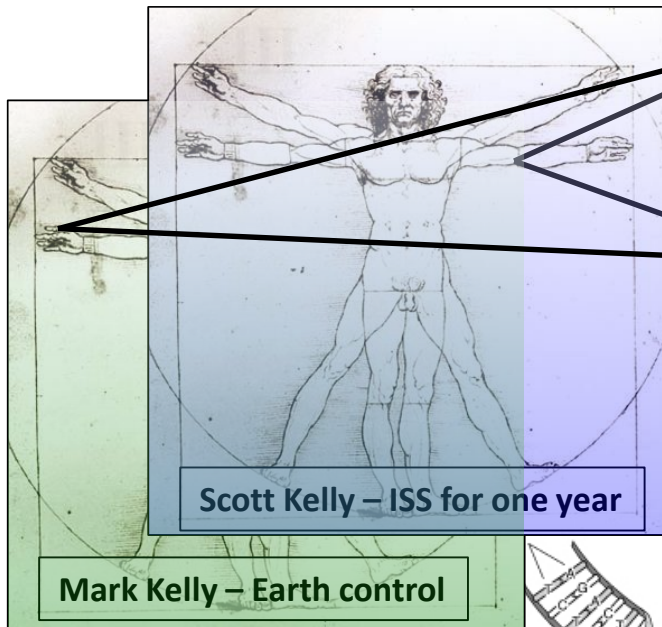
Microbiome



Targeted and Global Metabolomics



Cytokines



Scott Kelly - ISS for one year

Mark Kelly - Earth control

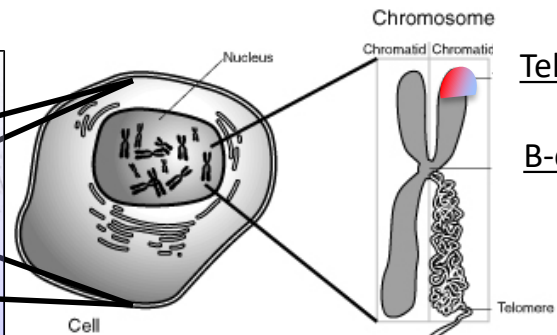
DNA Mutations & Structural Variation



DNA Hydroxy-methylation



DNA Methylation



Chromosome

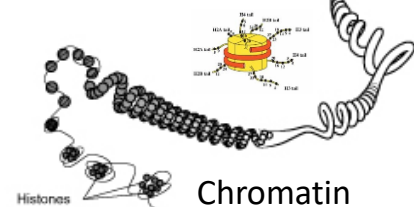
Chromatid Chromatic

Telomere Length

B-cells / T-cells

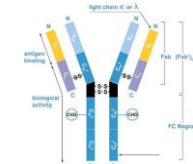


Antibody Titers



Histones

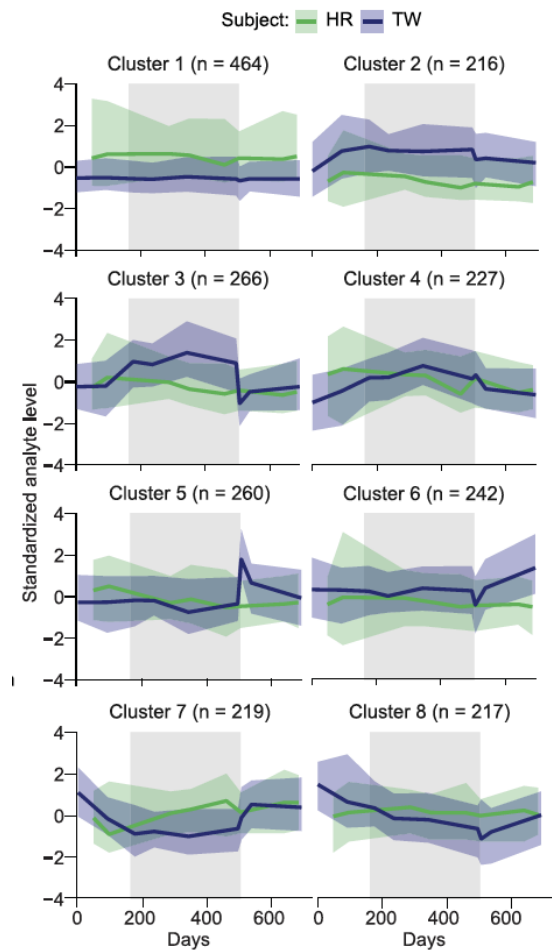
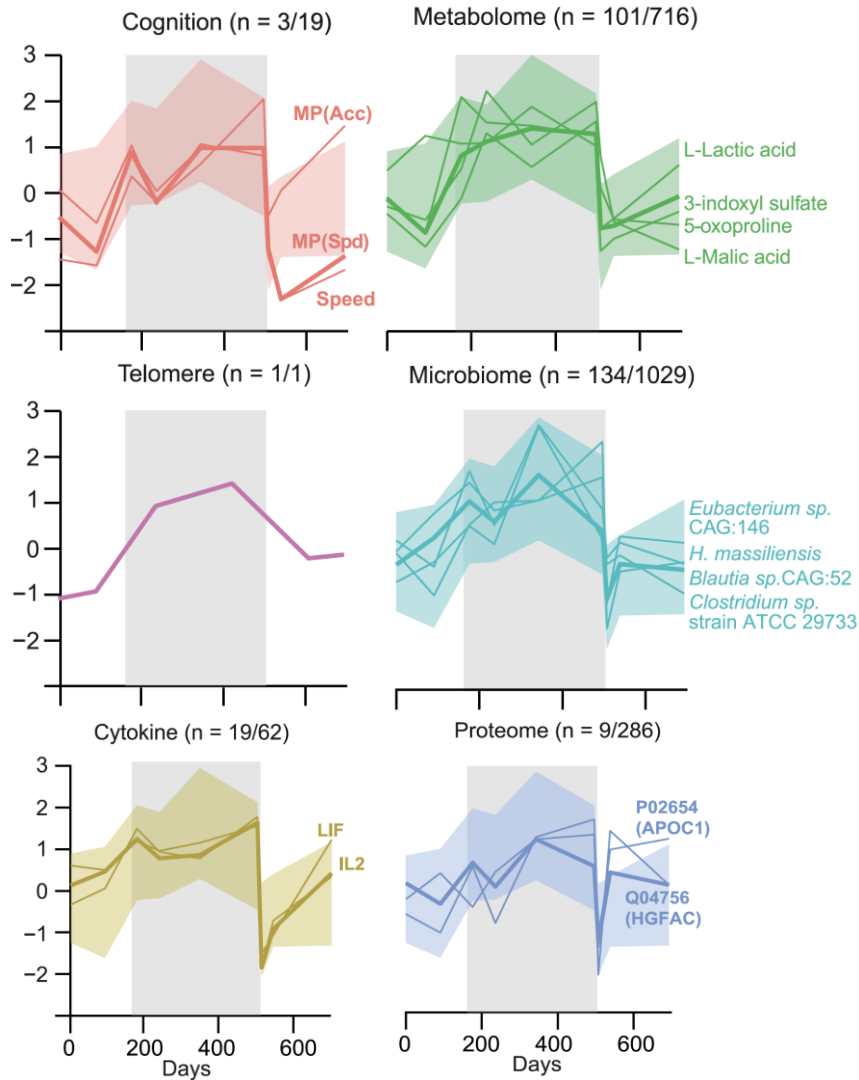
Chromatin



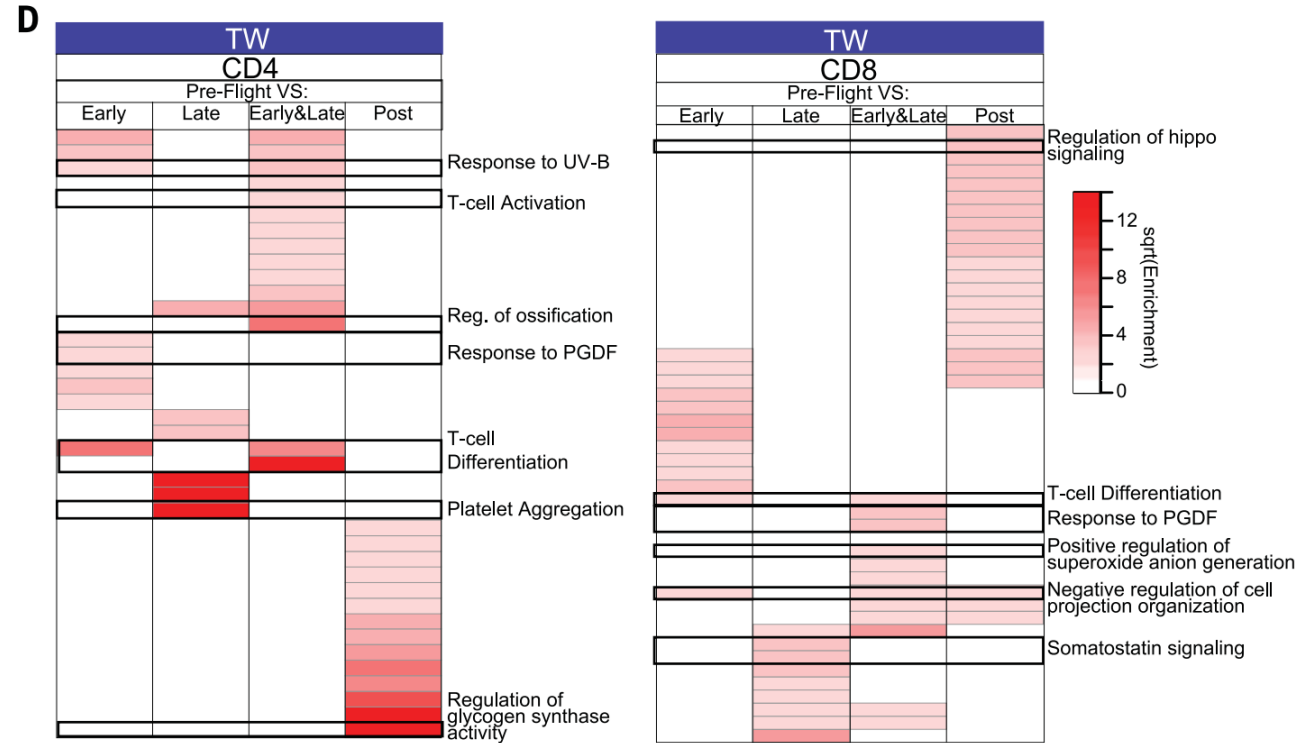
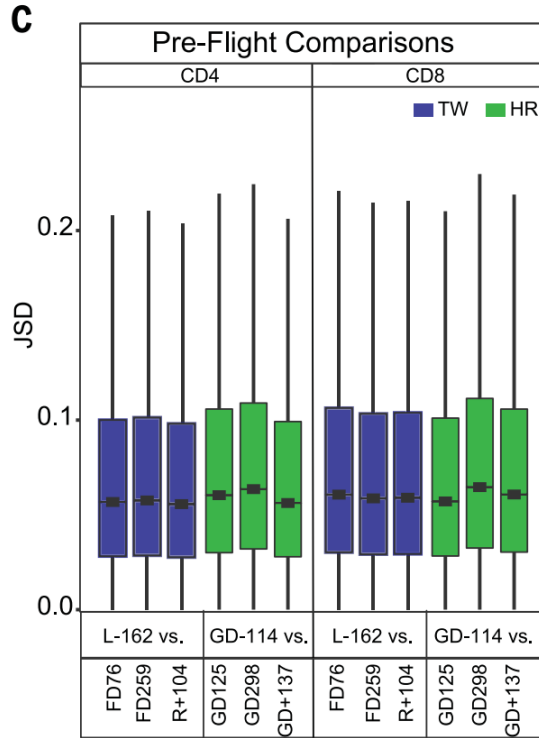
Proteomics

(small & large)
RNA expression
& RNA Methylation

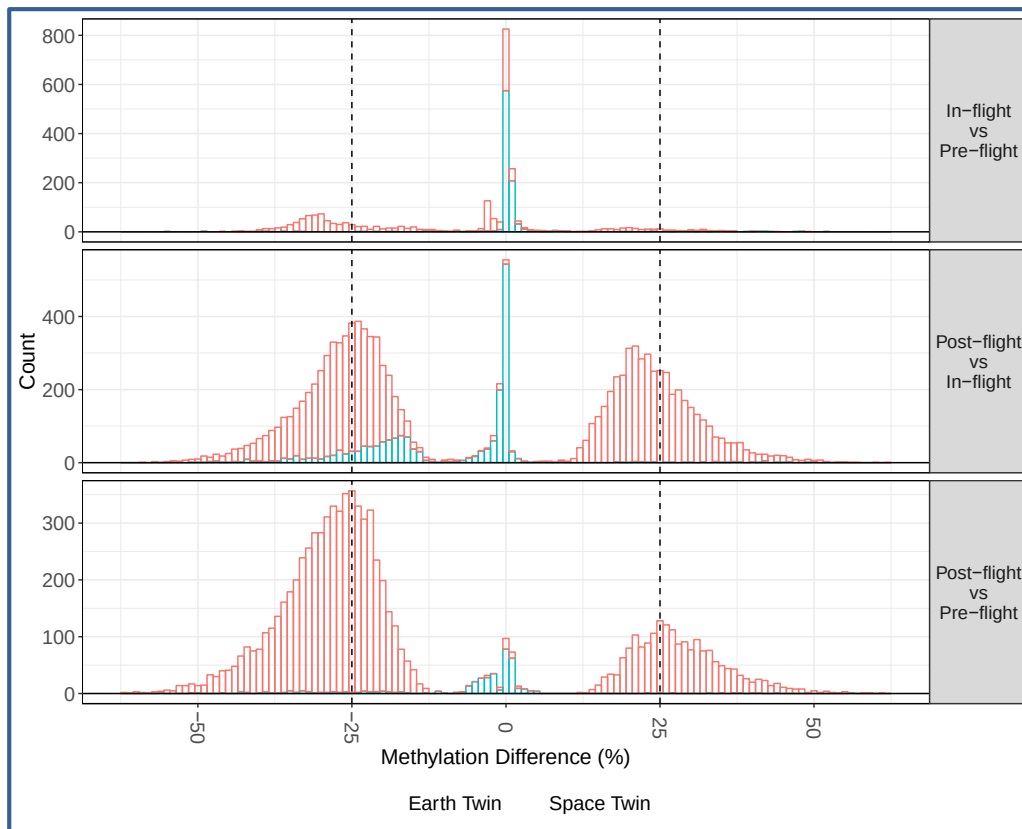
Multi-omics
analysis of human
spaceflight



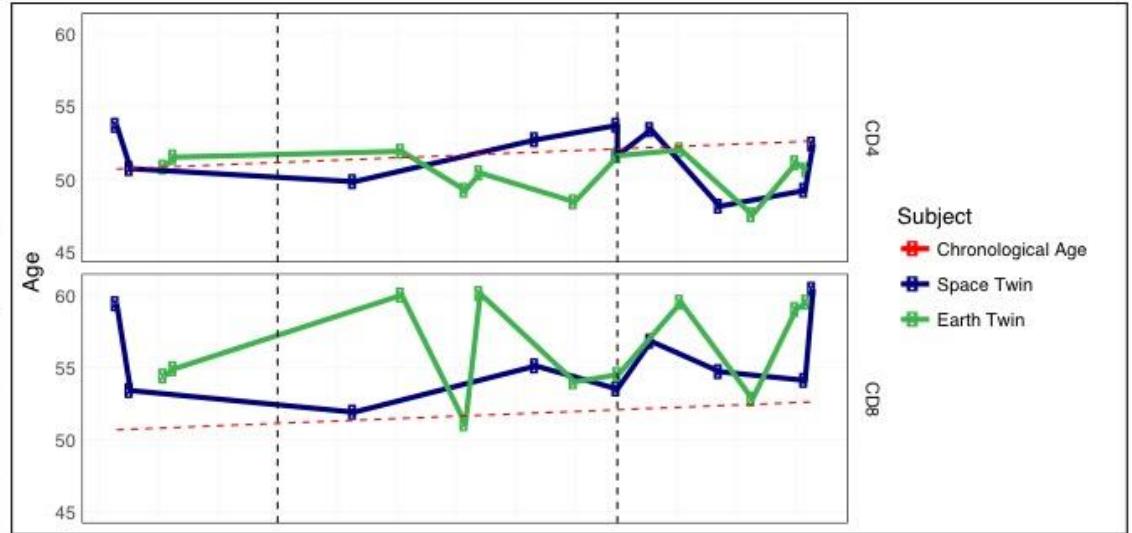
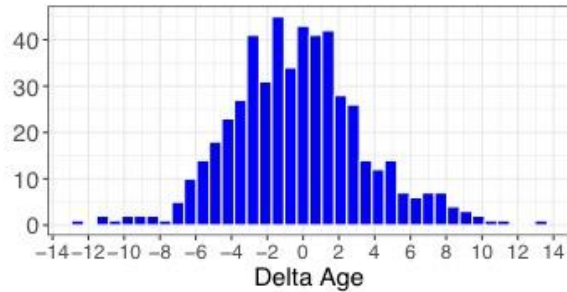
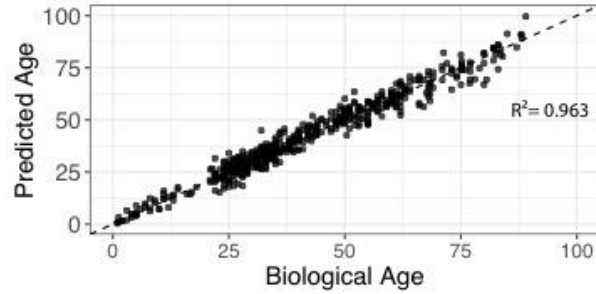
Significant shifts in the epigenome



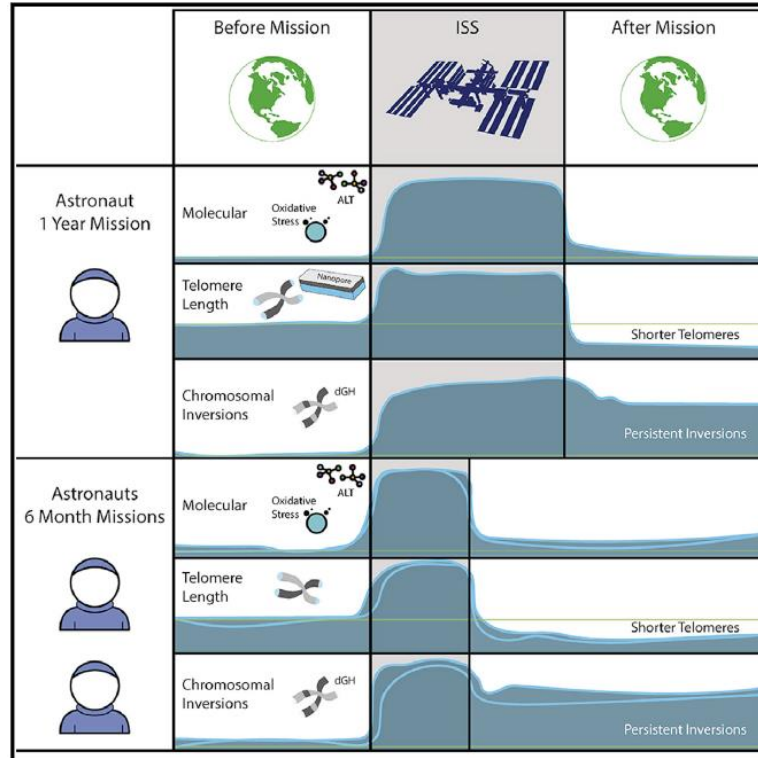
Dynamic on earth; stable in space



Age mostly stable



Telomere elongation validated in two other astronauts



SHARE



817



28



Matt Moniz (right) ascends the world's sixth-tallest peak, Mount Cho Oyu, during a 2014 climb. WILLIE BENEGAS

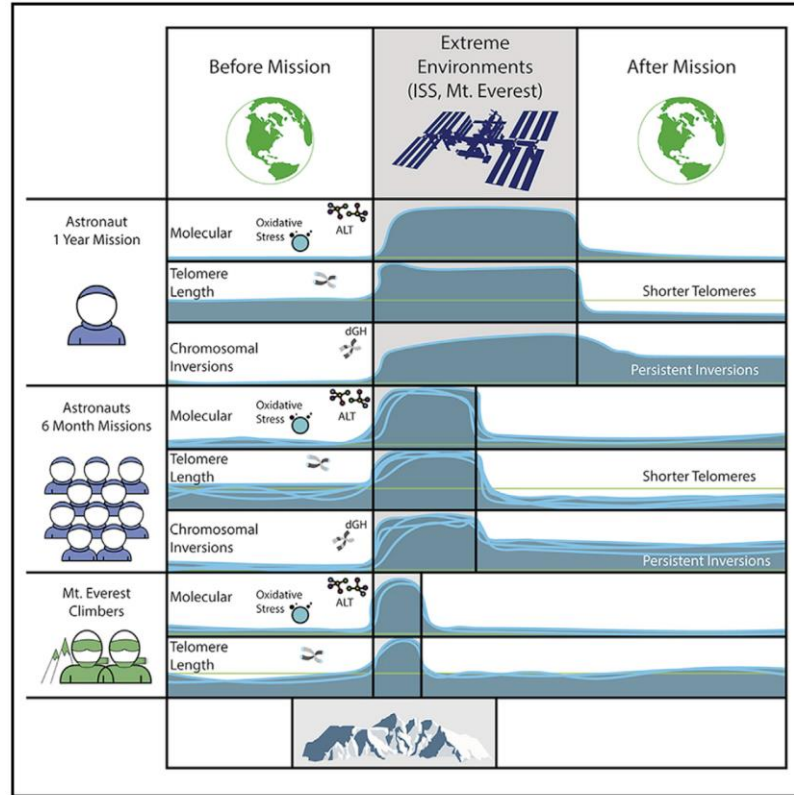
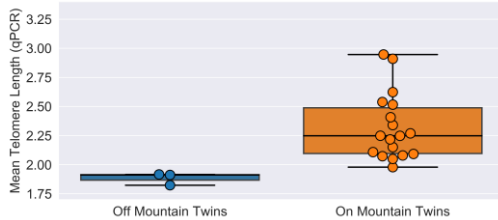
Two mountaineers are trying to recreate NASA's twin study—on Mount Everest

By **Vedrana Simicevic** | May. 4, 2018 , 12:15 PM

NASA's widely publicized twin study—which compared astronaut Scott Kelly's bodily functions to

<http://www.sciencemag.org/news/2018/05/two-mountaineers-are-trying-recreate-nasa-s-twin-study-mount-everest>

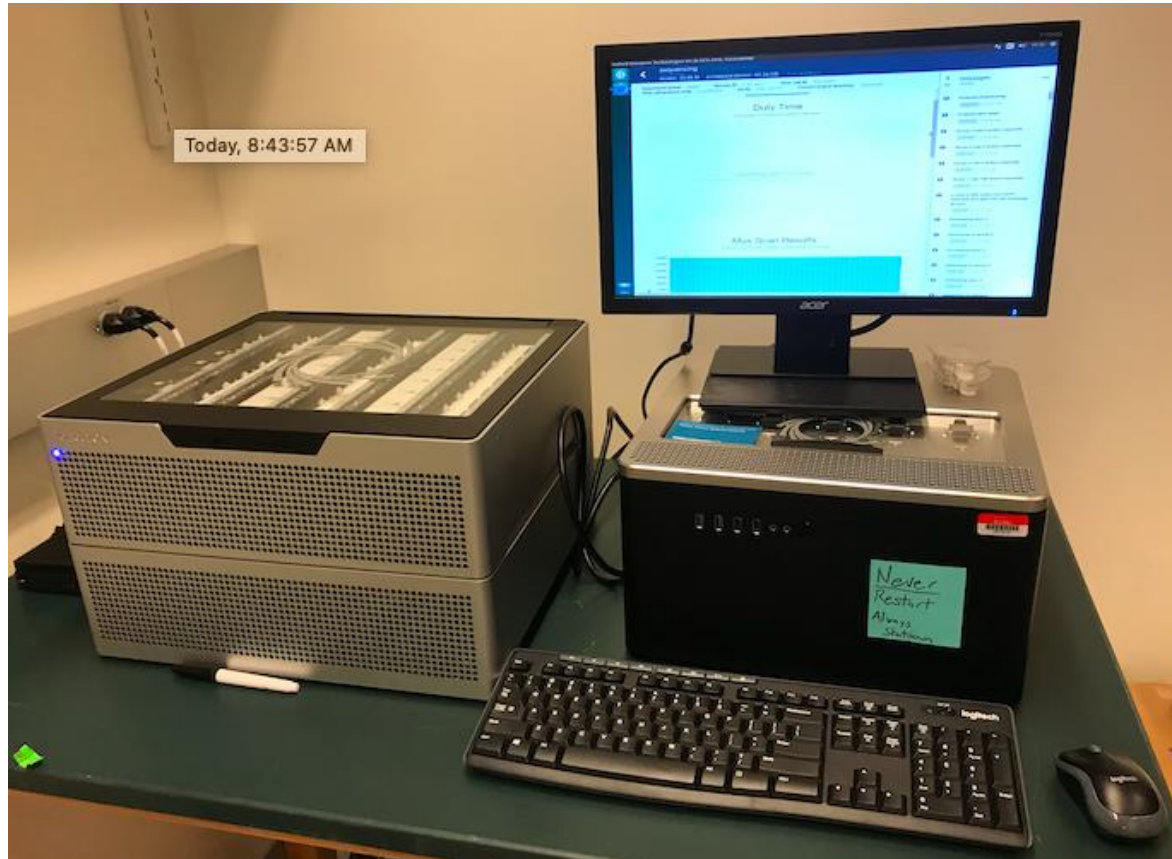
Then 11 other astronauts & on Mt. Everest



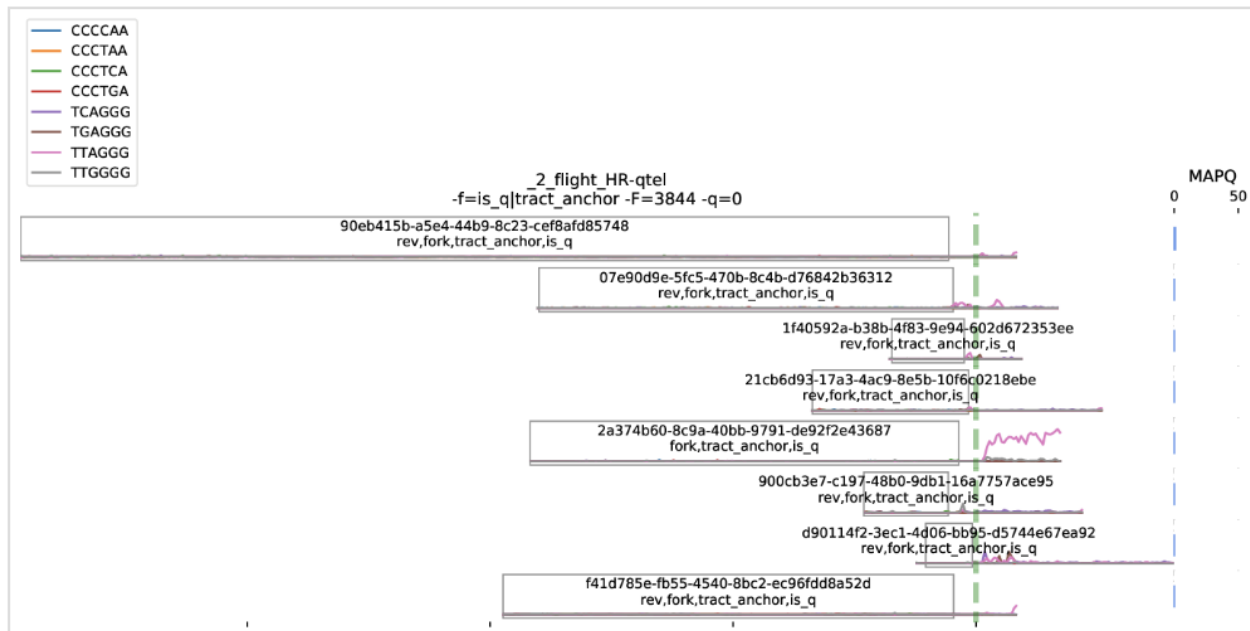
Luxton *et al.*, *Cell Reports*, 2020

[https://www.cell.com/cell-reports/fulltext/S2211-1247\(20\)31446-7](https://www.cell.com/cell-reports/fulltext/S2211-1247(20)31446-7)

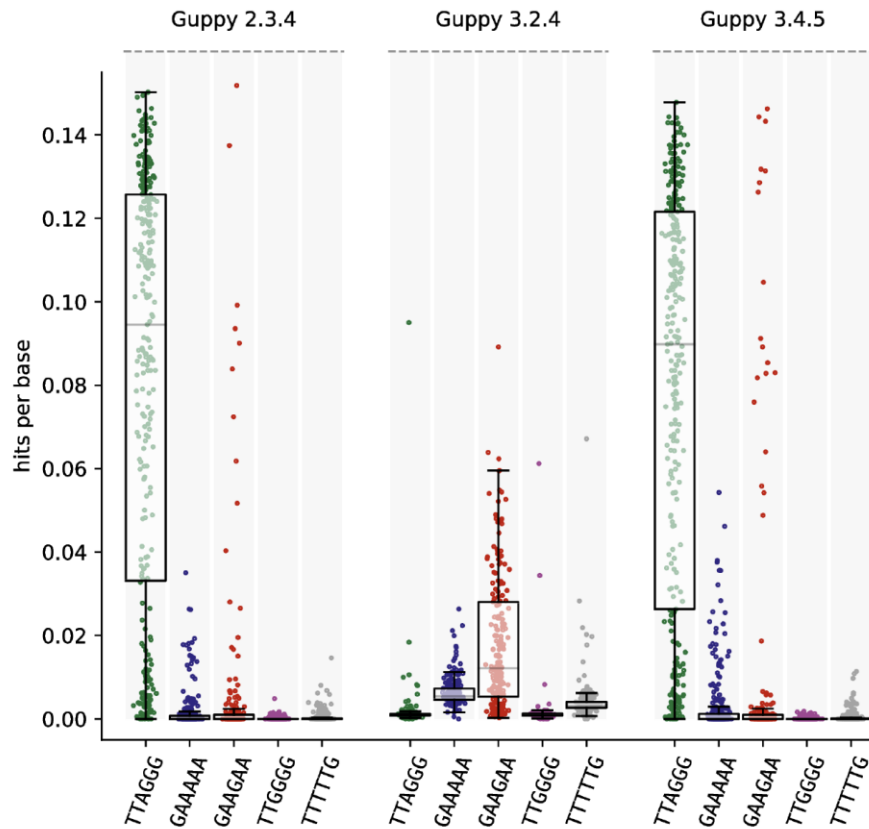
Break out the long reads! ONT PromethION



Mapping is a challenge (Minimap2) – TTAGGG not always clear



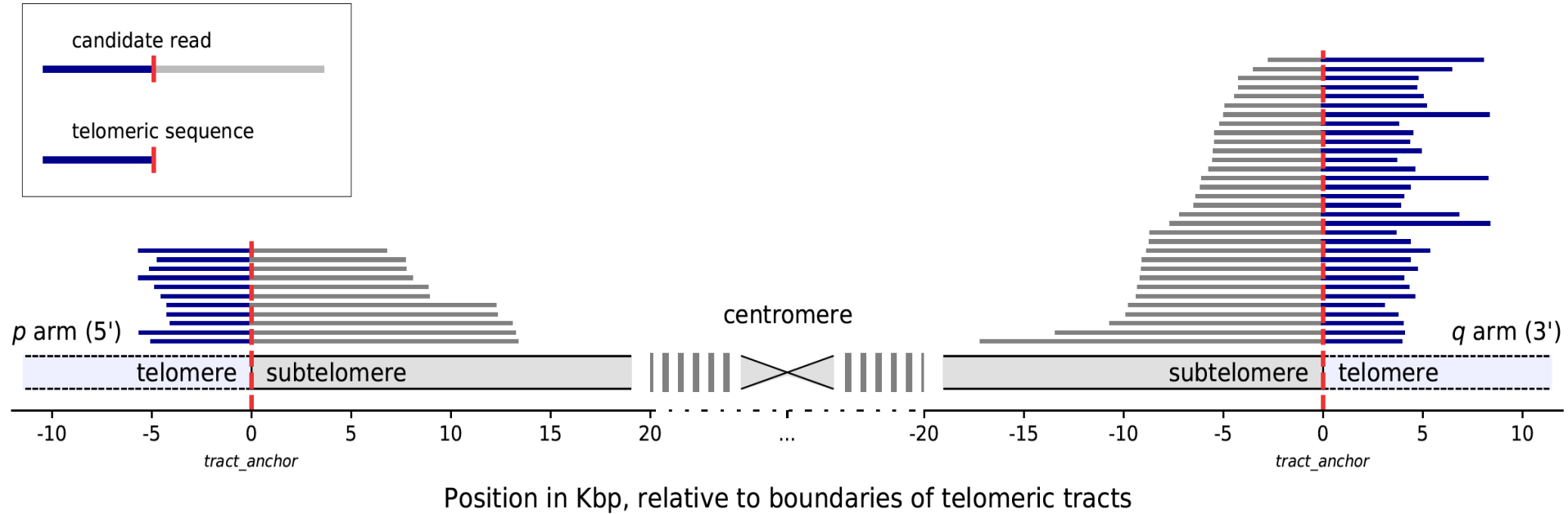
The base caller determined the motif



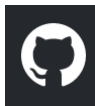
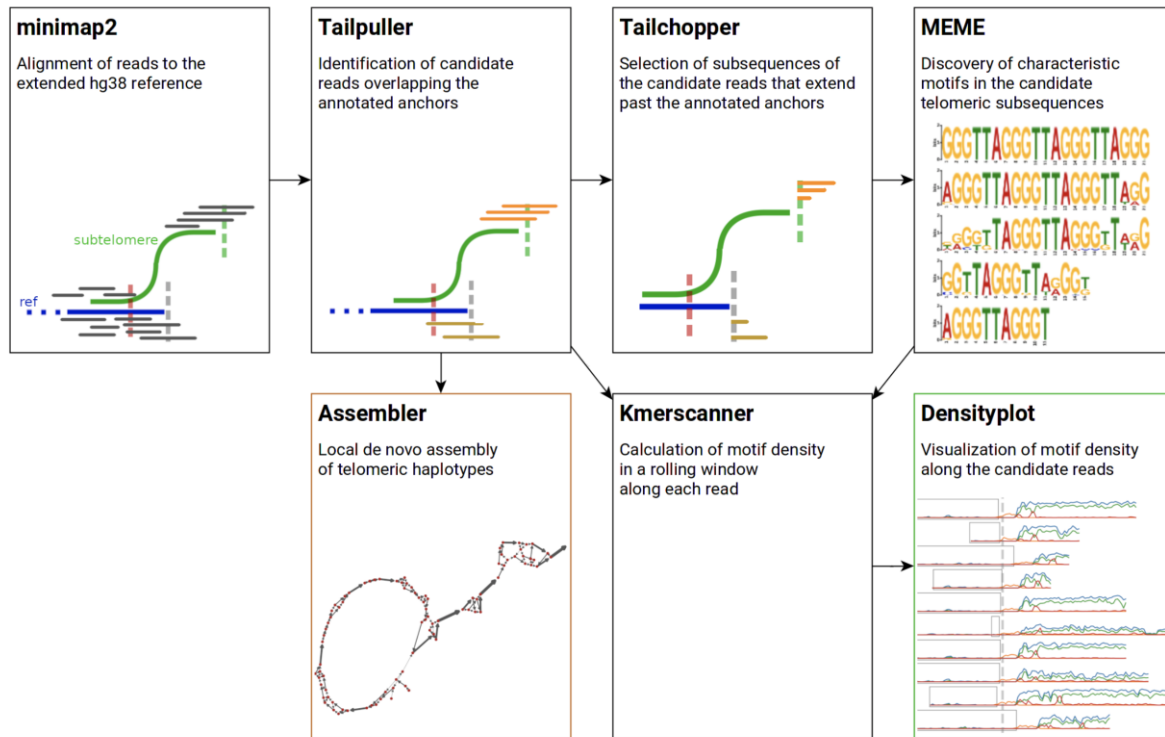
These aren't the telomeres you're looking for



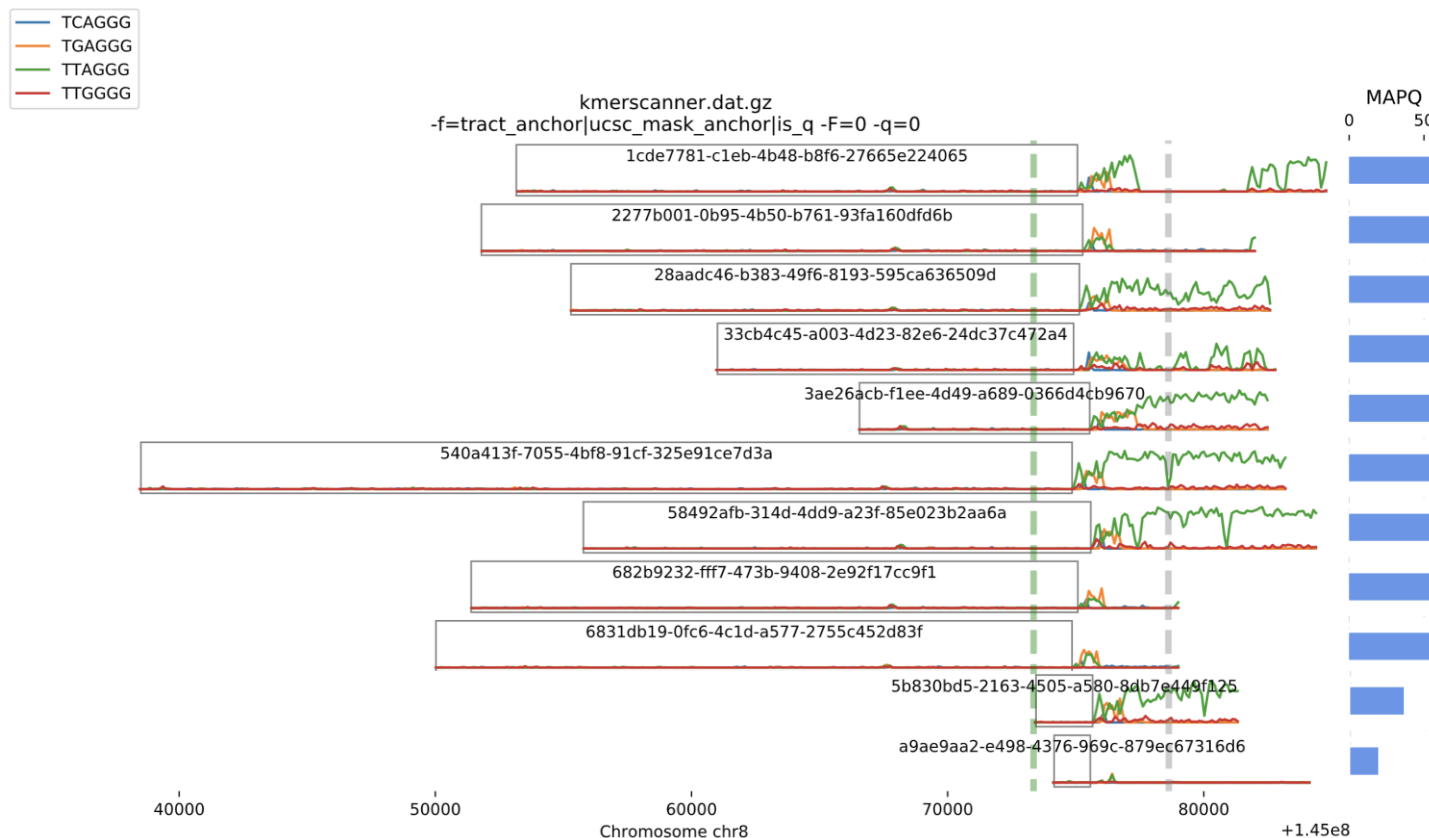
Line up the reads



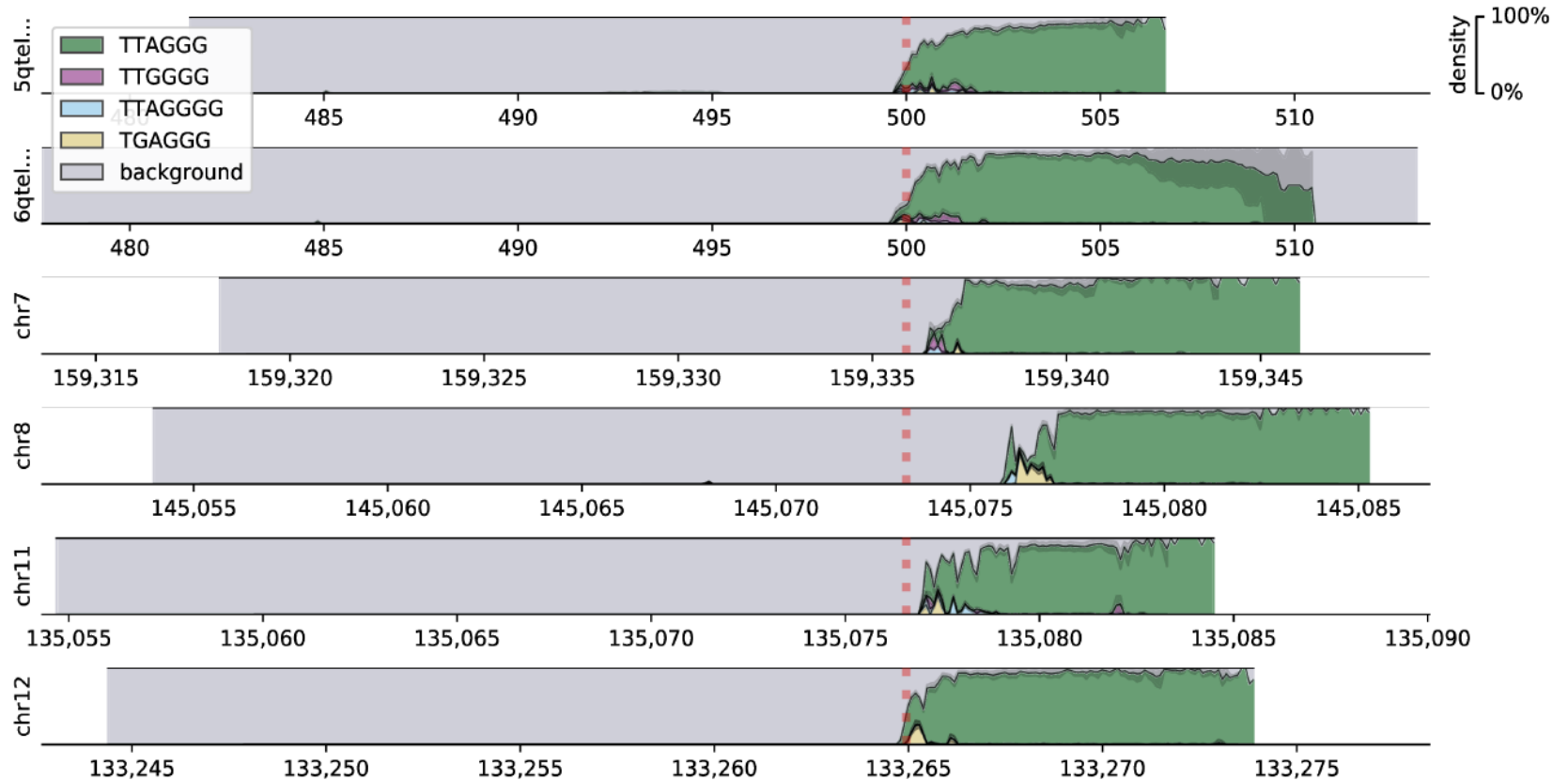
Looking for the Edge Cases (EdgeCase)



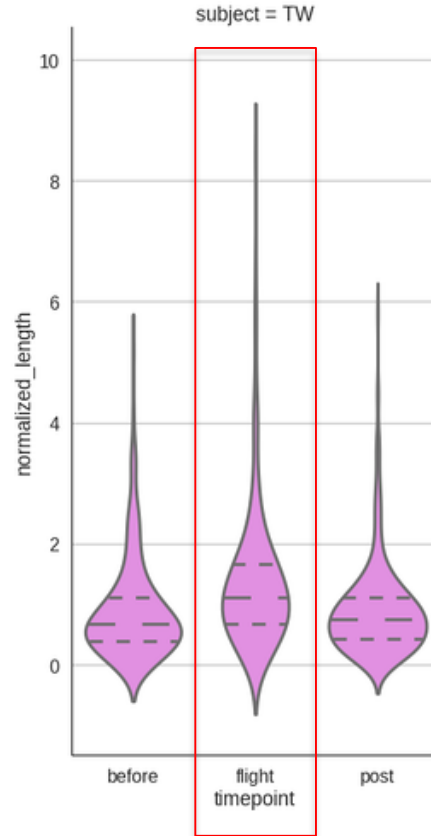
Now we can see it: nanopore data



And PacBio HiFi reads



Indeed the telomeres are longer



Telomere still have many stories to tell...



CSH PRESS GENOME RESEARCH

Uncover more genes and SNPs from single cells

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Haplotype diversity and sequence heterogeneity of human telomeres

Kirill Grigorev^{1,2,10}, Jonathan Foox^{1,2,3,10}, Daniela Bezdan^{1,2,3,4,5}, Daniel Butler¹, Jared J. Luxton^{6,7}, Jake Reed¹, Miles J. McKenna^{6,7}, Lynn Taylor⁶, Kerry A. George⁸, Cem Meydan^{1,2,3}, Susan M. Bailey^{6,7} and Christopher E. Mason^{1,2,3,9}

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OPEN ACCESS ARTICLE

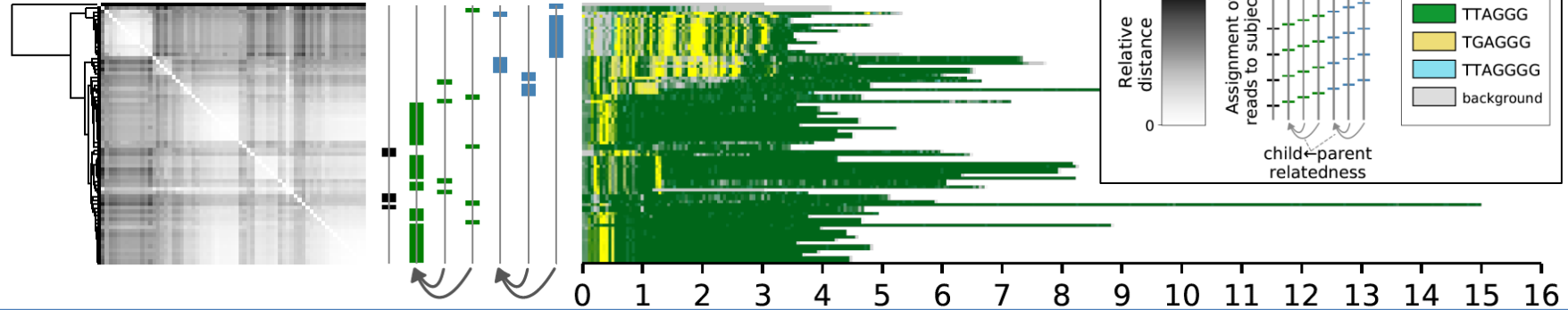
This Article

Published in Advance June 23, 2021, doi: 10.1101/gr.274639.120
Genome Res. 2021. 31: 1269-1279
© 2021 Grigorev et al.; Published

<https://genome.cshlp.org/content/31/7/1269>

Population-specific, non-canonical variants and haplotypes

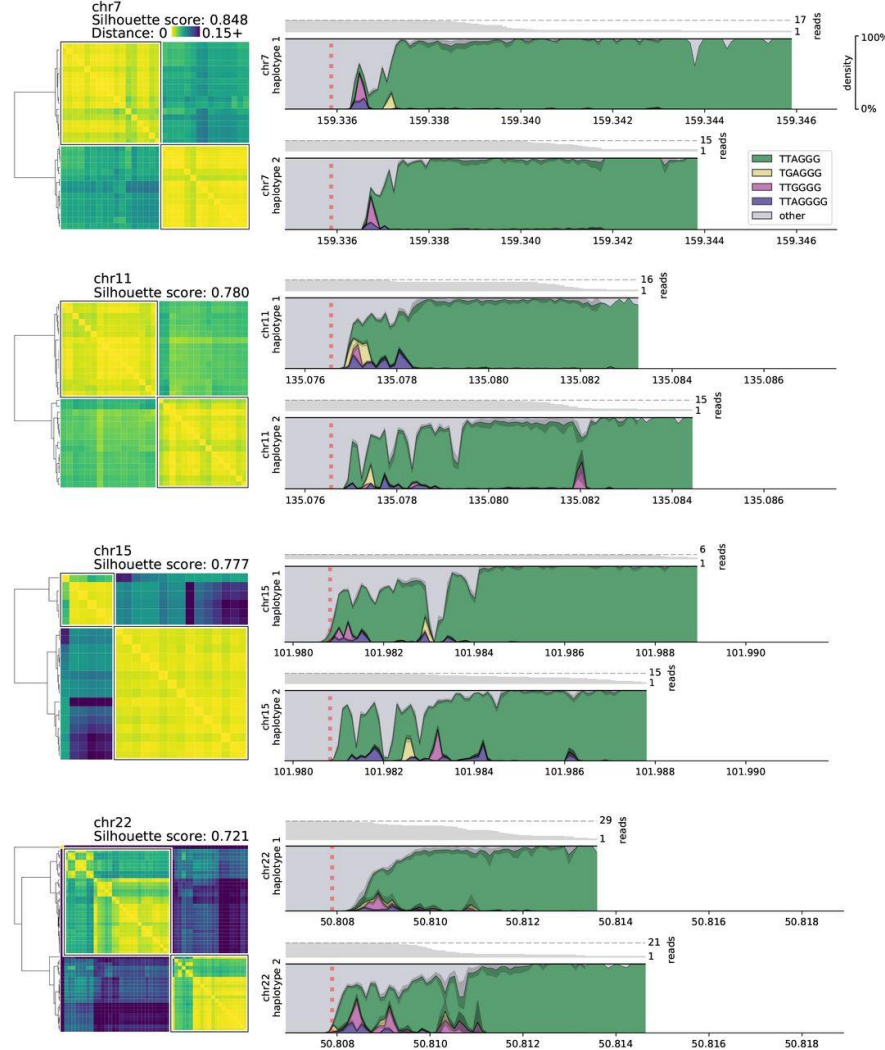
12q (chr12)



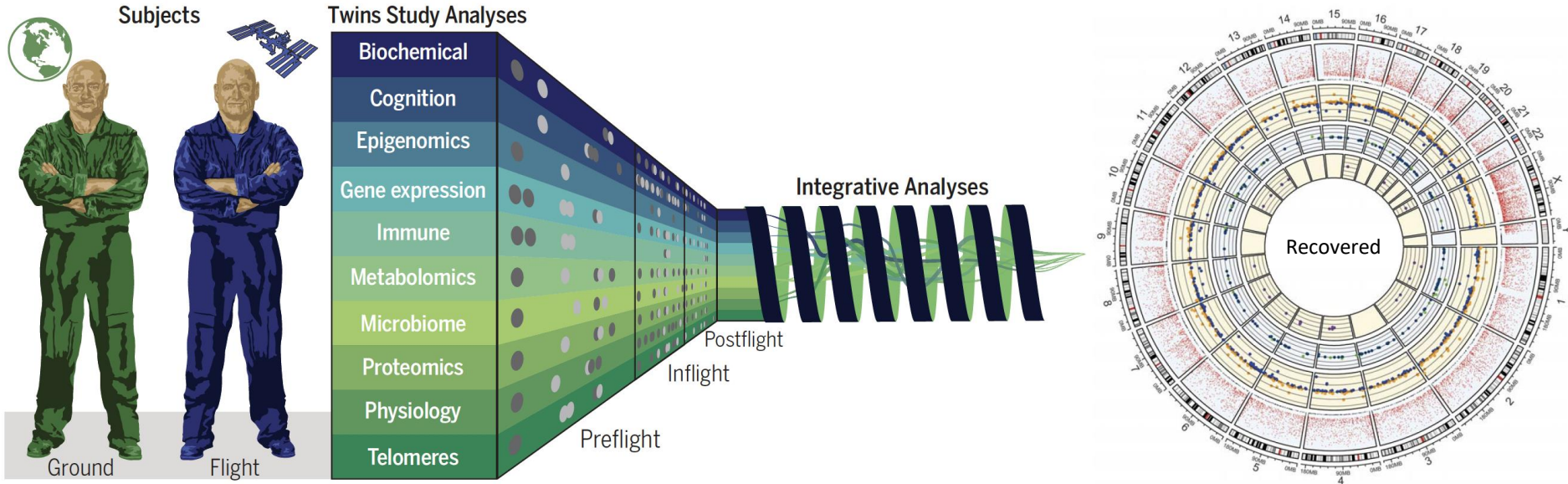
Pulling out more: Haplotypes to diplotypes

Levenshtein distances

$$\text{lev}_{a,b}(i, j) = \begin{cases} \max(i, j) \\ \min \begin{cases} \text{lev}_{a,b}(i-1, j) + 1 \\ \text{lev}_{a,b}(i, j-1) + 1 \\ \text{lev}_{a,b}(i-1, j-1) + 1_{(a_i \neq b_j)} \end{cases} \end{cases}$$

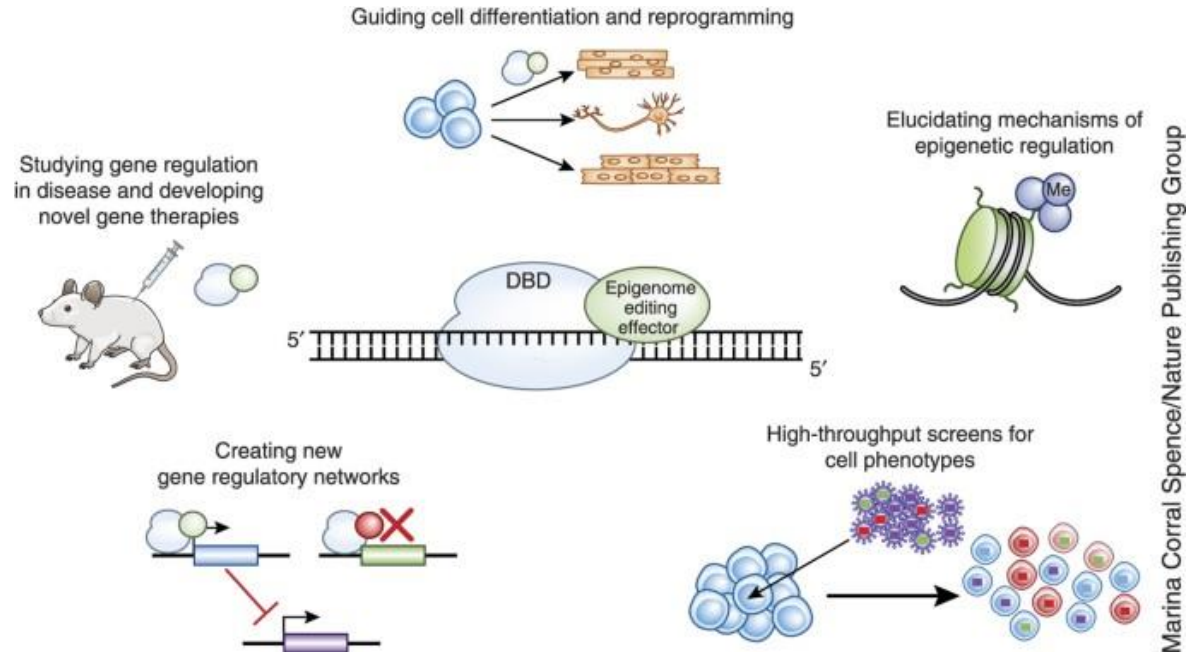


Overall good news

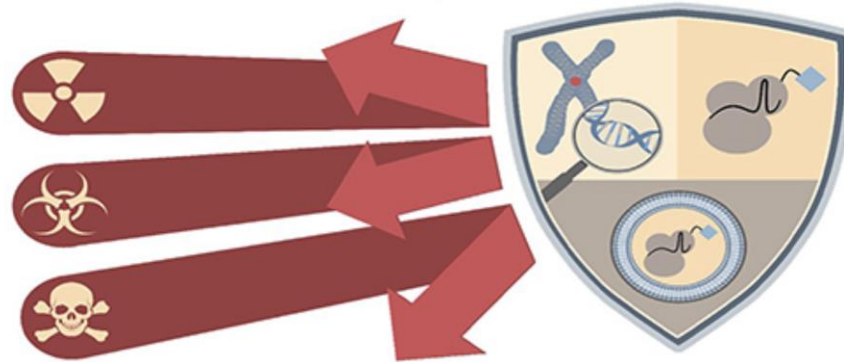


Can we just
tweak the
epigenome?

Tools to modify the epigenome (e.g. Cas9-TET1)

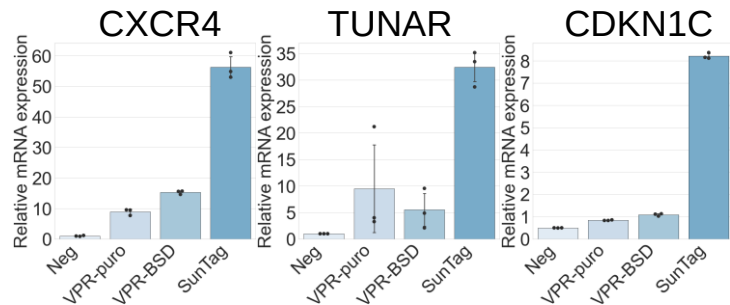


PReemptive Expression of Protective Alleles and Response Elements (PREPARE)

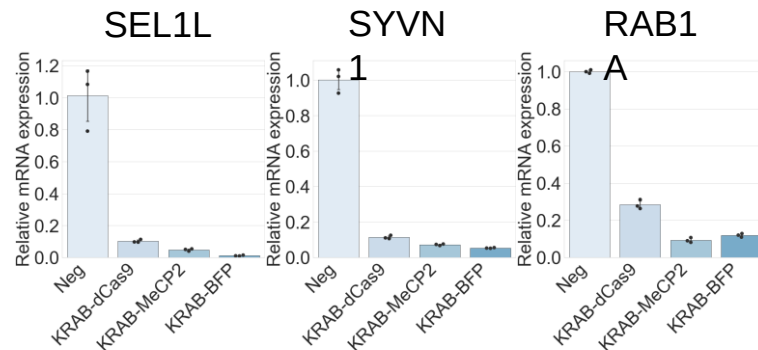


Epigenomic reprogramming for future space exploration

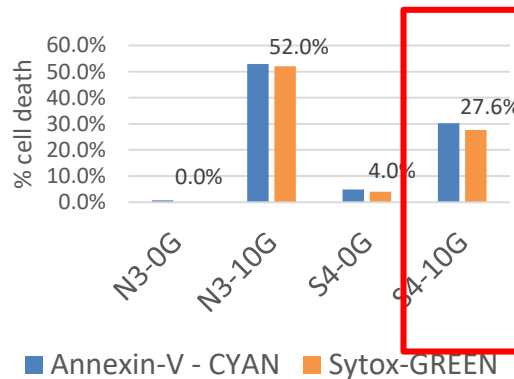
CRISPRa editor



CRISPRi editor

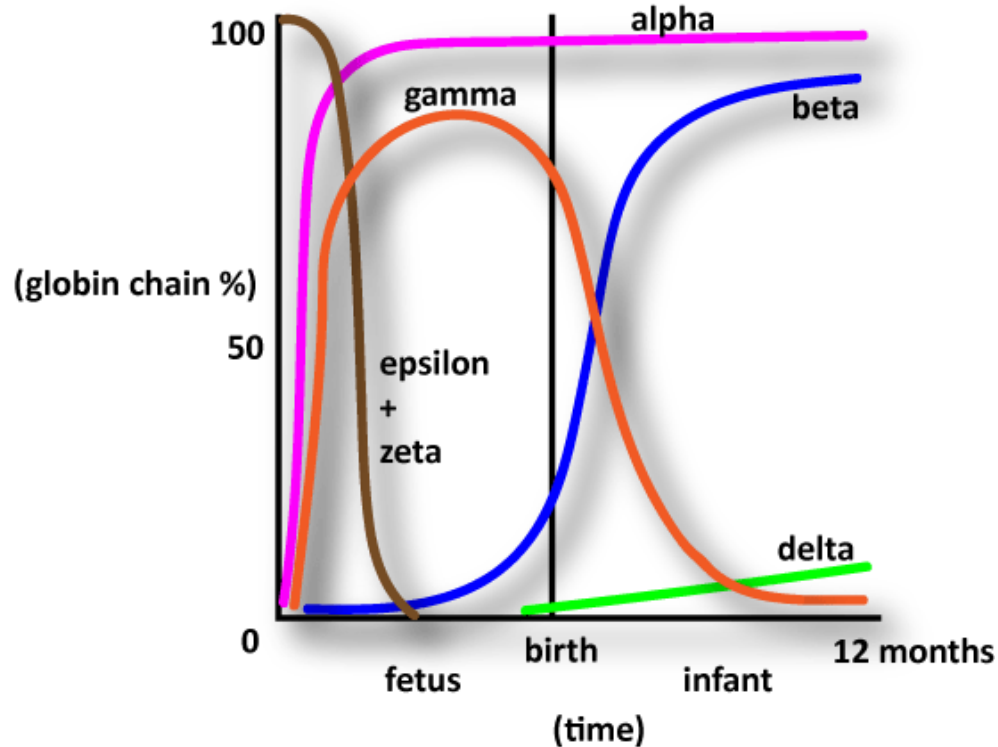


Cell death



Epigenome edited cells showed radioresistance.
JangKeun Kim

If alpha has a problem,
why not bring back gamma?



Some successes already for Transfusion-dependent β -thalassemia (TDT) and sickle cell disease (SCD)

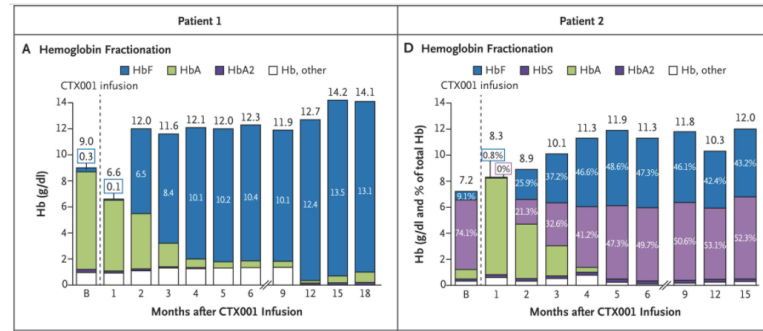
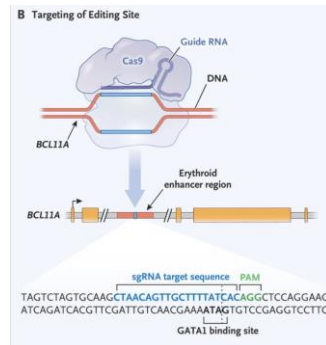
ORIGINAL ARTICLE BRIEF REPORT

January 21, 2021

N Engl J Med 2021; 384:252-260

CRISPR-Cas9 Gene Editing for Sickle Cell Disease and β -Thalassemia

Haydar Frangoul, M.D., David Altshuler, M.D., Ph.D., M. Domenica Cappellini, M.D., Yi-Shan Chen, Ph.D., Jennifer Domm, M.D., Brenda K. Eustace, Ph.D., Juergen Foell, M.D., Josu de la Fuente, M.D., Ph.D., Stephan Grupp, M.D., Ph.D., Rupert Handgretinger, M.D., Tony W. Ho, M.D., Antonis Kattamis, M.D., Andrew Kernysky, Ph.D., Julie Lekstrom-Himes, M.D., Amanda M. Li, M.D., Franco Locatelli, M.D., Markus Y. Mapara, M.D., Ph.D., Mariane de Montalembert, M.D., Damiano Rondelli, M.D., Akshay Sharma, M.B., B.S., Sujit Sheth, M.D., Sandeep Soni, M.D., Martin H. Steinberg, M.D., Donna Wall, M.D., Angela Yen, Ph.D., and Selim Corbacioglu, M.D.



Now an FDA-approved gene therapy



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FDA NEWS RELEASE

FDA Approves First Gene Therapies to Treat Patients with Sickle Cell Disease

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For Immediate Release: December 08, 2023

Today, the U.S. Food and Drug Administration approved two milestone treatments, Casgevy and Lyfgenia, representing the first cell-based gene therapies for the treatment of sickle cell disease (SCD) in patients 12 years and older. Additionally, one of these therapies, Casgevy, is the first FDA-approved treatment to utilize a type of novel genome editing technology, signaling an innovative advancement in the field of gene therapy.

Other places to use it (shameless plug)



The Next 500 Years

Engineering Life to Reach New Worlds

By Christopher E. Mason

An argument that we have a moral duty to explore other planets and solar systems – because human life on Earth has an expiration date.

<https://mitpress.mit.edu/books/next-500-years>

There are many RNA-mods as well:

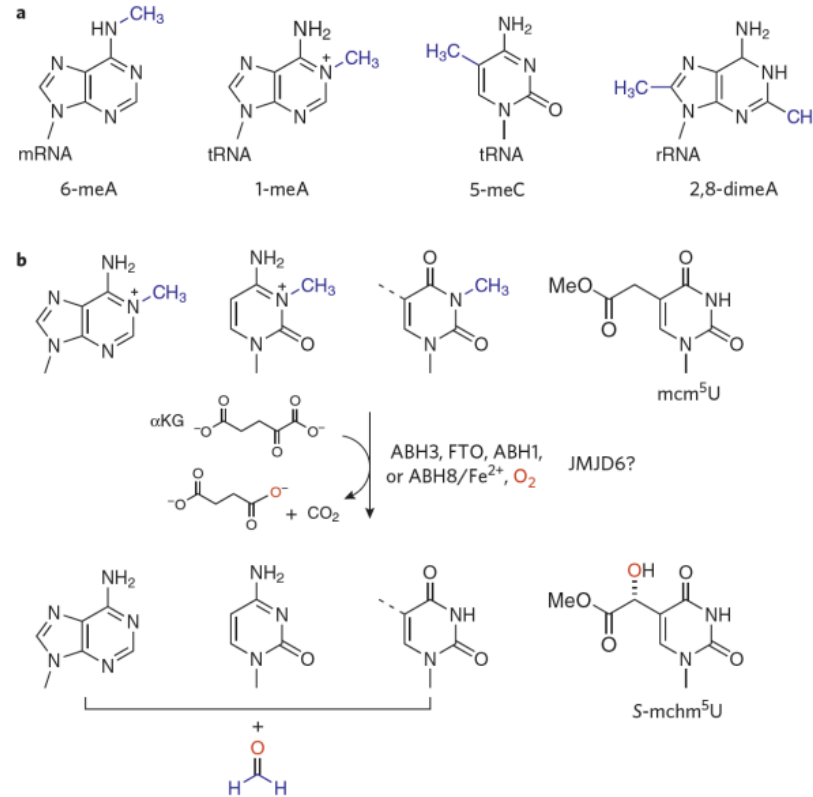
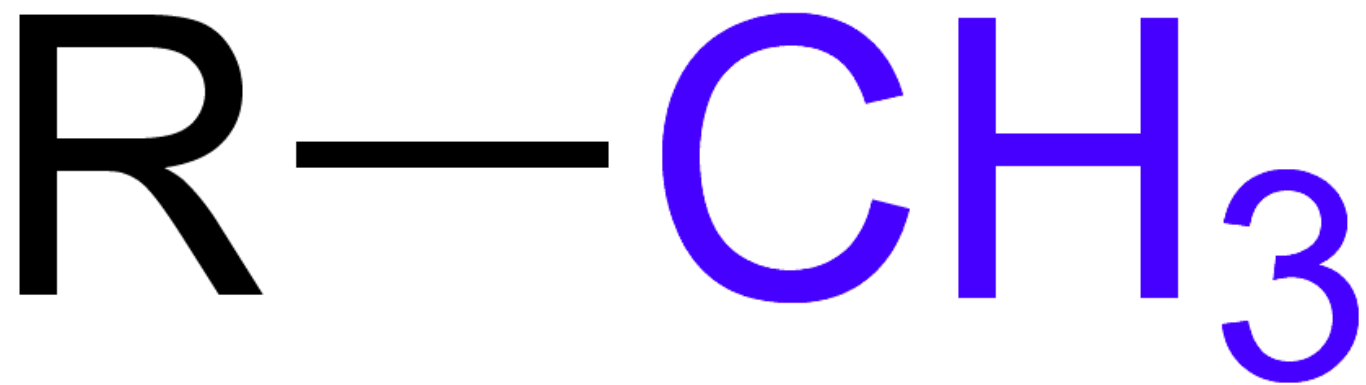
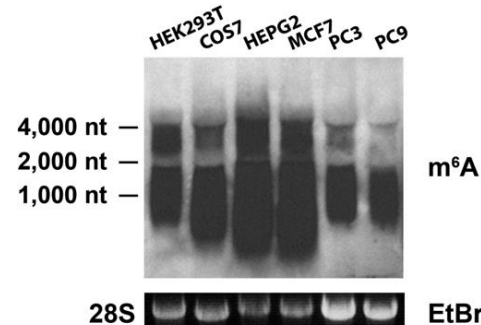
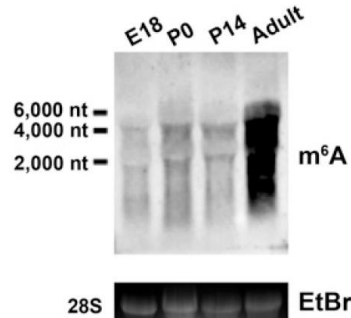
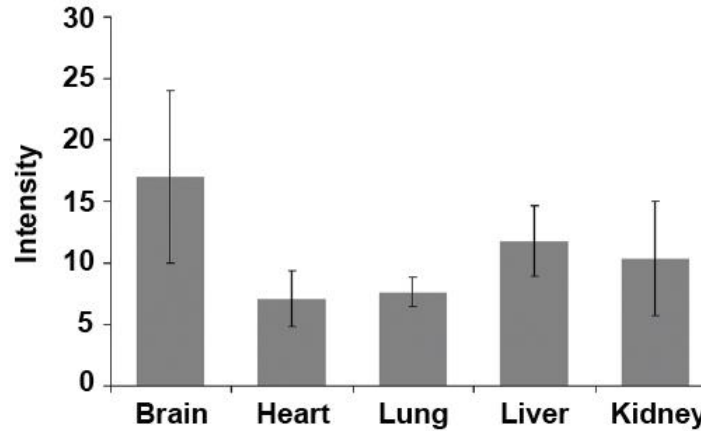
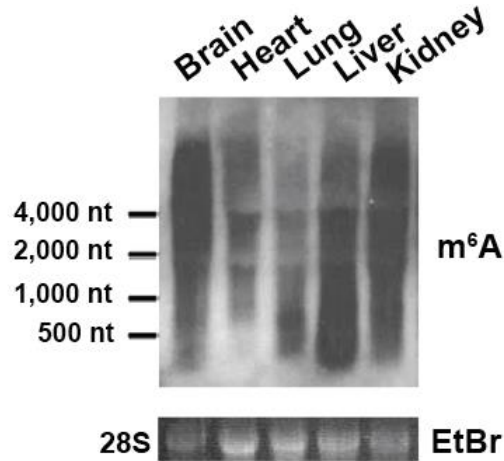


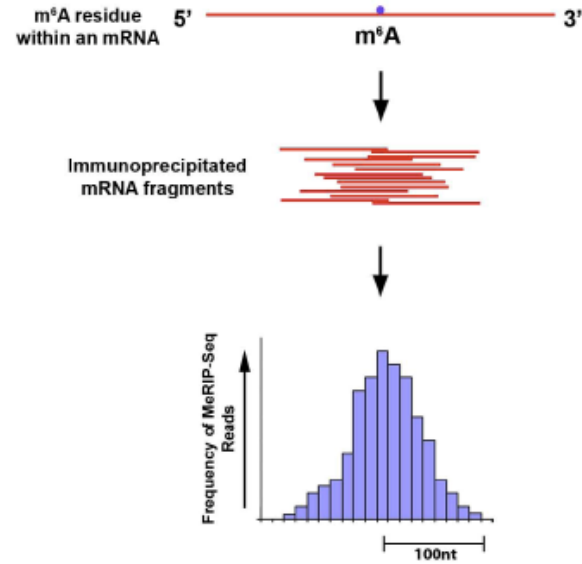
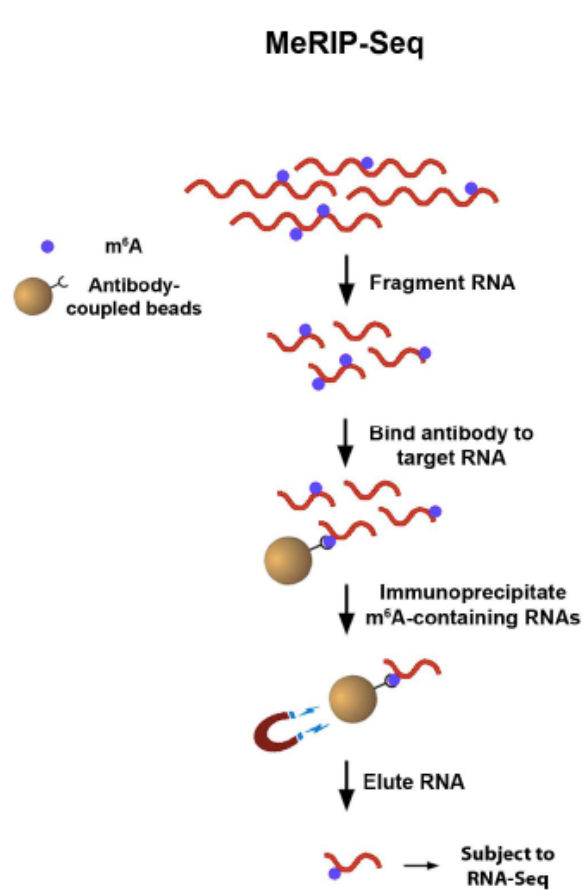
Figure 1 | Examples of RNA modification and demodification that may impact biological regulation. **(a)** Selected examples of RNA base methylation. **(b)** A group of dioxygenases that use iron, α -ketoglutarate and dioxygen to perform oxidation of modified RNA bases for demethylation or hypermethylation.



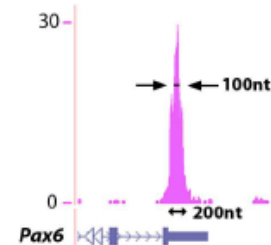
Methylation is important for methyl-6 adenosine (m^6A) in RNA, and is more prominent in brain & adults



A new method: MeRIP-Seq

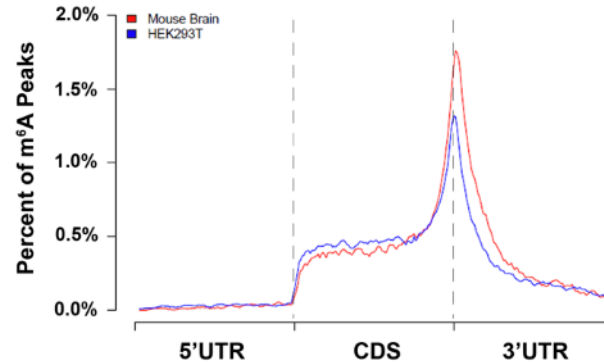


C

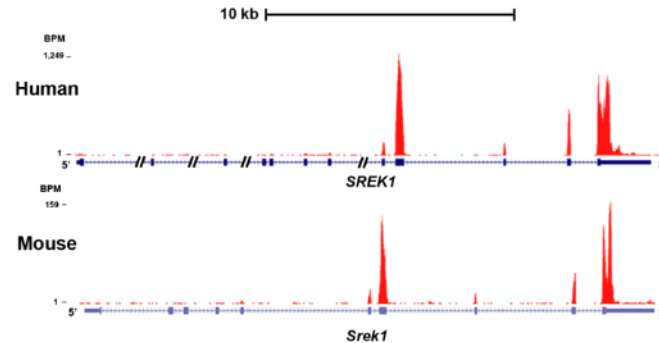


Conservations of signal and sites in >10,000 orthologous genes

D



E



Global mapping of RNA modifications started in 2012

The birth of the Epitranscriptome: deciphering the function of RNA modifications

Yogesh Saletore^{1,2,3}, Kate Meyer⁴, Jonas Korlach⁵, Igor D Vilfan⁵, Samie Jaffrey⁴ and Christopher E Mason^{1,2,*}

Abstract

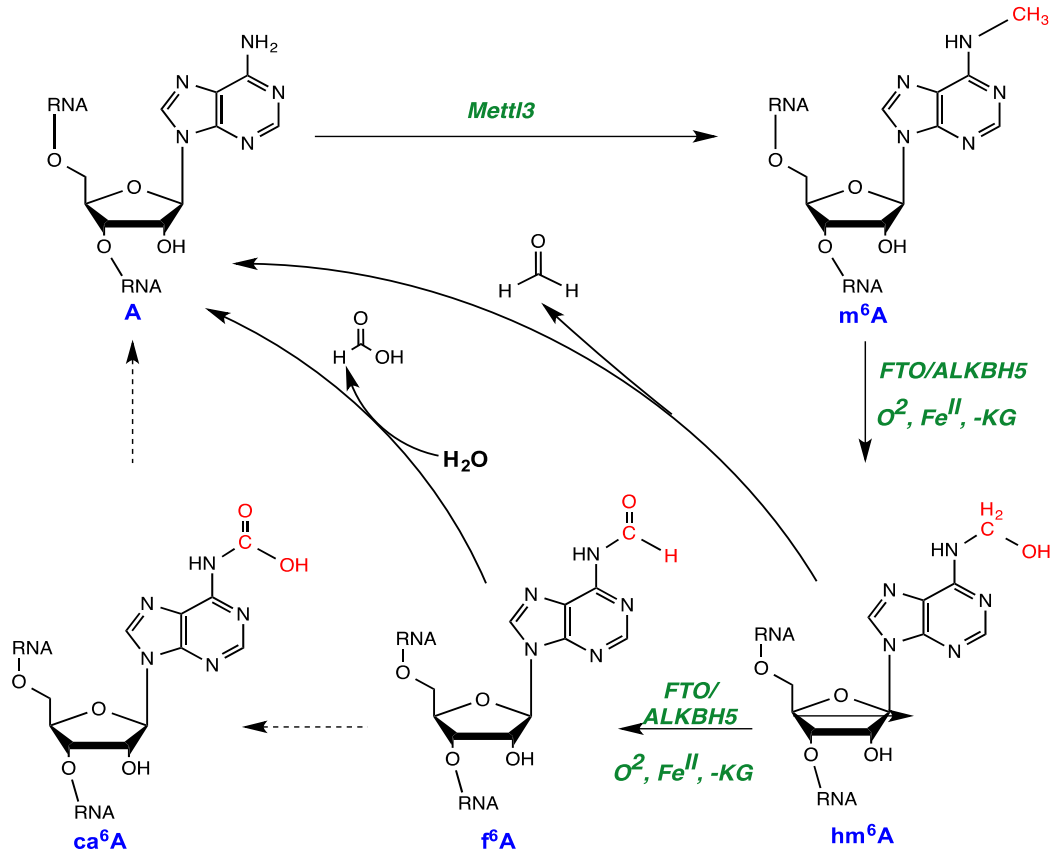
Recent studies have found methyl-6-adenosine in thousands of mammalian genes, and this modification is most pronounced near the beginning of the 3' UTR. We present a perspective on current work and new single-molecule sequencing methods for detecting RNA base modifications.

Keywords epigenetics, epigenomics, epitranscriptome, m⁶A, methyl-6-adenosine, methyladenosine, N⁶-methyladenosine, RNA modifications

Project [10]. Similarly, cell-specific, post-translational modifications of proteins, sometimes referred to collectively as the 'epiproteome' [11], are essential mechanisms necessary for the regulation of protein activity, folding, stability and binding partners. Elucidating the roles of protein and DNA modifications has had a major impact on our understanding of cellular signaling, gene regulation and cancer biology [12].

However, our understanding of an additional regulatory layer of biology that rests between DNA and proteins is still in its infancy; namely, the multitude of RNA modifications that together constitute the 'Epitranscriptome'. There are currently 107 known RNA base modifications, with the majority of these having been reported in tRNAs

RNA modifications give a new layer of cellular regulation



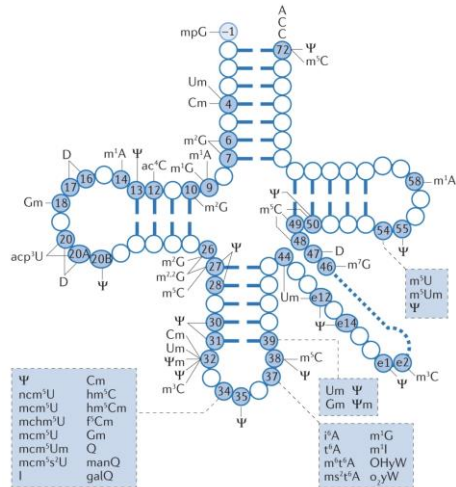
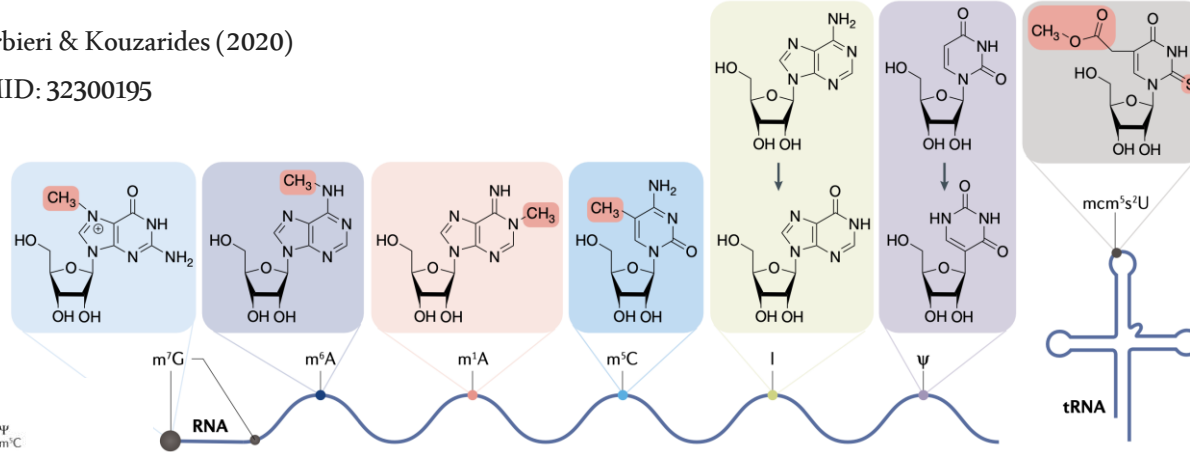
Over 400 RNA modifications known today

Barbieri & Kouzarides (2020)

PMID: 32300195

Syzuki (2021)

PMID: 33658722

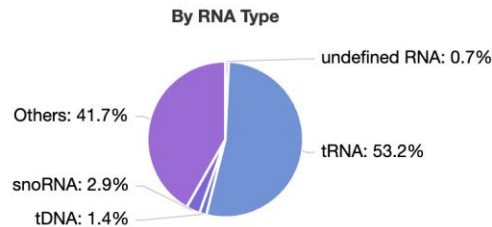
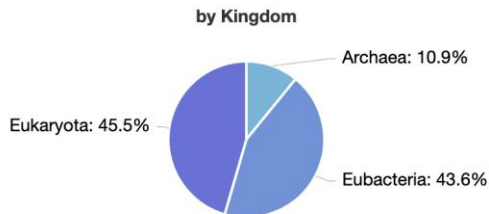


- ➡ 25% of all nucleotides in human tRNA are modified, but it can function without modifications
- ➡ Bacterial tRNA and rRNA contain significantly less modifications than human tRNA and rRNA and bacterial mRNA has no modifications
- ➡ RNA mods evolved to distinguish bacterial from mammalian RNA

MODomics DB

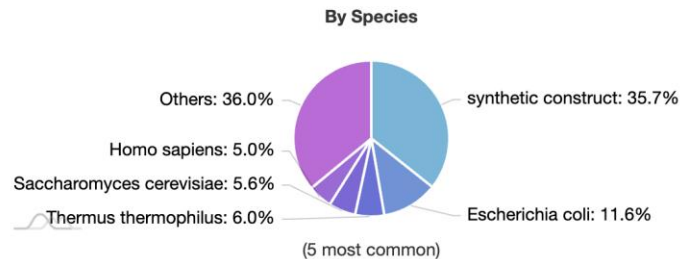
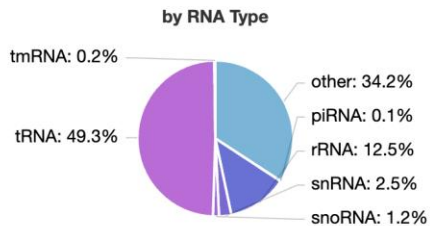
Modifications

MODOMICS hosts **421** different RNA **modified residues**



Sequences

MODOMICS hosts **1925** different RNA **Sequences**



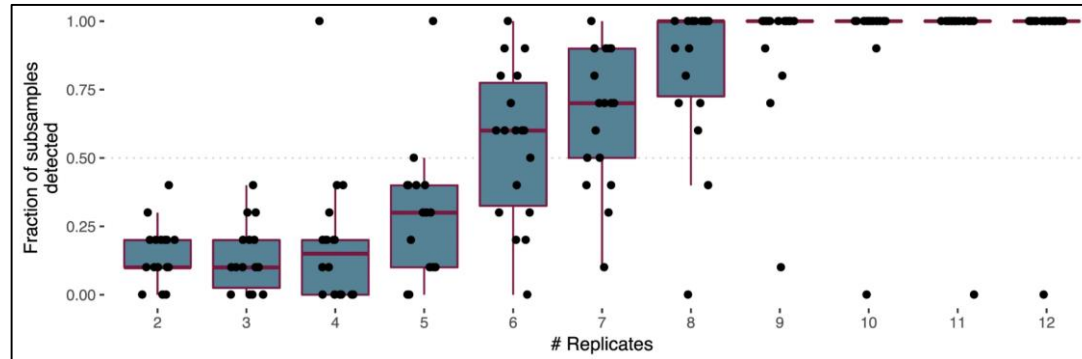
But, MeRIP-seq (m6A-seq) requires many replicates for reliable signal

Limits in the detection of m⁶A changes using MeRIP/m⁶A-seq

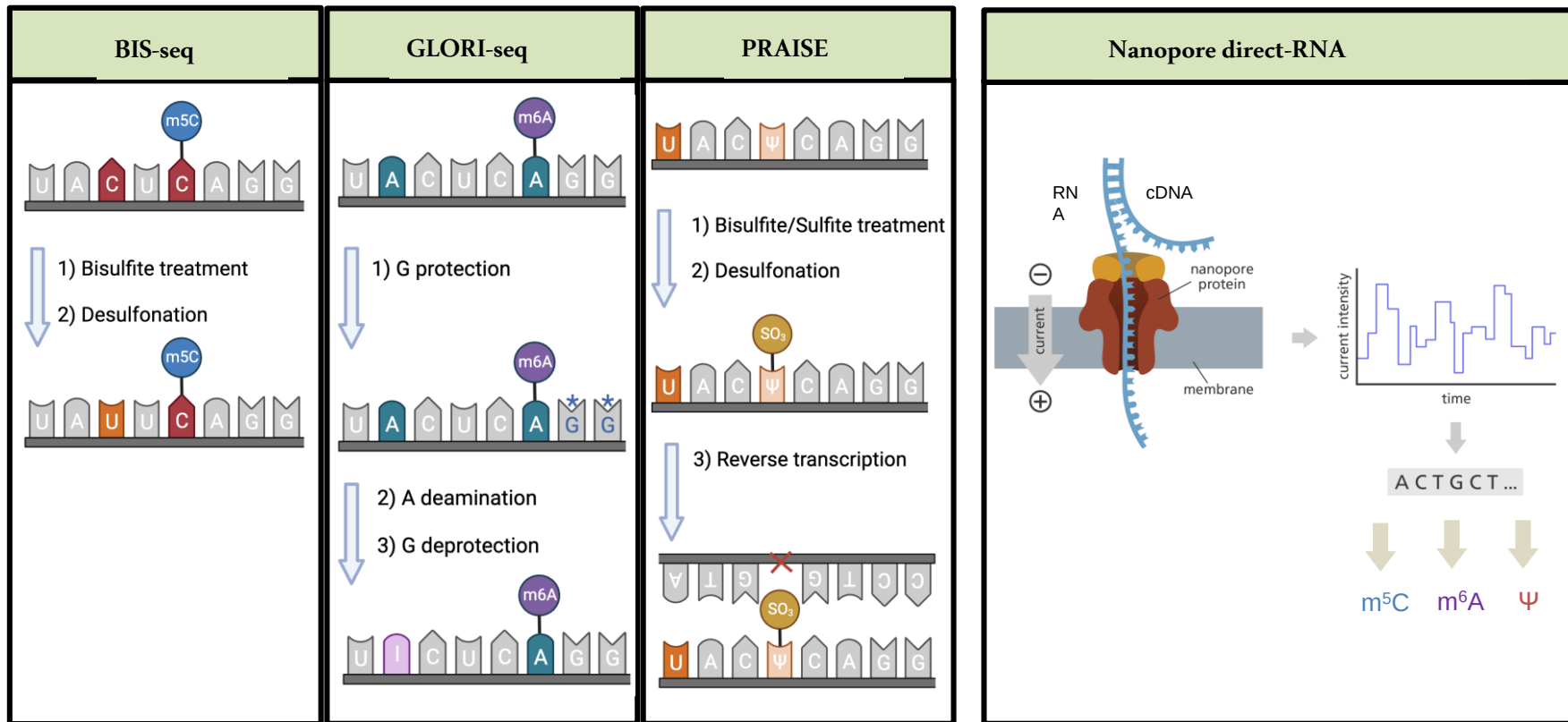
[Alexa B. R. McIntyre](#) , [Nandan S. Gokhale](#), [Leandro Cerchiatti](#), [Samie R. Jaffrey](#), [Stacy M. Horner](#)  & [Christopher E. Mason](#) 

[Scientific Reports](#) **10**, Article number: 6590 (2020) | [Cite this article](#)

19k Accesses | **116** Citations | **12** Altmetric | [Metrics](#)



We've now deployed several methods for detecting RNA modifications, but direct RNA-seq with nanopore is the most comprehensive; we get **all mods & long reads**



[nature](#) > [nature genetics](#) > [comment](#) > article

Comment | Published: 15 July 2021

A call for direct sequencing of full-length RNAs to identify all modifications

[Juan D. Alfonzo](#), [Jessica A. Brown](#), [Peter H. Byers](#), [Vivian G. Cheung](#) , [Richard J. Maraia](#) & [Robert L. Ross](#)

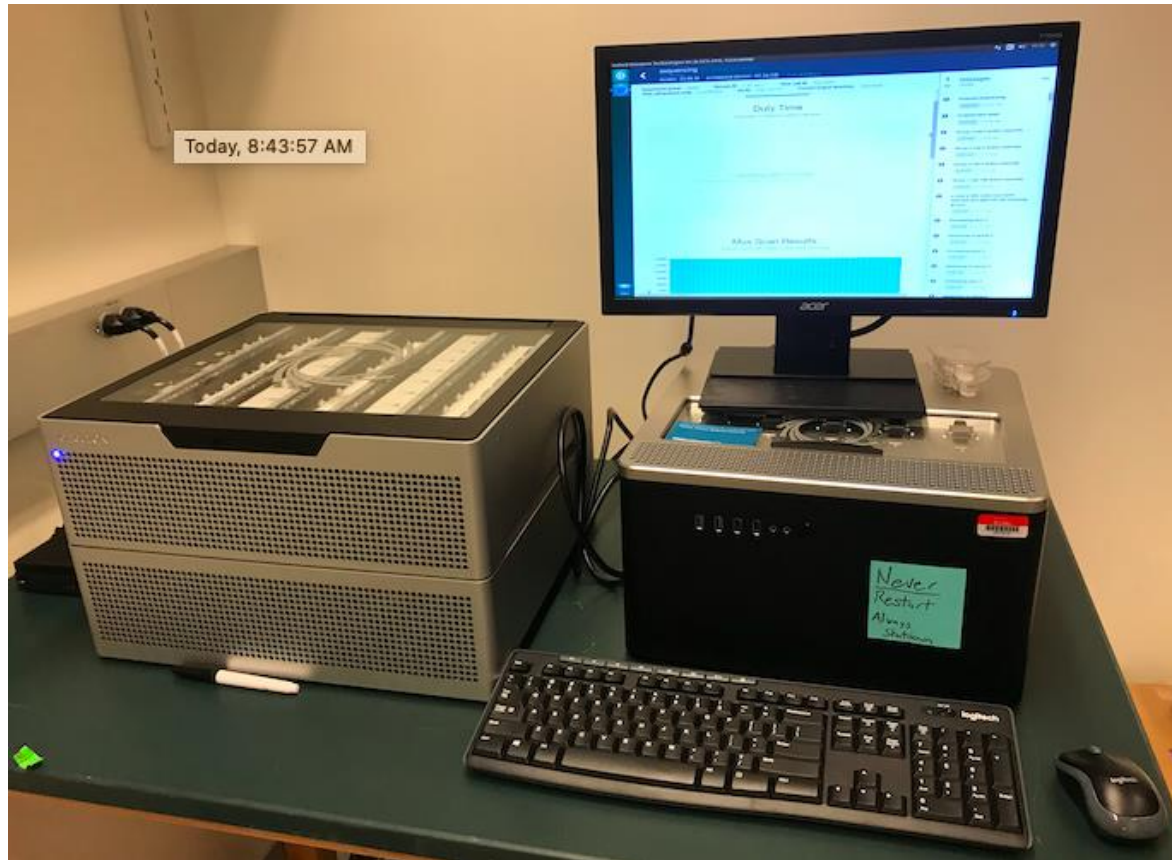
[Nature Genetics](#) **53**, 1113–1116 (2021) | [Cite this article](#)

14k Accesses | **28** Citations | **92** Altmetric | [Metrics](#)

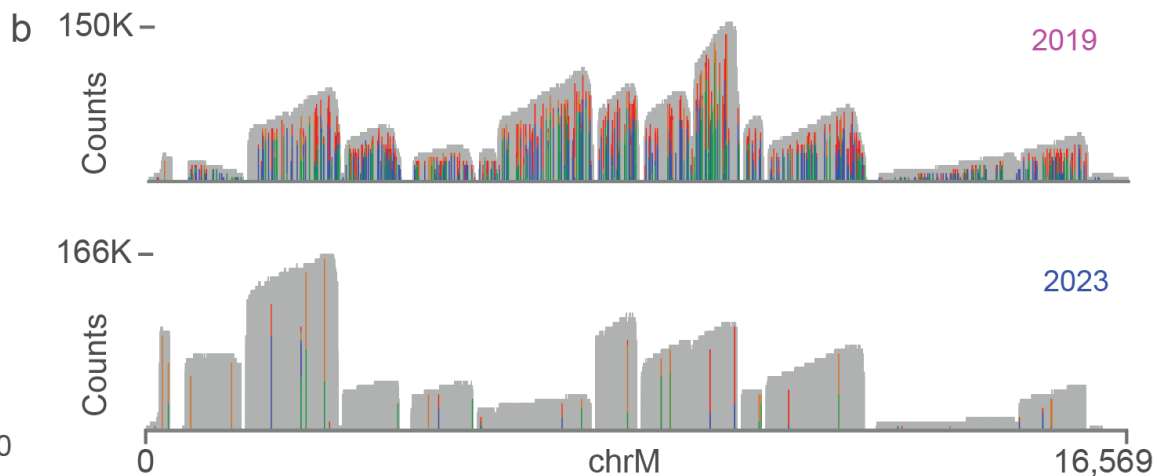
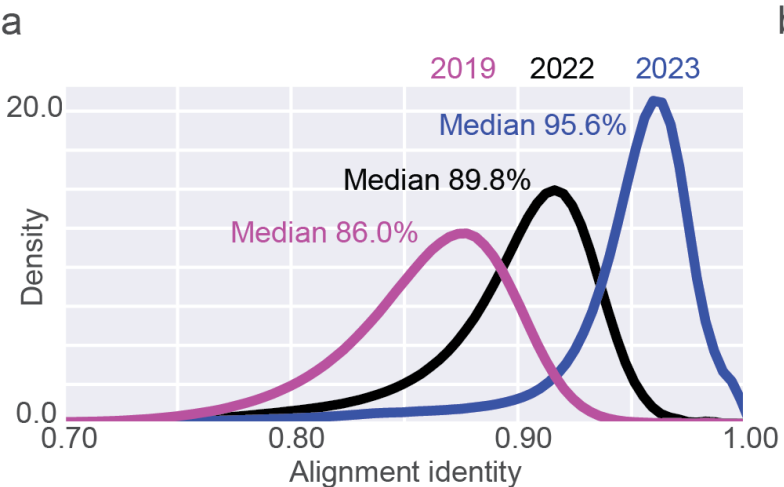
For most organisms, DNA sequences are available, but the complete RNA sequences are not. Here, we call for technologies to sequence full-length RNAs with all their modifications.

<https://www.nature.com/articles/s41588-021-00903-1>

Long reads! ONT PromethION

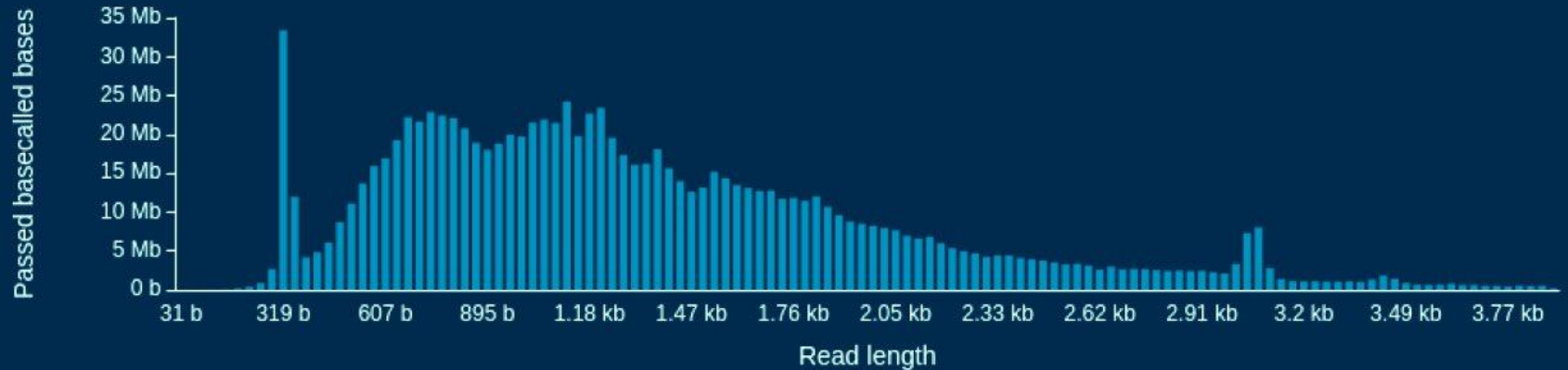


Direct RNA-sequencing from ONT finally reached maturity in 2023



Direct RNA runs can show a large range of transcript sizes

Estimated N50: 1.18 kb



New Oxford Nanopore Technologies (ONT) pricing enables efficiency at scale, if you have the cash for it

Flow Cell Pack Size	Number of Flow Cells	Cost per Flow Cell (USD)	Cost for Flow Cells (USD)	Cost of Libray Prep (USD)	Total Cost (USD)	Cost/Sample
1	4	\$855.00	\$3,420.00	\$399.33	\$3,819.33	\$954.83
3	12	\$855.00	\$10,260.00	\$1,198.00	\$11,458.00	\$954.83
8	32	\$820.00	\$26,240.00	\$3,194.67	\$29,434.67	\$919.83
24	96	\$785.00	\$75,360.00	\$9,584.00	\$84,944.00	\$884.83
48	192	\$745.00	\$143,040.00	\$19,168.00	\$162,208.00	\$844.83
128	512	\$680.00	\$348,160.00	\$51,114.67	\$399,274.67	\$779.83
256	1024	\$630.00	\$645,120.00	\$102,229.33	\$747,349.33	\$729.83
720	2880	\$600.00	\$1,728,000.00	\$287,520.00	\$2,015,520.00	\$699.83

SQK-RNA004, Direct RNA Sequencing Kit & Flow Cell
April 29, 2024 pricing from ONT web site